

Algebra 2

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1 Introduction

The course roughly covers the following three parts. You may refer to course website of Math 371 Spring 2020 at UPenn in my personal page for related materials.

Part I: Bilinear forms. Symmetric forms, Hermitian forms, and skew-symmetric forms. Orthogonality. Spectral Theorem. Conics and Quadrics. Key examples of classical groups, and their basic properties. Lie algebra (for such groups).

Part II: Group representations. Irreducible representations and unitary representations. Characters. Schur's Lemma. Modules over principal ideal domains. Free modules. Group rings. Noetherian rings. Structure of Abelian groups. Maschke's theorem. Constructions of representations, et cetera.

Part III: Field extensions, algebraic extensions and algebraic closures, splitting fields, separable and inseparable extensions, Galois extensions, Galois correspondences, cyclotomic extensions, solvability by radicals, et cetera.

2 Bilinear Forms

2.1 Basic Definitions

Definition 2.1 (Bilinear Form). *Let V be a vector space over a field $\mathbb{K}$. A map*

$$\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{K}$$

*is called a **bilinear form** if it is linear in both components. That is:*

1. $\langle au + v, w \rangle = a\langle u, w \rangle + \langle v, w \rangle$
2. $\langle u, av + w \rangle = a\langle u, v \rangle + \langle u, w \rangle$

for all $u, v, w \in V$ and $a \in \mathbb{K}$.

Definition 2.2 (Symmetric and Skew-Symmetric Forms). *A bilinear form $\langle \cdot, \cdot \rangle$ is called:*

1. **Symmetric** if $\langle v, w \rangle = \langle w, v \rangle$ for all $v, w \in V$.
2. **Skew-symmetric** (or **alternating**) if $\langle v, w \rangle = -\langle w, v \rangle$ for all $v, w \in V$.

There are following examples of bilinear forms.

Example 2.1 (Euclidean Space). *Let $V = \mathbb{R}^n$. The **standard inner product** defined by*

$$\langle \mathbf{x}, \mathbf{y} \rangle_{st} = \sum_{i=1}^n x_i y_i = \mathbf{x}^T \mathbf{y}$$

is a symmetric bilinear form. It allows us to define the length of vectors and the angle between non-zero vectors.

Example 2.2 (Minkowski Space). *Let $V = \mathbb{R}^{n+1}$. The **Lorentz form** is defined by*

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + \cdots + x_{n-1} y_{n-1} - x_{n+1} y_{n+1}.$$

This is a symmetric bilinear form used in special relativity.

Example 2.3 (Matrix Space). *Let $V = M_{m \times n}(\mathbb{R})$. Define*

$$\langle A, B \rangle = \text{tr}(A^T B).$$

This is a symmetric bilinear form on the space of matrices.

For infinite-dimensional spaces, we have the following example.

Example 2.4. *Let V be the space of continuous real-valued functions on $[0, 1]$. Define*

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx.$$

This is a symmetric bilinear form on V .

2.2 Gram Matrices and Congruency

Next we consider finite-dimensional vector spaces V over a field $\mathbb{K}$ with $\dim V = n < \infty$. Let $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis of V .

Definition 2.3 (Gram Matrix). *The **Gram matrix** of a bilinear form $\langle \cdot, \cdot \rangle$ with respect to the basis $\mathcal{B}$ is the matrix $G_{\langle \cdot, \cdot \rangle, \mathcal{B}} \in M_n(\mathbb{K})$ defined by:*

$$(G_{\langle \cdot, \cdot \rangle, \mathcal{B}})_{ij} = \langle \mathbf{v}_i, \mathbf{v}_j \rangle.$$

By expansion of bilinearity, we have the following important property.

Proposition 2.1 (Matrix Representation of Bilinear Forms). *If $\mathbf{u} = \sum x_i \mathbf{v}_i$ and $\mathbf{w} = \sum y_j \mathbf{v}_j$ are vectors in V with coordinate vectors $\mathbf{x}, \mathbf{y} \in \mathbb{K}^n$, then the value of the bilinear form can be computed via matrix multiplication:*

$$\langle \mathbf{u}, \mathbf{w} \rangle = \mathbf{x}^T G_{\langle \cdot, \cdot \rangle, \mathcal{B}} \mathbf{y}. \quad (1)$$

In fact the formula (1) can be used to define bilinear forms from arbitrary matrices A .

So if we consider the space of bilinear forms on V , it has a natural structure of $\mathbb{K}$ -vector space structure and it is isomorphic to the space of $n \times n$ matrices over $\mathbb{K}$. To summarize, we have the following proposition.

Proposition 2.2 (Matrix representation of bilinear forms). *Let $\mathbf{Bil}(V)$ denote the space of bilinear forms on V . Then $\mathbf{Bil}(V)$ is a vector space over $\mathbb{K}$, and the map*

$$\begin{aligned} \mathbf{Bil}(V) &\rightarrow M_n(\mathbb{K}) \\ \langle \cdot, \cdot \rangle &\mapsto G_{\langle \cdot, \cdot \rangle, \mathcal{B}} \end{aligned}$$

is a vector space isomorphism between the space of bilinear forms on V and the space of $n \times n$ matrices over $\mathbb{K}$.

The symmetric and skew-symmetric bilinear forms correspond to symmetric and skew-symmetric matrices, respectively.

Proposition 2.3. *Let $\langle \cdot, \cdot \rangle$ be a bilinear form on V and $A = G_{\langle \cdot, \cdot \rangle, \mathcal{B}}$ is the Gram matrix of $\langle \cdot, \cdot \rangle$ with respect to the basis $\mathcal{B}$. Then:*

1. $\langle \cdot, \cdot \rangle$ is symmetric if and only if A is a symmetric matrix, i.e. $A = A^T$.
2. $\langle \cdot, \cdot \rangle$ is skew-symmetric if and only if A is a skew-symmetric matrix, i.e. $A = -A^T$.

The dependence of Gram matrices on the choice of basis is described as follows.

Proposition 2.4 (Change of Basis). *Let $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and $\mathcal{B}' = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ be two bases of V . Let P be the transition matrix from $\mathcal{B}$ to $\mathcal{B}'$ (i.e., $\mathbf{w}_j = \sum_i P_{ij} \mathbf{v}_i$). Then the Gram matrices are related by:*

$$G_{\langle \cdot, \cdot \rangle, \mathcal{B}'} = P^T G_{\langle \cdot, \cdot \rangle, \mathcal{B}} P.$$

Proof. By definition:

$$\begin{aligned} (G_{\langle \cdot, \cdot \rangle, \mathcal{B}'})_{jk} &= \langle \mathbf{w}_j, \mathbf{w}_k \rangle = \left\langle \sum_i P_{ij} \mathbf{v}_i, \sum_l P_{lk} \mathbf{v}_l \right\rangle \\ &= \sum_i \sum_l P_{ij} \langle \mathbf{v}_i, \mathbf{v}_l \rangle P_{lk} \\ &= \sum_i \sum_l (P^T)_{ji} (G_{\langle \cdot, \cdot \rangle, \mathcal{B}})_{il} P_{lk} \\ &= (P^T G_{\langle \cdot, \cdot \rangle, \mathcal{B}} P)_{jk}. \end{aligned}$$

□

Definition 2.4 (Congruency). *Two square matrices A and B are called **congruent** if there exists an invertible matrix P such that $B = P^T A P$. It is straightforward to verify that congruency is an equivalence relation on the set of square matrices.*

Since all the invertible $n \times n$ matrices can appear as the change of basis matrix for an n -dimensional vector space, so two matrices are congruent if and only if they represent the same bilinear form under two bases.

Remark 2.1. *Coordinate change of bilinear forms corresponds to matrix congruency, whereas linear operators change coordinates via similarity ($P^{-1} A P$).*

Definition 2.5 (Isometry). *Let $(V_1, \langle \cdot, \cdot \rangle_1)$ and $(V_2, \langle \cdot, \cdot \rangle_2)$ be vector spaces equipped with bilinear forms. A linear map $f : V_1 \rightarrow V_2$ is called an **isometry** if*

$$\langle f(\mathbf{u}), f(\mathbf{v}) \rangle_2 = \langle \mathbf{u}, \mathbf{v} \rangle_1$$

for all $\mathbf{u}, \mathbf{v} \in V_1$.

Theorem 2.1. *Two finite-dimensional $\mathbb{K}$ -vector spaces with bilinear forms are isometric if and only if their Gram matrices (under any chosen bases) are congruent.*

3 Symmetric Forms

Throughout this subsection, assume $\langle \cdot, \cdot \rangle$ is a **symmetric** bilinear form on a vector space V over a field $\mathbb{K}$ (where $\text{char}(\mathbb{K}) \neq 2$, i.e. 2 is invertible in $\mathbb{K}$).

3.1 Diagonalization of Gram Matrices

The assumption on the characteristic is necessary for the following polarization identity.

Proposition 3.1 (Polarization Identity). *The bilinear form is completely determined by its quadratic form $q(\mathbf{v}) = \langle \mathbf{v}, \mathbf{v} \rangle$. Specifically:*

$$\langle \mathbf{v}, \mathbf{w} \rangle = \frac{1}{2} (\langle \mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w} \rangle - \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{w}, \mathbf{w} \rangle).$$

This implies that if $\langle \cdot, \cdot \rangle$ is not identically zero, there must exist some vector $\mathbf{v}$ such that $\langle \mathbf{v}, \mathbf{v} \rangle \neq 0$.

Theorem 3.1 (Diagonalization / Orthogonal Basis). *There exists a basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of V such that the Gram matrix of $\langle \cdot, \cdot \rangle$ is diagonal. That is, $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for $i \neq j$.*

Proof. We proceed by induction on $n = \dim V$.

1. **Base case:** If $\langle \cdot, \cdot \rangle \equiv 0$, any basis works. If $n = 1$, any basis works.
2. **Inductive step:** Assume the statement holds for dimensions $< n$. If $\langle \cdot, \cdot \rangle \equiv 0$, we are done. Otherwise, by the polarization identity, there exists $\mathbf{v}_1 \in V$ such that $\langle \mathbf{v}_1, \mathbf{v}_1 \rangle \neq 0$ (such a vector is called non-isotropic).

Define $W = \{\mathbf{w} \in V \mid \langle \mathbf{v}_1, \mathbf{w} \rangle = 0\}$. This is the orthogonal complement of the line spanned by $\mathbf{v}_1$. Consider the map $\phi : V \rightarrow \mathbb{K}$ given by $\mathbf{w} \mapsto \langle \mathbf{v}_1, \mathbf{w} \rangle$. Since $\langle \mathbf{v}_1, \mathbf{v}_1 \rangle \neq 0$, the map is non-zero, hence surjective. Thus $\dim W = \dim(\ker \phi) = n - 1$.

We claim $V = \text{span}(\mathbf{v}_1) \oplus W$. For any $\mathbf{v} \in V$, let

$$\mathbf{w} = \mathbf{v} - \frac{\langle \mathbf{v}, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} \mathbf{v}_1.$$

Then a direct check shows $\langle \mathbf{w}, \mathbf{v}_1 \rangle = 0$, so $\mathbf{w} \in W$. Thus $\mathbf{v} \in \text{span}(\mathbf{v}_1) + W$. The intersection is zero because $\mathbf{v}_1$ is not in W .

By the induction hypothesis, W admits an orthogonal basis $\{\mathbf{v}_2, \dots, \mathbf{v}_n\}$. Then $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthogonal basis for V .

□

Under this orthogonal basis, the Gram matrix is diagonal:

$$G = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix},$$

where $d_i = \langle \mathbf{v}_i, \mathbf{v}_i \rangle$.

Remark 3.1. The proof of the diagonalization can also be obtained from the theory of nondegeneracy criterion on subspaces (see Exercise 4.3).

3.2 Sylvester's Law of Inertia

When $\mathbb{K} = \mathbb{R}$, we can scale the basis vectors to normalize the diagonal entries coefficients to be 1, -1 , or 0. More precisely, we choose the basis vectors as follows:

- If $d_i > 0$, replace $\mathbf{v}_i$ by $\frac{1}{\sqrt{d_i}} \mathbf{v}_i$.
- If $d_i < 0$, replace $\mathbf{v}_i$ by $\frac{1}{\sqrt{-d_i}} \mathbf{v}_i$.
- If $d_i = 0$, leave $\mathbf{v}_i$ unchanged.

After this scaling, the Gram matrix becomes diagonal with entries in $\{1, -1, 0\}$. In fact, these numbers are determined by the bilinear form itself, independent of the choice of basis.

Theorem 3.2 (Sylvester's Law of Inertia). *Let $\langle \cdot, \cdot \rangle$ be a real symmetric bilinear form on V . There exists a basis under which the Gram matrix is diagonal with entries in $\{1, -1, 0\}$. Usually, the basis is ordered such that:*

$$G = \begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0_{n-p-q} \end{pmatrix}.$$

Furthermore, the integers p (index of positivity) and q (index of negativity) are invariants depending only on $\langle \cdot, \cdot \rangle$, not on the choice of basis.

The triple $(p, q, n - p - q)$ is called the **signature** of the form.

Proof of Uniqueness. The coordinate transformation allows us to write any symmetric matrix A as congruent to a diagonal matrix with diagonal entries $d_i \in \{1, -1, 0\}$. Suppose we have two such decompositions yielding indices (p, q) and (p', q') . Consider the subspaces corresponding to the basis vectors:

- Basis $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ gives p positive, q negative terms. Let $V^+ = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_p)$. Then $\dim V^+ = p$.
- Basis $\mathcal{B}' = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ gives p' positive, q' negative terms. Let $W^{\leq 0} = \text{span}(\mathbf{w}_{p'+1}, \dots, \mathbf{w}_n)$. Then $\dim W^{\leq 0} = n - p'$.

If $p > p'$, then $\dim V^+ + \dim W^{\leq 0} > n$. So the dimension of the intersection

$$\dim V^+ \cap W^{\leq 0} = \dim V^+ + \dim W^{\leq 0} - \dim(V^+ + W^{\leq 0}) \geq \dim V^+ + \dim W^{\leq 0} - \dim V > 0$$

Then there is a nonzero vector $\mathbf{x} \in V^+ \cap W^{\leq 0}$. Write $\mathbf{x} = \sum_{i=1}^p a_i \mathbf{v}_i = \sum_{j=p'+1}^n b_j \mathbf{w}_j$. Then $\langle \mathbf{x}, \mathbf{x} \rangle = \sum a_i^2 > 0$ (from V^+) and $\langle \mathbf{x}, \mathbf{x} \rangle = -\sum_{j=p'+1}^{p'+q'} b_j^2 \leq 0$ (from $W^{\leq 0}$). This is a contradiction. Thus $p \leq p'$. By symmetry, $p' \leq p$, so $p = p'$. A similar argument shows $q = q'$. $\square$

Corollary 3.1. *Two real symmetric matrices A and B of order n are congruent if and only if they have the same positive index of inertia and negative index of inertia.*

Remark 3.2. *Even though the signature $(p, q, n - p - q)$ is unique, the specific orthogonal basis achieving this signature is not unique. The subspaces V^+ contributing the positive part (or the subspaces for the negative part) are not unique. But the subspace corresponding to the zero part is unique. Try to define this subspace intrinsically. It is called the **radical** of the form.*

3.3 Positive Definite Forms

In the proof of Sylvester's law of inertia, we have constructed subspaces on which the quadratic form shows positive and negative behavior.

Definition 3.1 (Definiteness). *Let V be a real vector space and $\langle \cdot, \cdot \rangle$ be a symmetric bilinear form.*

1. $\langle \cdot, \cdot \rangle$ is **positive definite** (denoted $\langle \cdot, \cdot \rangle > 0$) if $\langle \mathbf{v}, \mathbf{v} \rangle > 0$ for all $\mathbf{v} \neq \mathbf{0}$.
2. $\langle \cdot, \cdot \rangle$ is **negative definite** (denoted $\langle \cdot, \cdot \rangle < 0$) if $\langle \mathbf{v}, \mathbf{v} \rangle < 0$ for all $\mathbf{v} \neq \mathbf{0}$.
3. $\langle \cdot, \cdot \rangle$ is **positive semi-definite** (denoted $\langle \cdot, \cdot \rangle \geq 0$) if $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ for all $\mathbf{v} \in V$.
4. $\langle \cdot, \cdot \rangle$ is **negative semi-definite** (denoted $\langle \cdot, \cdot \rangle \leq 0$) if $\langle \mathbf{v}, \mathbf{v} \rangle \leq 0$ for all $\mathbf{v} \in V$.

Proposition 3.2. *Let V be a finite-dimensional real vector space with a symmetric form $\langle \cdot, \cdot \rangle$. The form $\langle \cdot, \cdot \rangle$ is positive definite if and only if its index of positivity p equals $\dim(V)$.*

The proof for the uniqueness of signature also shows the following **intrinsic** characterization.

Proposition 3.3 (Characterization of Signature). *The positive index of inertia p of a symmetric form $\langle \cdot, \cdot \rangle$ on V can be characterized by:*

$$p = \max\{\dim W \mid W \subseteq V \text{ is a subspace where } \langle \cdot, \cdot \rangle|_W \text{ is positive definite}\}.$$

Similarly, the negative index q is the maximal dimension of a subspace where $\langle \cdot, \cdot \rangle$ is negative definite.

When the symmetric form is positive definite, we also call the corresponding Gram matrix a **positive definite matrix**. More properties of positive definite matrices are summarized in the exercises.

3.4 Euclidean Spaces

Definition 3.2 (Euclidean Space). A real vector space V equipped with a positive definite symmetric bilinear form $\langle \cdot, \cdot \rangle$ is called a **Euclidean space** or an **inner product space**.

See Exercise 4.3 for the definition of nondegeneracy and that if $(V, \langle \cdot, \cdot \rangle)$ is an inner product space, then it is non-degenerate and all its subspaces are also non-degenerate.

Proposition 3.4. Every Euclidean space of dimension n is isometric to the standard Euclidean space $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{st})$.

Example 3.1 (Polynomial Space). Let $V = P_n(\mathbb{R}) = \{f(x) \in \mathbb{R}[x] \mid \deg f \leq n\}$ (sometimes denoted $\mathbb{R}[x]_{\leq n}$). Define the bilinear form:

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx.$$

Since $\int_0^1 (f(x))^2 dx > 0$ for any non-zero polynomial, $\langle \cdot, \cdot \rangle$ is positive definite. Consider the standard basis $\{1, x, x^2, \dots\}$. The Gram matrix G under this basis has entries:

$$G_{ij} = \langle x^{i-1}, x^{j-1} \rangle = \int_0^1 x^{i+j-2} dx = \frac{1}{i+j-1}.$$

This matrix is known as the **Hilbert matrix**.

3.5 Gram-Schmidt process and QR Decomposition

While Sylvester's theorem guarantees an orthogonal basis, in Euclidean spaces we can construct an **orthonormal** basis algorithmically from any given basis.

Definition 3.3 (Orthonormal Basis). A basis $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ of a Euclidean space $(V, \langle \cdot, \cdot \rangle)$ is **orthonormal** if

$$\langle \mathbf{w}_i, \mathbf{w}_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

In terms of matrices, an orthonormal basis corresponds to orthogonal matrices.

Definition 3.4 (Orthogonal Matrix). A square real matrix Q is called an **orthogonal matrix** if its column vectors form an orthonormal basis of $\mathbb{R}^n$ under the standard inner product. Equivalently, Q is orthogonal if and only if $Q^T Q = I$ or $Q Q^T = I$, or all the row vectors of Q form an orthonormal basis of $\mathbb{R}^n$.

Theorem 3.3 (Gram-Schmidt Process). Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an arbitrary basis of a Euclidean space V . One can construct an orthonormal basis $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ such that

$$\text{span}(\mathbf{w}_1, \dots, \mathbf{w}_k) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$$

for all $k = 1, \dots, n$.

Algorithm. The construction proceeds inductively:

1. Set $\tilde{\mathbf{w}}_1 = \mathbf{v}_1$. Since basis vector are not zero, the inner product $\langle \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_1 \rangle > 0$. Normalize it to obtain $\mathbf{w}_1$:

$$\mathbf{w}_1 = \frac{\tilde{\mathbf{w}}_1}{\sqrt{\langle \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_1 \rangle}}.$$

2. Set $\tilde{\mathbf{w}}_2 = \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{w}_1 \rangle \mathbf{w}_1$. This vector satisfies $\langle \tilde{\mathbf{w}}_2, \mathbf{w}_1 \rangle = 0$. From the construction, we also know that

$$\text{span}(\tilde{\mathbf{w}}_2, \mathbf{w}_1) = \text{span}(\mathbf{v}_1, \mathbf{v}_2).$$

So $\tilde{\mathbf{w}}_2 \neq 0$. Normalize it:

$$\mathbf{w}_2 = \frac{\tilde{\mathbf{w}}_2}{\sqrt{\langle \tilde{\mathbf{w}}_2, \tilde{\mathbf{w}}_2 \rangle}}.$$

3. In general, for step k , define

$$\tilde{\mathbf{w}}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \langle \mathbf{v}_k, \mathbf{w}_j \rangle \mathbf{w}_j$$

Then

$$\langle \tilde{\mathbf{w}}_k, \mathbf{w}_i \rangle = 0 \text{ for } i < k,$$

Inductively we have

$$\text{span}(\mathbf{w}_1, \dots, \mathbf{w}_{k-1}), \tilde{\mathbf{w}}_k = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}), \tilde{\mathbf{w}}_k = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}, \mathbf{v}_k)$$

so $\tilde{\mathbf{w}}_k \neq 0$. Then normalize:

$$\mathbf{w}_k = \frac{\tilde{\mathbf{w}}_k}{\sqrt{\langle \tilde{\mathbf{w}}_k, \tilde{\mathbf{w}}_k \rangle}}.$$

□

To summarize, the Gram-Schmidt process constructs an orthonormal basis $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ from any given basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ such that the transition matrix from $\{\mathbf{w}_i\}$ to $\{\mathbf{v}_i\}$ is upper triangular with positive diagonal entries $\frac{1}{\sqrt{\langle \tilde{\mathbf{w}}_i, \tilde{\mathbf{w}}_i \rangle}}$. Upper triangularity follows from the fact that the subspaces spanned by the first k basis vectors are preserved. So the change of basis matrix has the form:

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ 0 & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & p_{nn} \end{pmatrix},$$

where $p_{ii} = \frac{1}{\sqrt{\langle \tilde{\mathbf{w}}_i, \tilde{\mathbf{w}}_i \rangle}} > 0$. The transition matrix from the orthonormal basis $\{\mathbf{w}_i\}$ back to the original basis $\{\mathbf{v}_i\}$ is then given by P^{-1} , which is also upper triangular with positive diagonal entries.

$$(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n) P^{-1}$$

In terms of matrices, this is called the **QR decomposition** of a matrix.

Definition 3.5 (QR Decomposition). *If V is the standard Euclidean space $\mathbb{R}^n$, then any basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ gather together to form an invertible matrix A . The orthonormal basis vectors $Q = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ form an orthogonal matrix. The decomposition above can be rewritten as*

$$A = QR,$$

where $R = P^{-1}$ is an upper triangular matrix with positive diagonal entries. This is called the **QR decomposition** of the matrix A .

The uniqueness of the QR decomposition is stated in Problem 4.2

4 Exercises

4.1 Useful practices

Please submit solutions to the following problems in this section. Some problems help you to review the material we have learned, and some problems introduce useful concepts and theorems not covered in class.

Exercise 4.1. Practice the Gram-Schmidt process and the QR decomposition. You can choose either one of the following two problems to solve.

1. Let $V = P_{\leq 2}(\mathbb{R})$ be the vector space of real polynomials with degree at most 2. Define an inner product on V by

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx$$

Given basis $1, x, x^2$ for V , use the Gram-Schmidt process to find an orthonormal basis for V .

2. Calculate the QR decomposition for the matrix

$$A = \begin{pmatrix} 3 & 2 & 100 \\ 4 & 0 & 0 \\ 0 & 0 & -5 \end{pmatrix}.$$

Exercise 4.2. Prove the uniqueness of the QR decomposition: if A is an $n \times n$ invertible real matrix then there exists a unique $n \times n$ orthogonal matrix Q and a unique $n \times n$ upper triangular matrix R with positive diagonal entries such that $A = QR$. (You only need to prove the uniqueness part; the existence part is given by the Gram-Schmidt process.)

Exercise 4.3. Let V be a finite-dimensional vector space over field F . Define a symmetric form on V to be a bilinear form $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$ similar as the real case. We call the symmetric form **non-degenerate** if for any $\mathbf{v} \in V$, $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ for all $\mathbf{w} \in V$ implies $\mathbf{v} = 0$.

1. Show that the symmetric form $\langle \cdot, \cdot \rangle$ is non-degenerate if and only if for some (and hence any) basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of V , the Gram matrix $A = (\langle \mathbf{v}_i, \mathbf{v}_j \rangle)_{1 \leq i, j \leq n}$ is invertible.
2. Assume V has a nondegenerate symmetric form $\langle \cdot, \cdot \rangle$. Let W be a subspace of V . Define the orthogonal complement of W to be

$$W^\perp = \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{w} \rangle = 0 \text{ for all } \mathbf{w} \in W\}.$$

Prove that $W \oplus W^\perp = V$ if and only if the restriction of $\langle \cdot, \cdot \rangle$ on W is non-degenerate.

3. When $\mathbb{R}$ is the base field, show that a positive definite symmetric form is non-degenerate.
4. For an inner product space V over $\mathbb{R}$, show that for any subspace W of V , $W \oplus W^\perp = V$.
5. In four-dimensional Minkowski space with the Lorentz form, find an one-dimensional subspace W such that W and $W^\perp$ do not form a direct sum of the whole space.

Exercise 4.4. In this problem, you will prove the **Cauchy-Schwarz inequality for Euclidean spaces** in an inner product space V . The **norm of an inner product space** is defined by $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$. The Cauchy-Schwarz inequality states that for any $\mathbf{u}, \mathbf{v} \in V$,

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \cdot \|\mathbf{v}\|.$$

Moreover, equality holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent. You can choose any one of the following two methods to prove it.

1. Assume $\mathbf{v} \neq 0$. Consider the quadratic function of λ

$$f(\lambda) = \langle \mathbf{u} - \lambda \mathbf{v}, \mathbf{u} - \lambda \mathbf{v} \rangle$$

Show that this function is non-negative and deduce the Cauchy-Schwarz inequality from this. Show that equality holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

2. There is another method to reduce the Cauchy-Schwarz inequality to two dimensional case. Assume $\mathbf{u}$ and $\mathbf{v}$ are linearly independent (otherwise the inequality is trivial). Let $W = \text{span}\{\mathbf{u}, \mathbf{v}\}$. Show that the Cauchy-Schwarz inequality holds in V if it holds in W . Then prove the Cauchy-Schwarz inequality in two-dimensional inner product space by directly considering the standard inner product on $\mathbb{R}^2$.

Exercise 4.5. We call a symmetric matrix A **positive definite** if it is the Gram matrix of any positive definite symmetric form.

1. Prove that a symmetric matrix A is positive definite if and only if there exists an invertible matrix P such that $A = P^T P$.
2. Show that if A is positive definite, then its determinant is positive.
3. Prove that a two by two symmetric matrix is positive definite if and only if it has positive trace and positive determinant.

Exercise 4.6. In the following, you will prove the **criterion for positive definiteness by principal minors**. A **principal minor** of a matrix A is the determinant of a square submatrix obtained by deleting certain rows and the corresponding columns. A **leading principal minor** is a principal minor obtained by deleting the last $n - k$ rows and columns for some k . In the following, show that a symmetric matrix A is positive definite if and only if all its leading principal minors are positive.

1. Show that if A is positive definite, then all its principal minors are positive. (Hint: consider the restriction of the corresponding symmetric form on the subspace spanned by the first k basis vectors.)
2. Use induction to show that if all leading principal minors of A are positive, then matrix A is positive definite. (Hint: use problem 4.3 and induction.)

Exercise 4.7. Let A be a real symmetric matrix where the diagonal elements are all 2, the elements on the two sub-diagonals are all -1 , and all other elements are 0. Prove that A is positive definite.

Exercise 4.8. Artin chapter 8 1.1. Show that a bilinear form $\langle \cdot, \cdot \rangle$ on a real vector space V is a sum of a symmetric form and a skew-symmetric form. (skew-symmetric means alternating)

Exercise 4.9. Let g be a bilinear form on a real vector space V . Prove that if g satisfies $g(\mathbf{x}, \mathbf{y}) = 0$ if and only if $g(\mathbf{y}, \mathbf{x}) = 0$, then g is either symmetric or alternating.

4.2 Optional problems

If you would like to try some additional problems, you can find them here and you do not need to submit them.

Exercise 4.10. Prove that the **Hilbert matrix** of order n ,

$$H_n = \left(\frac{1}{i+j-1} \right)_{n \times n}$$

is a positive definite matrix. (Hint: Use the symmetric form in Problem 4.1 (1).)

Exercise 4.11. Prove the **reversed Cauchy-Schwarz inequality in Minkowski space**:

For all $\mathbf{v} = (v_0, v_1, \dots, v_n), \mathbf{w} = (w_0, w_1, \dots, w_n) \in \mathbb{R}^{n+1}$ satisfying

$$v_0^2 - v_1^2 - \dots - v_n^2 > 0 \text{ and } w_0^2 - w_1^2 - \dots - w_n^2 > 0,$$

prove the following inequality

$$(v_0^2 - v_1^2 - \dots - v_n^2)(w_0^2 - w_1^2 - \dots - w_n^2) \leq (v_0 w_0 - v_1 w_1 - \dots - v_n w_n)^2$$

and determine the necessary and sufficient condition for equality to hold.

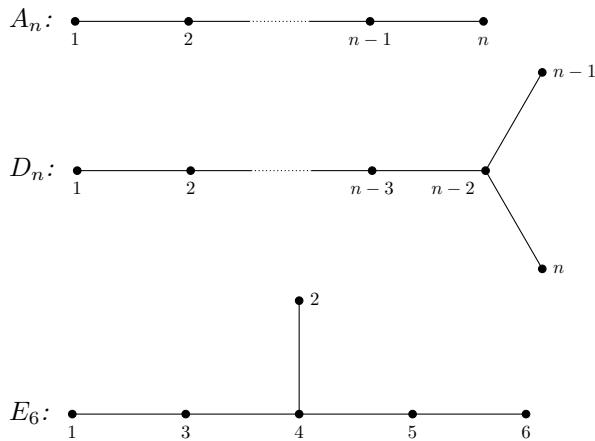
In terms of the Lorentz form $\langle \mathbf{v}, \mathbf{w} \rangle = v_0 w_0 - v_1 w_1 - \dots - v_n w_n$, the inequality can be rewritten as

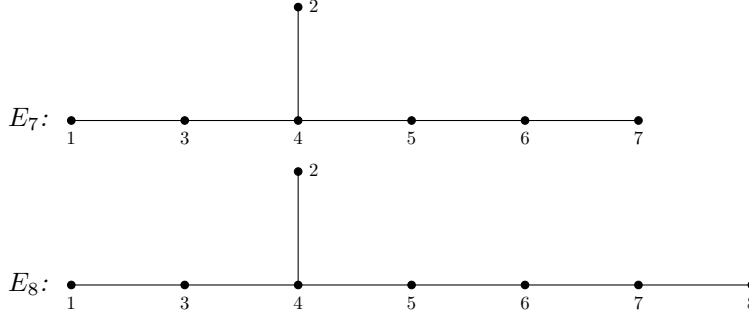
$$\langle \mathbf{v}, \mathbf{v} \rangle \langle \mathbf{w}, \mathbf{w} \rangle \leq \langle \mathbf{v}, \mathbf{w} \rangle^2.$$

when $\langle \mathbf{v}, \mathbf{v} \rangle > 0$ and $\langle \mathbf{w}, \mathbf{w} \rangle > 0$ (in physics $\mathbf{v}$ and $\mathbf{w}$ are called time-like vectors and this implies two time-like vectors have positive product under the Lorentz form).

Hint: Use similar method as in Problem 4.4 (2).

Exercise 4.12 (Challenge). In this problem, you will prove the **Cartan matrices** associated to **ADE Dynkin diagrams** are positive definite. For a graph Γ with vertices $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, consider the $n \times n$ real symmetric matrix defined by $A_\Gamma = (a_{ij})_{n \times n}$, where $a_{ij} = 2$ when $i = j$, $a_{ij} = -1$ when $i \neq j$ and $\mathbf{v}_i, \mathbf{v}_j$ are adjacent (connected by an edge), and $a_{ij} = 0$ otherwise. Prove that for the following graphs Γ , A_Γ is positive definite:





In fact, these graphs are exactly those connected and whose corresponding matrices are positive definite.

5 Geometry of Euclidean spaces: distance and projection

Cauchy–Schwartz inequality allows us to define angles and lengths in Euclidean spaces, which leads to the study of geometry in these spaces.

Definition 5.1. Let $(V, \langle \cdot, \cdot \rangle)$ be a Euclidean space.

1. For any $\mathbf{v} \in V$, the **norm** (or length) of $\mathbf{v}$ is defined as

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

2. For any non-zero vectors $\mathbf{u}, \mathbf{v} \in V$, the **angle** $\theta \in [0, \pi]$ between $\mathbf{u}$ and $\mathbf{v}$ is defined by

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

Note that the angle is well-defined due to the Cauchy-Schwartz inequality. When the angle is $\frac{\pi}{2}$ or equivalently when $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, we say that the two vectors are **orthogonal**. The norm function satisfies the following properties:

1. $\|\mathbf{v}\| \geq 0$ for all $\mathbf{v} \in V$, and $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.
2. For any scalar $c \in \mathbb{R}$ and vector $\mathbf{v} \in V$, $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$.
3. (**Triangle inequality**) For any $\mathbf{u}, \mathbf{v} \in V$, $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.
4. (**Pithagorean theorem**) For any $\mathbf{u}, \mathbf{v} \in V$, if $\mathbf{u} \perp \mathbf{v}$, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

By triangle inequality, we can define the distance on V by $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$. The distance function satisfies the following properties:

1. $d(\mathbf{u}, \mathbf{v}) \geq 0$ for all $\mathbf{u}, \mathbf{v} \in V$, and $d(\mathbf{u}, \mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$.
2. $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$ for all $\mathbf{u}, \mathbf{v} \in V$.
3. (**Triangle inequality**) For any $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, $d(\mathbf{u}, \mathbf{w}) \leq d(\mathbf{u}, \mathbf{v}) + d(\mathbf{v}, \mathbf{w})$.

In topology, a distance function satisfying the above three properties is called a **metric**, and such a space V with metric d is called a **metric space**. The distance function can induce distance between subsets of V as follows:

Definition 5.2. Let (V, d) be a metric space. For any two subsets A, B of V , the **distance** between A and B is defined as

$$d(A, B) = \inf\{d(\mathbf{a}, \mathbf{b}) \mid \mathbf{a} \in A, \mathbf{b} \in B\}.$$

We will consider natural subsets in inner product spaces, for example, in space of functions with inner products defined by integrals, this is useful when we want to approximate a function by polynomials or trigonometric functions. Or in space of data points or matrices, the distance function helps us to construct clustering or compressing algorithms. The most natural form of subsets are subspaces.

Proposition 5.1. Let $(V, \langle \cdot, \cdot \rangle)$ be a Euclidean space, and W be a subspace of V . For any $\mathbf{v} \in V$, there exists a unique vector $\mathbf{w}_0 \in W$ such that

$$d(\mathbf{v}, W) = d(\mathbf{v}, \mathbf{w}_0).$$

Moreover, the vector $\mathbf{w}_0$ satisfies that $\mathbf{v} - \mathbf{w}_0 \in W^\perp$.

Proof. By the Gram-Schmidt process, we can find an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ of W and extend it to an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_n\}$ of V . Then any vector $\mathbf{v} \in V$ can be written as

$$\mathbf{v} = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{u}_i \rangle \mathbf{u}_i.$$

Define

$$\mathbf{w}_0 = \sum_{i=1}^k \langle \mathbf{v}, \mathbf{u}_i \rangle \mathbf{u}_i \in W. \quad (2)$$

Then for any $\mathbf{w} \in W$, we have

$$\begin{aligned} d(\mathbf{v}, \mathbf{w})^2 &= \|\mathbf{v} - \mathbf{w}\|^2 \\ &= \|\mathbf{v} - \mathbf{w}_0 + \mathbf{w}_0 - \mathbf{w}\|^2 \\ &= \|\mathbf{v} - \mathbf{w}_0\|^2 + \|\mathbf{w}_0 - \mathbf{w}\|^2 \quad (\text{since } \mathbf{v} - \mathbf{w}_0 \perp \mathbf{w}_0 - \mathbf{w}) \\ &\geq \|\mathbf{v} - \mathbf{w}_0\|^2. \end{aligned}$$

Thus, $d(\mathbf{v}, W) = d(\mathbf{v}, \mathbf{w}_0)$. The equality holds if and only if $\mathbf{w} = \mathbf{w}_0$. $\square$

In Problem 4.3 of Exercise 4, we have shown that for any subspace W of an inner product space V , $V = W \oplus W^\perp$. Thus, any vector $\mathbf{v} \in V$ can be uniquely written as $\mathbf{v} = \mathbf{w} + \mathbf{w}^\perp$ with $\mathbf{w} \in W$ and $\mathbf{w}^\perp \in W^\perp$. The formula (2) also gives an effective way to obtain such a decomposition via orthonormal basis of W .

Definition 5.3. The map $\text{Proj}_W: \mathbf{v} \rightarrow \mathbf{w}_0$ is called a **projection map** onto W and it is a linear transformation satisfying $\text{Proj}_W \circ \text{Proj}_W = \text{Proj}_W$.

Later a projection map will be an example of symmetric or self-adjoint operator.

Remark 5.1. Notice that the distance function is invariant under translations, i.e., for any $\mathbf{u}, \mathbf{v}, \mathbf{a} \in V$, $d(\mathbf{u} + \mathbf{a}, \mathbf{v} + \mathbf{a}) = d(\mathbf{u}, \mathbf{v})$. Thus, Proposition 5.1 can also be used to describe the distance between a point and an affine subspace (a translation of a subspace).

6 Orthogonal Matrices

In QR decomposition, we have already seen orthogonal matrices. An invertible $n \times n$ real matrix A can be viewed as a basis of $\mathbb{R}^n$ by taking all the column vectors. The column vectors of an orthogonal matrix form an orthonormal basis of Euclidean space $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{\text{st}})$. Another interpretation of matrices is that they represent linear transformations

$$T_A: \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad \mathbf{v} \mapsto A\mathbf{v}.$$

Under this interpretation, orthogonal matrices represent linear transformations that preserve the inner product, i.e.,

Proposition 6.1. *Any $A \in M_n(\mathbb{R})$ is orthogonal if and only if*

$$\langle T_A(\mathbf{u}), T_A(\mathbf{v}) \rangle_{\text{st}} = \langle \mathbf{u}, \mathbf{v} \rangle_{\text{st}}$$

for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$.

Proof. For any $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we have

$$\langle T_A(\mathbf{u}), T_A(\mathbf{v}) \rangle_{\text{st}} = (A\mathbf{u})^T (A\mathbf{v}) = \mathbf{u}^T A^T A \mathbf{v}.$$

If A is orthogonal, then $A^T A = I$, so $\langle T_A(\mathbf{u}), T_A(\mathbf{v}) \rangle_{\text{st}} = \mathbf{u}^T \mathbf{v} = \langle \mathbf{u}, \mathbf{v} \rangle_{\text{st}}$. Conversely, if $\langle T_A(\mathbf{u}), T_A(\mathbf{v}) \rangle_{\text{st}} = \langle \mathbf{u}, \mathbf{v} \rangle_{\text{st}}$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, then $\mathbf{u}^T A^T A \mathbf{v} = \mathbf{u}^T \mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. When $\mathbf{u} = \mathbf{e}_i$ and $\mathbf{v} = \mathbf{e}_j$, this implies that the ij -th entry of $A^T A$ is δ_{ij} , so $A^T A = I$ and A is orthogonal. $\square$

Such linear transformations are called **isometries** under the standard inner product by Definition 2.5. More generally,

Proposition 6.2. *For an n -dimensional Euclidean space $(V, \langle \cdot, \cdot \rangle)$, the isometries on V are exactly those linear transformations whose matrix representations under any orthonormal basis are orthogonal matrices.*

The proof is similar as Proposition 6.2.

All the isometries of $V, \langle \cdot, \cdot \rangle$ form a group under composition. So we have a subgroup of $\text{GL}(n, \mathbb{R})$ consisting of orthogonal matrices.

Definition 6.1. *An **orthogonal group** of order n is defined as*

$$\text{O}(n, \mathbb{R}) := \{A \in M_n(\mathbb{R}) \mid A^T A = I\}$$

For Euclidean space $(V, \langle \cdot, \cdot \rangle)$, we also define the orthogonal group as the group of isometries on the space

$$\text{O}(V) := \{T \in \text{GL}(V) \mid \langle T(\mathbf{u}), T(\mathbf{v}) \rangle = \langle \mathbf{u}, \mathbf{v} \rangle \text{ for all } \mathbf{u}, \mathbf{v} \in V\}$$

The map determinant is a group homomorphism

$$\det: \text{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}^\times$$

For an orthogonal matrix A , its determinant is equal to ± 1 , and both ± 1 can be achieved. Thus, we introduce the following normal subgroup of $\text{O}(n, \mathbb{R})$ by the kernel of the determinant map.

Definition 6.2. A *special orthogonal group* of order n is defined as

$$\mathrm{SO}(n, \mathbb{R}) := \{A \in \mathrm{O}(n, \mathbb{R}) \mid \det(A) = 1\}$$

The special orthogonal group represents all orientation-preserving isometries of Euclidean space.

Example 6.1. First by direct computation of orthogonal basis in $\mathbb{R}^2$, we know that any orthogonal matrix in $\mathrm{O}(2, \mathbb{R})$ has the form

$$\mathrm{O}(2, \mathbb{R}) = \left\{ \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \mid \theta \in \mathbb{R} \right\} \sqcup \left\{ \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \mid \theta \in \mathbb{R} \right\}$$

Computing the determinants, we see that

$$\mathrm{SO}(2, \mathbb{R}) = \left\{ \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \mid \theta \in \mathbb{R} \right\}$$

represents all rotations by angle θ in $\mathbb{R}^2$.

The other coset represents all reflections in $\mathbb{R}^2$. One important observation is that the composition of two reflections is a rotation, i.e., for any two reflection matrices $P_1, P_2 \in \mathrm{O}(2, \mathbb{R}) \setminus \mathrm{SO}(2, \mathbb{R})$, we have $P_1 P_2 \in \mathrm{SO}(2, \mathbb{R})$. If the two reflection axes form an angle $\frac{\theta}{2}$, then $P_1 P_2$ is the rotation by angle θ .

The concept of reflections can be generalized to higher dimensions.

Definition 6.3 (reflections). A linear transformation $T: V \rightarrow V$ on a Euclidean space $(V, \langle \cdot, \cdot \rangle)$ is called a **reflection** if there exists a one-dimensional subspace $L \subset V$ such that

(1) $T(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in L$;

(2) $T(\mathbf{w}) = -\mathbf{w}$ for all $\mathbf{w} \in L^\perp$.

Conversely, given any nonzero vector $\mathbf{u} \in V$, we can define a reflection T with respect to the line spanned by $\mathbf{u}$ by

$$T(\mathbf{v}) = \mathbf{v} - 2 \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \mathbf{u}.$$

This vector $\mathbf{u}$ is called a **normal vector** of the reflection. Such a reflection is also denoted by $s_{\mathbf{u}}$.

From the coset decomposition of $\mathrm{O}(n, \mathbb{R})$ by $\mathrm{SO}(n, \mathbb{R})$, we know that

$$\mathrm{O}(n, \mathbb{R}) = \mathrm{SO}(n, \mathbb{R}) \cup P \cdot \mathrm{SO}(n, \mathbb{R}),$$

where $P = \begin{pmatrix} -1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$ is a reflection matrix with determinant -1 .

6.1 Spectral theorem

The classification of similar classes of matrices is equivalently the orbit decomposition of the conjugation action of $\text{GL}(n, \mathbb{R})$ on $M_n(\mathbb{R})$. The congruent classes of matrices are the orbit decomposition of the congruence action of $P \in \text{GL}(n, \mathbb{R})$ on $A \in M_n(\mathbb{R})$ by $P^T A P$. If we restrict the action to $\text{O}(n, \mathbb{R})$, we have the following definitions and theorem.

Definition 6.4.

- (1) For $A, B \in M_n(\mathbb{R})$, we say A, B are **orthogonally similar** if there exists $Q \in \text{O}(n, \mathbb{R})$ such that $A = Q^T B Q$ ($= Q^{-1} B Q$).
- (2) If A is orthogonally similar to a diagonal matrix, we say A is **orthogonally diagonalizable** (over $\mathbb{R}$).

We mainly deal with three special kinds of matrices:

1. symmetric matrices, i.e., $A^T = A$;
2. skew-symmetric matrices, i.e., $A^T = -A$;
3. orthogonal matrices, i.e., $A^T A = I$.

We study their orbits under orthogonal group actions. Among them, symmetric matrices have the best diagonalization property.

Theorem 6.1 (Spectral Theorem for Real Symmetric Matrices). *Real symmetric matrices are orthogonally diagonalizable.*

Proof. Let $A \in M_n(\mathbb{R})$ with $A = A^T$. We proceed by induction on n . First, we show that A has a real eigenvalue. Let $\lambda \in \mathbb{C}$ be an eigenvalue of A , and let $A\mathbf{v} = \lambda\mathbf{v}$ for some $\mathbf{v} \in \mathbb{C}^n \setminus \{0\}$. Then

$$\overline{\mathbf{v}}^T A \mathbf{v} = \lambda \overline{\mathbf{v}}^T \mathbf{v},$$

where $\overline{\mathbf{v}}^T \mathbf{v} > 0$. Since

$$(\overline{\mathbf{v}}^T A \mathbf{v})^T = \overline{\mathbf{v}}^T A \mathbf{v},$$

it follows that $\lambda \in \mathbb{R}$. Since $A - \lambda I$ is a singular real matrix, the solution $\mathbf{v}$ to $(A - \lambda I)\mathbf{v} = 0$ can be chosen in $\mathbb{R}^n$.

Assuming $(\mathbf{v}, \mathbf{v}) = 1$, extend $\mathbf{v} = \mathbf{v}_1$ to an orthonormal basis of $\mathbb{R}^n$, denoted by

$$Q_1 = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$$

Then

$$\begin{aligned} A Q_1 &= (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) \\ &= (\lambda \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) \\ &= (\mathbf{v}_1 \cdots \mathbf{v}_n) \begin{pmatrix} \lambda & * \\ 0 & * \end{pmatrix} \end{aligned}$$

Thus,

$$A = Q_1 \begin{pmatrix} \lambda & * \\ * & * \end{pmatrix} Q_1^{-1} = Q_1 \begin{pmatrix} \lambda & * \\ * & * \end{pmatrix} Q_1^T$$

Since A is symmetric, the above decomposition actually takes the form

$$A = Q_1 \begin{pmatrix} \lambda & 0 \\ 0 & * \end{pmatrix} Q_1^{-1}.$$

The result follows by induction on n . □

In the proof, we can also obtain a useful proposition of invariant subspace for real linear transformations. This is essentially because irreducible polynomials over $\mathbb{R}$ have degree at most 2.

Proposition 6.3 (Invariant subspace of real linear transformations). *Let $T: V \rightarrow V$ be a linear transformation on an $n \geq 1$ -dimensional real vector space V . Then there exists a one-dimensional or two-dimensional T -invariant subspace of V .*

Proof. Let $A \in M_n(\mathbb{R})$. Then A has a complex eigenvalue $\lambda = a + b\sqrt{-1} \in \mathbb{C}$ with eigenvector $\mathbf{v} \in \mathbb{C}^n \setminus \{0\}$. Write $\mathbf{v} = \mathbf{u} + \sqrt{-1}\mathbf{w}$ with $\mathbf{u}, \mathbf{w} \in \mathbb{R}^n$. Then

$$A(\mathbf{u} + \sqrt{-1}\mathbf{w}) = (a + b\sqrt{-1})(\mathbf{u} + \sqrt{-1}\mathbf{w}).$$

Equating real and imaginary parts, we have

$$\mathbf{u} = a\mathbf{u} - b\mathbf{w}, \quad \mathbf{w} = b\mathbf{u} + a\mathbf{w}.$$

So the subspace spanned by $\mathbf{u}$ and $\mathbf{w}$ is a A -invariant subspace in $\mathbb{R}^n$. For general V , choose a basis of V and identify V with $\mathbb{R}^n$. The result follows. $\square$

The notion of symmetric matrices can be generalized to self-adjoint operators on inner product spaces and we obtain a more intrinsic proof for the spectral theorem.

Definition 6.5. *Let V be a vector space equipped with a symmetric or skew-symmetric bilinear form g . A linear map $T: V \rightarrow V$ is called **self-adjoint** if*

$$g(T(\mathbf{u}), \mathbf{v}) = g(\mathbf{u}, T(\mathbf{v}))$$

holds for all $\mathbf{u}, \mathbf{v} \in V$.

Proposition 6.4. *Let V be a vector space with a symmetric or skew-symmetric bilinear form g . Suppose T is a self-adjoint linear map. If $W \subset V$ is an T -invariant subspace, then $W^\perp$ is also an T -invariant subspace.*

Proof. For any $\mathbf{u} \in W^\perp$ and $\mathbf{w} \in W$, we have

$$g(T(\mathbf{u}), \mathbf{w}) = g(\mathbf{u}, T(\mathbf{w})) \in g(\mathbf{u}, W) = 0.$$

Thus $T(\mathbf{u}) \in W^\perp$. $\square$

Now assume (V, g) is an inner product space, and $T: V \rightarrow V$ is a self-adjoint linear map. Let $\mathcal{B} = \{e_1, \dots, e_n\}$ be an orthonormal basis of V , and let A be the matrix representation of T with respect to $\mathcal{B}$, i.e.,

$$T(e_1, \dots, e_n) = (e_1, \dots, e_n)A$$

From $g(T(x), y) = g(x, T(y))$, we have:

$$\begin{pmatrix} T(e_1) \\ \vdots \\ T(e_n) \end{pmatrix} \cdot_g (e_1, \dots, e_n) = \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} \cdot_g (T(e_1), \dots, T(e_n))$$

That is,

$$A^T \begin{pmatrix} T(e_1) \\ \vdots \\ T(e_n) \end{pmatrix} \cdot_g (e_1, \dots, e_n) = \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} \cdot_g (e_1, \dots, e_n)A$$

Since $\mathcal{B}$ is an orthonormal basis,

$$\begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} \cdot_g (e_1, \dots, e_n) = G_{g, \mathcal{B}} = I_n,$$

Thus $A = A^T$. So we have the following proposition.

Proposition 6.5 (Self-adjoint operators and symmetric matrices). *Let (V, g) be an inner product space and $T: V \rightarrow V$ be a linear map. Then T is self-adjoint if and only if the matrix representation of T with respect to any orthonormal basis is symmetric.*

Theorem 6.2 (Spectral Theorem for Self-adjoint Operators). *Let (V, g) be a finite-dimensional inner product space over $\mathbb{R}$. Then any self-adjoint operator $T: V \rightarrow V$ is orthogonally diagonalizable, i.e., there exists an orthonormal basis of V consisting of eigenvectors of T .*

Proof. Proposition 6.3 shows that T has a one-dimensional or two-dimensional invariant subspace $W_1 \subset V$. If W_1 is two dimensional, under orthonormal basis of W_1 , the matrix representation of $T|_{W_1}$ is symmetric

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

The characteristic polynomial is

$$p(\lambda) = \lambda^2 - (a + c)\lambda + (ac - b^2).$$

The discriminant is

$$(a + c)^2 - 4(ac - b^2) = (a - c)^2 + 4b^2 \geq 0.$$

Thus $T|_{W_1}$ has a real eigenvalue, and hence W_1 has a one-dimensional invariant subspace. Therefore we can assume $\dim_{\mathbb{R}} W_1 = 1$. The orthogonal complement $W_1^\perp$ is also invariant by Proposition 6.4. Using induction on $\dim_{\mathbb{R}} V$, we can find an orthonormal basis of V consisting of eigenvectors of T . $\square$

Next, we look at the conjugacy classes of orthogonal group.

Theorem 6.3 (Conjugacy classes of orthogonal matrices). *Suppose $A \in O(n, \mathbb{R})$. Then A is orthogonally similar to*

$$\text{diag}\left\{\begin{pmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{pmatrix}, \begin{pmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{pmatrix}, \dots, 1, \dots, -1, \dots\right\},$$

Or equivalently, for any orthogonal transformation $T: V \rightarrow V$ on an n -dimensional Euclidean space $(V, \langle \cdot, \cdot \rangle)$, there exists an orthonormal basis of V such that the matrix representation of T with respect to this basis is block diagonal with blocks of the form in $SO(2, \mathbb{R})$, 1 or -1 .

Proof. We use the same method as in the proof of Theorem 6.2. By Proposition 6.3, there exists a one-dimensional or two-dimensional A -invariant subspace $W_1 \subset \mathbb{R}^n$. If $\dim_{\mathbb{R}} W_1 = 1$, then W_1 is spanned by an eigenvector of A with eigenvalue 1 or -1 since A is orthogonal. If $\dim_{\mathbb{R}} W_1 = 2$, then under an orthonormal basis of W_1 , the matrix representation of $A|_{W_1}$ is in $O(2, \mathbb{R})$. When the matrix is in $SO(2, \mathbb{R})$, it is of the form

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

When the matrix is not in $\text{SO}(2, \mathbb{R})$, it is of the form

$$\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}.$$

The characteristic polynomial is

$$p(\lambda) = \lambda^2 - 1.$$

So the eigenvalues are 1 and -1 . Thus, in any case, W_1 has a one-dimensional invariant subspace. The orthogonal complement $W_1^\perp$ is also invariant by Proposition 6.4. Using induction on n , we can find an orthonormal basis of $\mathbb{R}^n$ such that the matrix representation of A with respect to this basis is block diagonal with blocks of the form in $\text{SO}(2, \mathbb{R})$, 1 or -1 . $\square$

Example 6.2. For $A \in \text{SO}(3, \mathbb{R})$, it is orthogonally similar to

$$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Let

$$Q^T A Q = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and $Q = (v_1, v_2, v_3)$, then

$$A(v_1, v_2, v_3) = (v_1, v_2, v_3) \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

So the action of A on $\mathbb{R}^3$ is a rotation in the plane $\text{span}_{\mathbb{R}}\{v_1, v_2\}$.

Thus we have the following corollary in geometry.

Corollary 6.1. The composition of two rotations of three-dimensional Euclidean space along two intersecting lines is a rotation.

If we also generalize the concept of rotation to higher dimensions as orthogonal transformations whose matrix representations under some orthonormal basis has one block in $\text{SO}(2, \mathbb{R})$ and all others are 1 blocks, then we have the following corollary.

Corollary 6.2. The composition of two reflections of an n -dimensional Euclidean space along two hyperplanes is a rotation.

7 Singular Value Decomposition and Low Rank Approximation

In previous sections, we gave the structure theorem for self-adjoint operators on a given linear space, which is also known as the spectral theorem for self-adjoint operators. If $T: V \rightarrow W$ is a linear map between two different spaces, how do we find the canonical form of T ?

The theory of equivalence canonical forms for matrices tells us that, given a matrix $A \in M_{m \times n}(\mathbb{R})$, there exist $P \in \text{GL}(n, \mathbb{R})$, $Q \in \text{GL}(m, \mathbb{R})$ such that:

$$Q^{-1} A P = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$$

where $r = \text{rank } A$. Now we attach an inner product structure to the linear space, so we hope that P, Q can be chosen as orthogonal matrices.

Theorem 7.1 (Singular Value Decomposition). *Let $A \in M_{m \times n}(\mathbb{R})$. There exist $P \in O(n)$, $Q \in O(m)$ such that:*

$$A = QDP^T,$$

where

$$D = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{pmatrix}_{m \times n}, \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0.$$

The σ_i are called the **singular values** of A .

Remark 7.1. Suppose $A = QDP^T$ is the singular value decomposition of A . Let $Q = (w_1, \dots, w_m)$, $P = (v_1, \dots, v_n)$, then

$$A = \sum_{i=1}^r \sigma_i w_i v_i^T,$$

This is a commonly used form of the singular value decomposition.

Proof. We first prove the uniqueness of the singular values: If $A = QDP^T$, then

$$A^T A = P \text{diag}\{\sigma_1^2, \dots, \sigma_n^2\} P^T,$$

so $\sigma_1^2, \dots, \sigma_n^2$ are the eigenvalues of $A^T A$, which are uniquely determined by A after sorting.

Next, we prove the existence of the singular value decomposition: Since $A^T A$ is symmetric, there exists $P \in O(n)$ such that

$$P^{-1}(A^T A)P = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix},$$

where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$. Let $P = (v_1, \dots, v_n)$, then $A^T A v_i = \lambda_i v_i$.

We show that $Av_1, \dots, Av_n$ are mutually orthogonal:

$$\begin{aligned} \langle Av_i, Av_j \rangle_{\mathbb{R}^m} &= (Av_i)^T (Av_j) \\ &= v_i^T (A^T A v_j) \\ &= \lambda_j v_i^T v_j \\ &= \lambda_j \langle v_i, v_j \rangle_{\mathbb{R}^n}. \end{aligned}$$

Thus, it is 0 when $i \neq j$, and λ_i when $i = j$.

Assume $\lambda_1 \geq \cdots \geq \lambda_r > 0$, $\lambda_{r+1} = \cdots = \lambda_n = 0$. Let

$$w_i = \frac{Av_i}{\|Av_i\|_{\mathbb{R}^m}} = \frac{Av_i}{\sqrt{\lambda_i}}, \quad i = 1, \dots, r$$

Then $w_1, \dots, w_r$ are orthonormal. If we denote $\sigma_i := \sqrt{\lambda_i}$, then

$$(Av_1, \dots, Av_n) = (w_1, \dots, w_n) \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{pmatrix}_{m \times n}$$

That is,

$$AP = QD, \quad P \in O(n), \quad Q \in O(m).$$

□

7.1 Low-Rank Approximation and Application: Image Compression

Given a matrix $A = (a_{ij})_{m \times n}$, where a_{ij} represents the grayscale of the pixel at (i, j) , the amount of data needed to record this is mn , which is very large when m, n are large. Using SVD, we can write it as

$$A = \sum_{i=1}^{\min(m,n)} \sigma_i w_i v_i^T$$

Take $k \ll \min(m, n)$, then we have the approximation

$$A_k = \sum_{i=1}^k \sigma_i w_i v_i^T,$$

The storage data amount is now $k(m + n + 1)$.

Intuitively, A_k should be "very close" to A . Next, we describe this strictly. Define the **Frobenius inner product** on the matrix space $M_{m \times n}(\mathbb{R})$:

$$\langle A, B \rangle_F = \text{tr}(A^T B).$$

Then the distance between matrices A, B is

$$\sqrt{\langle A - B, A - B \rangle_F} = \sqrt{\sum_{i,j} (a_{ij} - b_{ij})^2}.$$

Under Frobenius inner product, we have the following important theorem about the **low-rank approximation** using SVD.

Theorem 7.2 (Eckart-Young, Schmidt). A_k is the closest matrix to A among matrices of rank $\leq k$, i.e.,

$$\|A - A_k\|_F = \min_{\text{rank } B \leq k} \|A - B\|_F.$$

Corollary 7.1. Let $A = QDP^T$ and $D = \text{diag}(\sigma_1, \dots, \sigma_n)$ be the singular value decomposition of matrix A , then

$$\|A\|_F = \sqrt{\sigma_1^2 + \cdots + \sigma_n^2},$$

and

$$\|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \cdots + \sigma_n^2}.$$

Before proving Theorem 7.2, we first present some important properties used in the proof.

Lemma 7.1. *Given matrices $A, B \in M_{m \times n}(\mathbb{R})$, for any $P \in O(n), Q \in O(m)$, we have*

$$\langle QAP^T, QBP^T \rangle_F = \langle A, B \rangle_F.$$

Lemma 7.2. *Given matrix $A \in M_{m \times n}(\mathbb{R})$, then*

$$\sigma_1(A) = \max_{0 \neq v \in \mathbb{R}^n} \frac{|Av|_{\mathbb{R}^m}}{|v|_{\mathbb{R}^n}}.$$

Proof. Consider the symmetric matrix $M = A^T A$. According to the Min-Max principle for eigenvalues of symmetric matrices, we have

$$\lambda_1(A^T A) = \max_{0 \neq v \in \mathbb{R}^n} \frac{\langle v, A^T A v \rangle_{\mathbb{R}^n}}{\langle v, v \rangle_{\mathbb{R}^n}} = \max_{0 \neq v \in \mathbb{R}^n} \frac{\langle Av, Av \rangle_{\mathbb{R}^m}}{\langle v, v \rangle_{\mathbb{R}^n}},$$

Thus,

$$\sigma_1(A) = \sqrt{\lambda_1(A^T A)} = \max_{0 \neq v \in \mathbb{R}^n} \frac{|Av|_{\mathbb{R}^m}}{|v|_{\mathbb{R}^n}}.$$

□

Lemma 7.3.

(1) *For any matrix $A \in M_{m \times n}(\mathbb{R})$, we have*

$$\sigma_\ell(A - A_k) = \sigma_{k+\ell}(A)$$

(2) *For any matrix $A, B \in M_{m \times n}(\mathbb{R})$, where $\text{rank } B \leq k$, then*

$$\sigma_\ell(A - B) \geq \sigma_{k+\ell}(A)$$

Proof. (1): Assume the singular value decomposition of A is denoted as

$$A = \sum_{i=1}^{\min\{m,n\}} \sigma_i w_i v_i^T,$$

then

$$A - A_k = \sum_{i=k+1}^{\min\{m,n\}} \sigma_i w_i v_i^T,$$

so $\sigma_\ell(A - A_k) = \sigma_{k+\ell}(A)$.

(2): Assume the singular value decomposition of A is denoted as

$$A = \sum_{i=1}^{\min\{m,n\}} \sigma_i w_i v_i^T.$$

We first prove the case $\ell = 1$: Let $W = \text{span}_{\mathbb{R}}\{v_1, \dots, v_{k+1}\}$. Since $\text{rank } B \leq k$, we have $\dim \ker B \geq n - k$, thus $\ker B \cap W \neq \emptyset$. Take $0 \neq v \in \ker B \cap W$, assume without loss of generality that $v = \sum_{i=1}^{k+1} a_i v_i$, then by Lemma 7.2,

$$\sigma_1(A - B) \geq \frac{|(A - B)v|_{\mathbb{R}^m}}{|v|_{\mathbb{R}^n}} = \frac{|Av|_{\mathbb{R}^m}}{|v|_{\mathbb{R}^n}} = \frac{\sqrt{\sum_{i=1}^{k+1} \sigma_i^2 a_i^2}}{\sqrt{\sum_{i=1}^{k+1} a_i^2}} \geq \sigma_{k+1}(A).$$

Now we prove the general case: By (1), we have

$$\begin{aligned}\sigma_\ell(A - B) &= \sigma_1((A - B) - (A - B)_{\ell-1}) \\ &= \sigma_1(A - (B + (A - B)_{\ell-1}))\end{aligned}$$

Since $\text{rank } B \leq k, \text{rank}(A - B)_{\ell-1} \leq \ell - 1$, we have

$$\text{rank}(B + (A - B)_{\ell-1}) \leq k + \ell - 1,$$

By the previous case,

$$\sigma_1(A - (B + (A - B)_{\ell-1})) \geq \sigma_{k+\ell}(A).$$

□

Now we give the proof of Theorem 7.2:

Proof.

$$\begin{aligned}\|A - B\|_F^2 &= \sum_{i=1}^n \sigma_i^2(A - B) \\ &\geq \sum_{i=1}^n \sigma_{i+k}^2(A) \\ &= \sum_{i=1}^n \sigma^2(A - A_k)^2 \\ &= \|A - A_k\|_F^2.\end{aligned}$$

□

7.2 Application: Low-dimensional Fitting (PCA)

Problem 7.1. Given m experiments, where each experiment yields an n -dimensional datum, assume $m \gg n$, then we obtain the following matrix:

$$A = \begin{pmatrix} \alpha_1^T \\ \alpha_2^T \\ \vdots \\ \alpha_m^T \end{pmatrix}.$$

How can we determine if $\alpha_1, \dots, \alpha_m$ lie near some lower-dimensional linear subspace?

Proposition 7.1. Given the singular value decomposition of a matrix $A \in M_{m \times n}(\mathbb{R})$ with $m > n$:

$$A = \sum_{i=1}^n \sigma_i w_i v_i^T.$$

Let $W_k = \text{span}_{\mathbb{R}}\{v_1, \dots, v_k\}$. Then W_k minimizes the following value:

$$\sum_{i=1}^m (\text{dist}(\alpha_i, W))^2,$$

where W is a k -dimensional subspace of $\mathbb{R}^n$.

Proof. For any k -dimensional subspace W , take an orthonormal basis $\{u_1, \dots, u_k\}$ of W . Then

$$\begin{aligned} (\text{dist}(\alpha_i, W))^2 &= \|\alpha_i - \text{Proj}_W \alpha_i\|^2 \\ &= \|\alpha_i - \sum_{j=1}^k \langle \alpha_i, u_j \rangle u_j\|^2 \\ &= \|\alpha_i^T - ((\alpha_i, u_1), \dots, (\alpha_i, u_k)) \begin{pmatrix} u_1^T \\ \vdots \\ u_k^T \end{pmatrix}\|_2^2. \end{aligned}$$

Thus,

$$\sum_{i=1}^m (\text{dist}(\alpha_i, W))^2 = \|A - \begin{pmatrix} \langle \alpha_1, u_1 \rangle & \dots & \langle \alpha_1, u_k \rangle \\ \vdots & & \vdots \\ \langle \alpha_m, u_1 \rangle & \dots & \langle \alpha_m, u_k \rangle \end{pmatrix} \begin{pmatrix} u_1^T \\ \vdots \\ u_k^T \end{pmatrix}\|_F^2.$$

On the other hand, since $\alpha = \sum_{i=1}^n \langle \alpha, v_i \rangle v_i$, we can write

$$\alpha^T = (\langle \alpha, v_1 \rangle, \dots, \langle \alpha, v_n \rangle) P,$$

where $P = (v_1, \dots, v_n)$. Thus

$$A = QDP^T,$$

where the i -th row of QD is $(\langle \alpha_i, v_1 \rangle, \dots, \langle \alpha_i, v_n \rangle)$. Therefore

$$QD \begin{pmatrix} v_1^T \\ \vdots \\ v_k^T \\ 0 \\ \vdots \\ 0 \end{pmatrix} = A_k.$$

According to Theorem 7.2, W_k achieves the minimum of $\sum_{i=1}^m (\text{dist}(\alpha_i, W))^2$. $\square$

Proposition 7.2. Let $\mu = \frac{1}{m} \sum_{i=1}^m (\alpha_i^T)$, and $\bar{\mu} = (1, \dots, 1)^T \cdot \mu$. Perform singular value decomposition on $B = A - \bar{\mu}$ to obtain $B = QDP^T$, where $P = (v_1, \dots, v_n)$. Then

$$\mu + \text{span}_{\mathbb{R}}(v_1, \dots, v_k)$$

is the k -dimensional affine plane that minimizes the sum of squared distances to $\alpha_1, \dots, \alpha_m$.

7.3 Application: Least Squares Method

To predict y from n -dimensional data $(a_1, \dots, a_n)$,

$$y = x_1 a_1 + \dots + x_n a_n.$$

After multiple experiments, we obtain a system of equations

$$\begin{pmatrix} \alpha_1^T \\ \alpha_2^T \\ \vdots \\ \alpha_m^T \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

When $m \gg n$, this system of equations most likely has no solution. How do we find the closest solution? That is, minimizing the sum of squared errors between the model predicted values and the m experiment results:

$$\sum_{i=1}^m (\alpha_i^T \cdot x - b)^2$$

Let W be the column space of A . Then the distance between b and W is

$$|b - \text{Proj}_W b|.$$

Since Ax ranges over W , there exists x such that $Ax = \text{Proj}_W b$. However, such x is not unique; they differ by elements in $\ker A$. Usually, we require minimizing the length of x to give a unique solution, which is called the optimal least squares solution.

Next, we introduce how to use singular value decomposition to provide the least squares solution. Given a matrix $A \in M_{m \times n}(\mathbb{R})$ and its singular value decomposition

$$A = QDP^T,$$

where

$$D = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{pmatrix}$$

Let

$$A^\dagger = PD^\dagger Q^T,$$

where

$$D^\dagger = \begin{pmatrix} \sigma_1^{-1} & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \sigma_2^{-1} & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r^{-1} & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{pmatrix}$$

Then we have the following theorem:

Theorem 7.3. For $Ax = b$, $\hat{x} = A^\dagger b$ is the *optimal least squares solution*.

Proof. First we prove that $A\hat{x} - b \perp \text{im } A$. That is, we verify

$$A^T(A(A^\dagger b) - b) = 0.$$

By direct calculation,

$$\begin{aligned} PD^T Q^T (QDP^T PD^\dagger Q^T - I_m) b &= (PD^T DD^\dagger Q^T - PD^T Q^T) b \\ &= (PD^T Q^T - PD^T Q^T) b \\ &= 0. \end{aligned}$$

Next, we verify that $\ker A \perp A^\dagger b$. Since $\ker A = \ker(QDP^T) = \ker D$, and $A^\dagger b = PD^\dagger Q^T b$, we need to verify

$$D^\dagger Q^T b \perp \ker D,$$

which is evident from the expressions of D and $D^\dagger$. □

Remark 7.2. *If one wants to use*

$$y = x_1 a_1 + \cdots + x_n a_n + c$$

to fit the data, consider

$$\begin{pmatrix} \alpha_1^T, 1 \\ \alpha_2^T, 1 \\ \vdots \\ \alpha_m^T, 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ c \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

Let

$$\tilde{A} = \begin{pmatrix} \alpha_1^T, 1 \\ \alpha_2^T, 1 \\ \vdots \\ \alpha_m^T, 1 \end{pmatrix},$$

and consider $\tilde{A}^\dagger$.

8 Exercises

8.1 Useful Exercises

Exercise 8.1. *Prove that any skew-symmetric matrix $A \in M_n(\mathbb{R})$ can be orthogonally similar to a block diagonal matrix with blocks of the form*

$$\begin{pmatrix} 0 & -\lambda \\ \lambda & 0 \end{pmatrix}$$

and possibly a 0 block if n is odd. Use this to show that any skew-symmetric matrix over $\mathbb{R}$ is congruent to a block diagonal matrix with blocks of the form

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

and possibly some 0 block.

Exercise 8.2. *Let V be the linear space consisting of all skew-symmetric real matrices of order n .*

1. *For any $A \in V$, prove that $I + A$ is invertible.*
2. *For any $A \in V$, define $f(A) = (I - A)(I + A)^{-1}$. Prove that $f(A)$ is an orthogonal matrix.*
3. *Give a characterization of the image of $f: V \rightarrow O(n)$ in terms of eigenvalues, that is, which matrices can be written in the form $(I - A)(I + A)^{-1}$ for some $A \in V$.*

Exercise 8.3. Let A be 2×2 real symmetric matrix

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

Write down an orthogonal matrix Q which diagonalizes A in terms of a, b, c .

Exercise 8.4. Consider the groups $O(2)$, its subgroup $SO(2)$ and group $SO(3)$. Determine whether the following statements are correct. If correct, prove it; if incorrect, provide a counterexample:

1. Two elements in the group $O(2)$ are conjugate if and only if they have the same trace.
2. Two elements in the group $SO(2)$ are conjugate in the group $SO(2)$ if and only if they have the same trace.
3. Two elements in the group $SO(2)$ are conjugate in the group $O(2)$ if and only if they have the same trace.
4. Two elements in the group $SO(3)$ are conjugate if and only if they have the same trace.

Exercise 8.5 (Cartan–Dieudonné theorem). Prove that any orthogonal transformation of Euclidean space $(V, \langle \cdot, \cdot \rangle)$ can be expressed as a composition of at most $\dim V$ reflections.

(The nontrivial part of the original theorem is to show this also holds for any non-degenerate symmetric bilinear form over a field of characteristic not equal to 2.)

Exercise 8.6 (Courant–Fischer–Weyl Min-Max Principle). You may choose to prove either part (1) or part (2).

1. Let $(E, \langle \cdot, \cdot \rangle)$ be an n -dimensional real inner product space. Suppose T is a self-adjoint transformation on E with real eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. Prove that the eigenvalues of T can be characterized by the following min-max method:

$$\lambda_k = \min \{ \max \{ \langle T(x), x \rangle : x \perp W_k, |x| = 1 \} : W_k \subset E \text{ is a subspace, } \dim W_k = k - 1 \}$$

Here, for a fixed $(k - 1)$ -dimensional subspace W_k , we first compute the maximum value

$$\max \{ \langle T(x), x \rangle : x \perp W_k, |x| = 1 \}.$$

Then we vary W_k over all $(k - 1)$ -dimensional subspaces and take the minimum of these maximum values.

2. Alternatively, you may prove the following special case: Let A be an $n \times n$ real symmetric matrix and v be an arbitrary n -dimensional real column vector, where $|v|$ denotes the vector length under the standard inner product. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be all eigenvalues of A . Prove that:

$$|Av| \leq \max\{|\lambda_1|, |\lambda_2|, \dots, |\lambda_n|\}|v|.$$

8.2 Optional problems

You do not need to hand in these problems, but you are encouraged to discuss and try them.

Exercise 8.7 (Outer automorphisms of $SO(n, \mathbb{R})$). An automorphism of a group G is called an inner automorphism if it is of the form $g \mapsto hgh^{-1}$ for some fixed $h \in G$. An automorphism which is not inner is called an outer automorphism. Consider the automorphism of $SO(n, \mathbb{R})$ defined by $A \mapsto PAP^{-1}$ where $P \in O(n, \mathbb{R})$ with $\det P = -1$. Is this an inner automorphism or an outer automorphism? Prove your answer. (The answer may depend on n .)

Exercise 8.8 (Challenge Problem). You will obtain a standard form for Lorentz transformations on $\mathbb{R}^4$. Let e_i ($i = 1, \dots, 4$) be the standard basis for $\mathbb{R}^4$. Consider the symmetric bilinear on $\mathbb{R}^4$ defined by

$$\langle x, y \rangle = x_1y_1 + x_2y_2 + x_3y_3 - x_4y_4.$$

A basis f_i ($i = 1, \dots, 4$) of $\mathbb{R}^4$ is called orthonormal if

$$\langle f_1, f_1 \rangle = \langle f_2, f_2 \rangle = \langle f_3, f_3 \rangle = 1, \quad \langle f_4, f_4 \rangle = -1, \quad \langle f_i, f_j \rangle = 0 \text{ if } i \neq j.$$

Suppose T is a Lorentz transformation on $\mathbb{R}^4$, that is, T is a linear transformation such that

$$\langle Tx, Ty \rangle = \langle x, y \rangle$$

for all $x, y \in \mathbb{R}^4$. Prove that there exists an orthonormal basis of $\mathbb{R}^4$ such that the matrix of T is block diagonal with blocks of the following types:

1. A block of order 1 with entry ± 1 .

2. A block of order 2 of the form

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

3. a block of order 2 of the form

$$\pm \begin{pmatrix} \cosh \theta & \sinh \theta \\ \sinh \theta & \cosh \theta \end{pmatrix} \quad \text{or} \quad \pm \begin{pmatrix} \cosh \theta & \sinh \theta \\ -\sinh \theta & -\cosh \theta \end{pmatrix}.$$

4. A block A of order 3 with eigenvalue $\lambda = \pm 1$ so that $(A - \lambda I)^3 = 0$ but $(A - \lambda I)^2 \neq 0$.

Exercise 8.9. If the Lorentz transformation T in Problem 8.8 is replaced by a transformation satisfying

$$\langle Tx, y \rangle = -\langle x, Ty \rangle$$

can you obtain a similar result? State the result and prove it.

Exercise 8.10 (Cauchy Interlacing Theorem). Let A be an $n \times n$ real symmetric matrix, and let B be an $m \times m$ principal submatrix of A , where $m < n$. If the eigenvalues of A are $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, and the eigenvalues of B are $\mu_1 \geq \mu_2 \geq \dots \geq \mu_m$, then for all $1 \leq i \leq m$, we have

$$\lambda_i \geq \mu_i \geq \lambda_{i+n-m}.$$

(Hint: Use the Courant-Fischer-Weyl min-max principle from Problem 8.6.)

Exercise 8.11 (Sylvester's Criterion). Use the Cauchy interlacing theorem to prove Sylvester's criterion: A symmetric matrix is positive definite if and only if all its leading principal minors are positive.

9 Hermitian Forms and Unitary Matrices

Definition 9.1. Let V be a finite-dimensional vector space over $\mathbb{C}$. A map $h: V \times V \rightarrow \mathbb{C}$ is called a Hermitian form, if it satisfies:

$$(1) \quad h(\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2, \mathbf{y}) = \overline{\lambda_1} h(\mathbf{x}_1, \mathbf{y}) + \overline{\lambda_2} h(\mathbf{x}_2, \mathbf{y});$$

$$(2) \ h(\mathbf{x}, \mathbf{y}) = \overline{h(\mathbf{y}, \mathbf{x})}.$$

Remark 9.1. Condition (2) guarantees that $h(\mathbf{x}, \mathbf{x})$ is real for all $\mathbf{x}$.

Example 9.1. The map

$$h: \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$$

$$(\mathbf{x}, \mathbf{y}) \mapsto \sum_{i=1}^n \bar{x}_i y_i$$

is called the standard Hermitian form on $\mathbb{C}^n$.

Problem 9.1. Similar to symmetric bilinear forms and quadratic forms on real vector spaces, how can we recover h from $h(\mathbf{x}, \mathbf{x})$?

Lemma 9.1. For a Hermitian form $h: V \times V \rightarrow \mathbb{C}$, we have

$$\operatorname{Re} h(\mathbf{x}, \mathbf{y}) = \frac{1}{2}(h(\mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}) - h(\mathbf{x}, \mathbf{x}) - h(\mathbf{y}, \mathbf{y}))$$

$$\operatorname{Im} h(\mathbf{x}, \mathbf{y}) = \frac{1}{2}(h(\sqrt{-1}\mathbf{x} + \mathbf{y}, \sqrt{-1}\mathbf{x} + \mathbf{y}) - h(\mathbf{x}, \mathbf{x}) - h(\mathbf{y}, \mathbf{y})).$$

Moreover, $\operatorname{Re} h$ is symmetric, and $\operatorname{Im} h$ is skew-symmetric (alternating).

Given a Hermitian form $h: V \times V \rightarrow \mathbb{C}$, and a basis $\alpha_1, \dots, \alpha_n$, the Gram matrix $(h(\alpha_i, \alpha_j)) = H$ satisfies $H = \overline{H}^T$. Such a matrix H is called a *Hermitian matrix*. For convenience, we denote $H^* = \overline{H}^T$ hereafter. For a Hermitian matrix H , it is also true that $\operatorname{Re} H$ is symmetric and $\operatorname{Im} H$ is skew-symmetric.

Definition 9.2. For $A, B \in M_n(\mathbb{C})$, if there exists $P \in \operatorname{GL}_n(\mathbb{C})$ such that $PAP^* = B$, then A and B are said to be congruent (or Hermitian congruent).

Now given a Hermitian form $h: V \times V \rightarrow \mathbb{C}$, and two bases $(\alpha_1, \dots, \alpha_n), (\beta_1, \dots, \beta_n)$ satisfying $\{\beta_1, \dots, \beta_n\} = (\alpha_1, \dots, \alpha_n)P$, then

$$\begin{aligned} (h(\beta_i, \beta_j))^* &= \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix} \cdot \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \\ &= \overline{P}^T \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \cdot (\alpha_1, \dots, \alpha_n)P \\ &= \overline{P}^T (h(\alpha_i, \alpha_j)) P \end{aligned}$$

So the Gram matrices of the same Hermitian form under different bases are congruent.

Theorem 9.1. Any Hermitian matrix is congruent to a diagonal matrix, and the diagonal entries must be real numbers.

Definition 9.3. A Hermitian form h on a vector space V over $\mathbb{C}$ is called:

- Positive definite ($h > 0$), if $h(v, v) > 0$ for all $v \neq 0$;
- Positive semi-definite ($h \geq 0$), if $h(v, v) \geq 0$ for all v ;

- Negative definite ($h < 0$), if $h(v, v) < 0$ for all $v \neq 0$;
- Negative semi-definite ($h \leq 0$), if $h(v, v) \leq 0$ for all v .

Theorem 9.2. For a Hermitian matrix H , the following are equivalent:

- (1) H is positive definite;
- (2) H is congruent to the identity matrix;
- (3) $H = PP^*$, where $P \in \text{GL}_n(\mathbb{C})$;
- (4) All leading principal minors of H are positive;
- (5) All principal minors of H are positive.

Theorem 9.3. For a Hermitian matrix H , the following are equivalent:

- (1) H is positive semi-definite;
- (2) H is congruent to

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix};$$

- (3) $H = PP^*$, where $P \in M_n(\mathbb{C})$;
- (4) All principal minors of H are nonnegative.

Definition 9.4. Given a finite-dimensional $\mathbb{C}$ -vector space V and a Hermitian form $h: V \times V \rightarrow \mathbb{C}$, if $h > 0$, then (V, h) is called a Hermitian inner product space, or a unitary space.

For convenience, we denote $\langle x, y \rangle := h(x, y)$.

- (1) Length (Norm): For $x \in V$, $|x| := \sqrt{\langle x, x \rangle}$.
- (2) Angle: For $x, y \in V \setminus \{0\}$, there exists $\theta(x, y) \in [0, \frac{\pi}{2}]$ such that

$$\cos(\theta(x, y)) = \frac{|\langle x, y \rangle|}{|x||y|}.$$

Lemma 9.2 (Cauchy-Schwarz Inequality).

$$|\langle x, y \rangle| \leq |\langle x, x \rangle| |\langle y, y \rangle|.$$

Proof. Let $\langle x, y \rangle = re^{\sqrt{-1}\theta}$. Consider $|te^{\sqrt{-1}\theta}x + y|^2$, where $t \in \mathbb{R}$. Then

$$t^2|x|^2 + 2rt + |y|^2 \geq 0, \quad \forall t \in \mathbb{R},$$

Thus the discriminant is ≤ 0 , and equality holds if and only if x, y are linearly dependent over $\mathbb{C}$. $\square$

Lemma 9.3 (Triangle Inequality). $\forall x, y \in V$, $|x + y| \leq |x| + |y|$.

Proof. By definition,

$$|x + y|^2 = |x|^2 + \langle x, y \rangle + \langle y, x \rangle + |y|^2.$$

By Cauchy-Schwarz inequality,

$$|\langle x, y \rangle| \leq |x||y|, \quad |\langle y, x \rangle| \leq |x||y|,$$

Thus

$$|x + y|^2 \leq |x|^2 + 2|x||y| + |y|^2 = (|x| + |y|)^2.$$

□

Lemma 9.4 (Parallelogram Identity). $\forall x, y \in V$, we have

$$|x + y|^2 + |x - y|^2 = 2(|x|^2 + |y|^2).$$

Definition 9.5. Given a unitary space (V, h) , a basis $\{\alpha_1, \dots, \alpha_n\}$ is called an **orthonormal basis** if $\langle \alpha_i, \alpha_j \rangle = \delta_{ij}$.

Remark 9.2. Similar to the inner product space case, starting from any basis, one can obtain an orthonormal basis via the Gram-Schmidt process.

Theorem 9.4. Given a unitary space (V, h) and two orthonormal bases $\{\alpha_1, \dots, \alpha_n\}$ and $\{\beta_1, \dots, \beta_n\}$. Assume $(\alpha_1, \dots, \alpha_n) = (\beta_1, \dots, \beta_n)P$, then

$$PP^* = I_n.$$

Definition 9.6. A matrix $P \in M_n(\mathbb{C})$ is called a unitary matrix if $PP^* = I_n$.

Lemma 9.5. If $P \in M_n(\mathbb{C})$ is a unitary matrix, then $\bar{P}, P^{-1}, P^*$ are all unitary matrices.

Definition 9.7. Two complex matrices $A, B \in M_n(\mathbb{C})$ are called unitarily similar if there exists a unitary matrix $U \in M_n(\mathbb{C})$ such that $A = U^*BU$.

Proposition 9.1. Any complex square matrix is unitarily similar to an upper triangular matrix.

Proof. By induction. Let $A \in M_n(\mathbb{C})$, induct on n .

Let λ be an eigenvalue of A , $v \in \mathbb{C}^n$ be a λ -eigenvector. Assume without loss of generality $\|v\| = 1$. Extend v to an orthonormal basis of $\mathbb{C}^n$, say $v_1, \dots, v_n$. Let $U = (v_1 \cdots v_n)$ be a unitary matrix. Then

$$AU = (Av_1 \cdots Av_n) = (v_1 \cdots v_n) \begin{pmatrix} \lambda & * \\ 0 & * \end{pmatrix}$$

$$\text{That is } U^*AU = \begin{pmatrix} \lambda & * \\ 0 & * \end{pmatrix}.$$

The result follows by induction. □

Definition 9.8. A square matrix $N \in M_n(\mathbb{C})$ is called **normal** if $NN^* = N^*N$.

Lemma 9.6. If a normal matrix N is unitarily similar to $\begin{pmatrix} N_1 & N_2 \\ 0 & N_3 \end{pmatrix}$, where N_1, N_3 are square matrices, then it must be that $N_2 = 0$, and both N_1 and N_3 are normal.

Proof. Suppose $\begin{pmatrix} N_1 & N_2 \\ 0 & N_3 \end{pmatrix} = U^*NU$ for a unitary matrix U . Then U^*NU commutes with $(U^*NU)^* = U^*N^*U$, implying U^*NU is normal. Thus

$$\begin{pmatrix} N_1 & N_2 \\ 0 & N_3 \end{pmatrix} \begin{pmatrix} N_1^* & 0 \\ N_2^* & N_3^* \end{pmatrix} = \begin{pmatrix} N_1^* & 0 \\ N_2^* & N_3^* \end{pmatrix} \begin{pmatrix} N_1 & N_2 \\ 0 & N_3 \end{pmatrix}$$

Comparing top-left blocks, $N_1N_1^* + N_2N_2^* = N_1^*N_1 \Rightarrow \text{tr}(N_2N_2^*) = 0 \Rightarrow N_2 = 0$. $\square$

Theorem 9.5 (Spectral Theorem for Normal Matrices). *A normal matrix is unitarily similar to a diagonal matrix.*

Proof. Let $A \in M_n(\mathbb{C})$ be normal. Then A is unitarily similar to an upper triangular matrix

$$\begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & * \end{pmatrix}$$

By Lemma 9.6, it must be a diagonal matrix. $\square$

Example 9.2. *Real symmetric, real skew-symmetric, orthogonal, Hermitian, unitary, and skew-Hermitian matrices are all normal matrices.*

Corollary 9.1.

1. *Any Hermitian matrix (including real symmetric matrices) is unitarily similar to $\text{diag}(\lambda_1, \dots, \lambda_n)$, where λ_i are real numbers.*
2. *Any unitary matrix (including orthogonal matrices) is unitarily similar to $\text{diag}(\lambda_1, \dots, \lambda_n)$, where $\lambda_i = e^{\sqrt{-1}\theta_i}$, $\theta_i \in \mathbb{R}$.*
3. *Any skew-Hermitian matrix (including skew-symmetric matrices) is unitarily similar to $\text{diag}(\lambda_1, \dots, \lambda_n)$, where $\lambda_i = \sqrt{-1}g_i$, $g_i \in \mathbb{R}$.*

Proof.

1. Suppose a Hermitian matrix is unitarily similar to $\text{diag}(\lambda_1, \dots, \lambda_n)$. Since the diagonal matrix is also Hermitian $\Rightarrow \lambda_i$ are real.
2. Suppose a unitary matrix is unitarily similar to a diagonal matrix $\text{diag}(\lambda_1, \dots, \lambda_n)$. Similarity preserves unitarity $\Rightarrow$ the modulus of each λ_i is 1.
3. Similar argument.

$\square$

Definition 9.9. *Let $(V, \langle \cdot, \cdot \rangle)$ be an n -dimensional unitary space (i.e., the Gram matrix of $\langle \cdot, \cdot \rangle$ is a positive definite Hermitian matrix). A linear transformation $\varphi: V \rightarrow V$ is called a **normal transformation**, **Hermitian transformation**, **unitary transformation**, or **skew-Hermitian transformation**, if the matrix representation of φ under an orthonormal basis of V is a matrix of the corresponding type.*

Restating the Spectral Theorem from the viewpoint of linear transformations:

Lemma 9.7. Let φ be a normal transformation on a unitary space V . If W is an invariant subspace of φ , then the orthogonal complement $W^\perp$ is also an invariant subspace.

Theorem 9.6 (Spectral Theorem, Transformation Form). Let V be an n -dimensional unitary space, and $\varphi: V \rightarrow V$ be a normal transformation. Then there exists an orthonormal basis $v_1, \dots, v_n$ of V such that each v_i is an eigenvector of φ . Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of φ , and let W_i be the eigenspace of φ corresponding to λ_i . Then

$$\varphi = \lambda_1 \pi_1 + \dots + \lambda_k \pi_k,$$

where $\pi_i: V \rightarrow W_i$ is the orthogonal projection. This is called the **spectral decomposition** of φ .

Theorem 9.7. Given an n -dimensional unitary space V and a linear transformation $\varphi: V \rightarrow V$, there exists a unique linear transformation $\varphi^*: V \rightarrow V$ such that for any $\alpha, \beta \in V$,

$$\langle \varphi^*(\alpha), \beta \rangle = \langle \alpha, \varphi(\beta) \rangle.$$

We call φ^* the **adjoint** of φ . Let A and B be the matrices of φ and φ^* under an orthonormal basis of V , respectively, then $B = A^* = \overline{A}^T$.

Proof. Assume $\langle \varphi^*(\alpha), \beta \rangle = \langle \alpha, \varphi(\beta) \rangle, \forall \alpha, \beta$. Take an orthonormal basis $v_1, \dots, v_n$. Then:

$$\begin{pmatrix} \varphi^*(v_1) \\ \vdots \\ \varphi^*(v_n) \end{pmatrix} = B \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, \quad (\varphi(v_1), \dots, \varphi(v_n)) = (v_1, \dots, v_n)A$$

The matrix of inner products on the left is $B^T \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \cdot (v_1, \dots, v_n) = B^T$, and the matrix of

inner products on the right is $\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \cdot (v_1, \dots, v_n) \overline{A} = \overline{A}$. Thus $B^T = \overline{A} \Rightarrow B = A^*$.

Conversely, to construct φ^* , one only needs to pick an orthonormal basis. Let A be the matrix of φ , then we use A^* to obtain φ^* . $\square$

Theorem 9.8. Let V_1, V_2 be n -dimensional unitary spaces, and $\gamma: V_1 \rightarrow V_2$ be a linear map. The following are equivalent:

1. γ preserves the inner product;
2. γ preserves the length of vectors (isometry);
3. $\gamma: V_1 \rightarrow V_2$ is a linear isomorphism and preserves the inner product;
4. γ maps any orthonormal basis of V_1 to an orthonormal basis of V_2 ;
5. γ maps some orthonormal basis to an orthonormal basis;
6. The matrix representation of γ with respect to orthonormal bases of V_1 and V_2 is a unitary matrix.

Example 9.3. When $n = 1$, a unitary matrix is just a complex number with modulus 1, i.e., $e^{\sqrt{-1}\theta}$ for some $\theta \in \mathbb{R}$. Thus, a unitary transformation on a 1-dimensional unitary space is just a rotation by angle θ . Or in other words, the group of unitary transformations on a 1-dimensional unitary space is isomorphic to the real special orthogonal group $\text{SO}(2)$.

When $n = 2$, we will discuss in detail the group $\text{SU}(2) = \{A \in M_2(\mathbb{C}) : AA^* = I_2, \det A = 1\}$ in the next lecture.

10 Application of spectral theorem: conics and quadrics

Definition 10.1. A conic in $\mathbb{R}^2$ is the locus of points (x, y) satisfying a quadratic equation

$$Ax_1^2 + Bx_1x_2 + Cx_2^2 + Dx_1 + Ex_2 + F = 0,$$

where $A, B, C, D, E, F \in \mathbb{R}$ and not all of A, B, C are zero.

The conic is called *nondegenerate* if it is not two lines, one line, a point, or empty set. If we allow rigid motion transformations of $\mathbb{R}^2$, $\mathbf{x} \mapsto P\mathbf{x} + \mathbf{b}$ where $P \in \text{O}(2, \mathbb{R})$ and $\mathbf{b} \in \mathbb{R}^2$, then we can simplify the equation of a conic. Using the results on symmetric bilinear forms and quadratic forms, we can show the classification theorem for conics:

Theorem 10.1. Any conic with nondegenerate quadratic part can be transformed by rigid motion into one of the following standard forms:

1. *Ellipse:* $\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} = 1$, where $a, b > 0$;
2. *Hyperbola:* $\frac{x_1^2}{a^2} - \frac{x_2^2}{b^2} = 1$, where $a, b > 0$;
3. *Parabola:* $x_1^2 = 2px_2$, where $p \neq 0$.

Proof. The quadratic part $Ax_1^2 + Bx_1x_2 + Cx_2^2$ defines a symmetric bilinear form on $\mathbb{R}^2$ by Gram matrix

$$\begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}.$$

Then there exists an orthogonal matrix $P \in \text{O}(2, \mathbb{R})$ such that

$$P^T \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix} P = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix},$$

where λ_1, λ_2 are the eigenvalues of the above matrix. Thus, under the change of variables $\mathbf{x} = P\mathbf{y}$, the conic equation becomes

$$\lambda_1 y_1^2 + \lambda_2 y_2^2 + D'y_1 + E'y_2 + F = 0,$$

for some $D', E' \in \mathbb{R}$. Now we consider the following cases:

1. If λ_1, λ_2 have the same sign, then we can complete the square to obtain the standard form of an ellipse.
2. If λ_1, λ_2 have opposite signs, then we can complete the square to obtain the standard form of a hyperbola.

3. If one of λ_1, λ_2 is zero, then we can complete the square to obtain the standard form of a parabola.

□

Definition 10.2. A quadric in $\mathbb{R}^n$ is the locus of points $\mathbf{x} = (x_1, x_2, \dots, x_n)$ satisfying a quadratic equation

$$\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j + \sum_{i=1}^n b_i x_i + c = 0,$$

where $a_{ij}, b_i, c \in \mathbb{R}$ and not all of a_{ij} are zero.

We call a quadric *nondegenerate* if it is not a union of hyperplanes, or a cone over lower dimensional quadric, or empty set. Similar to the conic case, using the results on symmetric bilinear forms and quadratic forms, we can show the classification theorem for three-dimensional quadrics:

Theorem 10.2. Any quadric in $\mathbb{R}^3$ with nondegenerate quadratic part can be transformed by rigid motion into one of the following standard forms:

1. Ellipsoid: $\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} + \frac{x_3^2}{c^2} = 1$, where $a, b, c > 0$;
2. Hyperboloid of one sheet: $\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} - \frac{x_3^2}{c^2} = 1$, where $a, b, c > 0$;
3. Hyperboloid of two sheets: $-\frac{x_1^2}{a^2} - \frac{x_2^2}{b^2} + \frac{x_3^2}{c^2} = 1$, where $a, b, c > 0$;
4. Elliptic paraboloid: $\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} = 2px_3$, where $a, b, p > 0$;
5. Hyperbolic paraboloid: $\frac{x_1^2}{a^2} - \frac{x_2^2}{b^2} = 2px_3$, where $a, b, p > 0$.

11 Skew-symmetric Bilinear Forms and Symplectic Matrices

The classification of skew-symmetric bilinear forms on finite-dimensional vector spaces is similar to that of symmetric bilinear forms. We use one unified approach to summarize the idea of the proofs here.

Definition 11.1 (Nondegenerate Forms). If $B: V \times V \rightarrow F$ is a bilinear (or Hermitian) form on a finite-dimensional vector space V over a field F , then B is called nondegenerate if for any nonzero $v \in V$, there exists some $u \in V$ such that $B(v, u) \neq 0$ (or equivalently, $B(u, v) \neq 0$). Or equivalently, the Gram matrix of B under any basis of V is invertible.

We will always assume B is symmetric or skew-symmetric bilinear form on V over a field F with $\text{char}(F) \neq 2$, or Hermitian form on V over $\mathbb{C}$. Then the next definition of orthogonal complement has no ambiguity. Let W be a subspace of V . The *orthogonal complement* of W with respect to B is defined as

Definition 11.2 (Radical). The radical of B is defined as

$$\text{Rad}(B) = \{v \in V : B(v, u) = 0, \forall u \in V\} = V^\perp.$$

Then have an orthogonal decomposition theorem.

Proposition 11.1. *The vector space is decomposed into the orthogonal direct sum*

$$V = \text{Rad}(B) \oplus W,$$

where W is a subspace of V such that the restriction of B on W is nondegenerate.

The next step is to classify nondegenerate forms and we have the following important theorem generalizing the theorem in Exercise 4.3.

Theorem 11.1. *Let B be a nondegenerate symmetric, or skew-symmetric (or Hermitian) form on a finite-dimensional vector space V over a field F (or over $\mathbb{C}$). Fix a subspace W of V . Then the following are equivalent:*

- (1) *The restriction of B on W is nondegenerate;*
- (2) *$V = W \oplus W^\perp$;*
- (3) *The restriction of B on $W^\perp$ is nondegenerate.*

The proof is similar to that of Exercise 4.3.

Then we can classify skew-symmetric bilinear forms as follows.

Theorem 11.2. *Let B be a skew-symmetric bilinear form on a finite-dimensional vector space V over a field F whose characteristic is not 2. Then there exists a basis of V such that the Gram matrix of B under this basis is*

$$\begin{pmatrix} O_r & I_r & O \\ -I_r & O_r & O \\ O & O & O \end{pmatrix},$$

where $2r = \text{rank}(B)$. Or equivalently, there exists a basis $\{\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_r, \gamma_1, \dots, \gamma_{n-2r}\}$ of V such that

$$B(\alpha_i, \beta_j) = \delta_{ij}, \quad B(\beta_i, \alpha_j) = -\delta_{ij}, \quad B(\alpha_i, \alpha_j) = B(\beta_i, \beta_j) = B(\gamma_i, \cdot) = 0.$$

Proof. We reduce to nondegenerate case by the orthogonal decomposition by the radical. Then we prove by induction on the dimension of V . The base case $\dim V = 0$ is trivial. For $\dim V > 0$, we can find $u, v \in V$ such that $B(u, v) \neq 0$. Rescale u such that $B(u, v) = 1$. Let $W = \text{span}\{u, v\}$. Then the restriction of B on W is nondegenerate. By the previous theorem, we have the orthogonal decomposition $V = W \oplus W^\perp$. By induction hypothesis, we can find a basis of $W^\perp$ such that the Gram matrix of the restriction of B on $W^\perp$ has the desired form. Combining with $\{u, v\}$ gives the desired basis of V . $\square$

Corollary 11.1. *The rank of any skew-symmetric matrix is even.*

Corollary 11.2. *Any two nondegenerate skew-symmetric bilinear forms on a $2n$ -dimensional vector space over a field F are equivalent.*

Definition 11.3 (Symplectic basis). *A basis $\{\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n\}$ of a $2n$ -dimensional vector space V over a field F with a nondegenerate skew-symmetric bilinear form B is called a symplectic basis if*

$$B(\alpha_i, \beta_j) = \delta_{ij}, \quad B(\beta_i, \alpha_j) = -\delta_{ij}, \quad B(\alpha_i, \alpha_j) = B(\beta_i, \beta_j) = 0.$$

Definition 11.4 (Symplectic Matrix). A matrix $A \in M_{2n}(F)$ is called a symplectic matrix if it satisfies

$$A^T J A = J,$$

where

$$J = \begin{pmatrix} O_n & I_n \\ -I_n & O_n \end{pmatrix}.$$

Proposition 11.2. The set of all symplectic matrices in $M_{2n}(F)$ forms a group under matrix multiplication, denoted by $\text{Sp}(2n, F)$.

One nontrivial fact about symplectic matrices is that their determinants are always 1.

Theorem 11.3. If $A \in \text{Sp}(2n, F)$, then $\det(A) = 1$.

Proof. Taking determinants on both sides of the equation $A^T J A = J$, we have

$$(\det A)^2 \det J = \det J.$$

Since $\det J = 1$, we have $(\det A)^2 = 1$. Thus $\det A = \pm 1$. To show $\det A = 1$, we consider the symplectic form on $\wedge^{2n} F^{2n}$ induced by J . Then we have

$$(\det A)\omega = \omega,$$

where ω is a nonzero element in $\wedge^{2n} F^{2n}$. Thus $\det A = 1$. □

Another nontrivial factor about symplectic matrices is that they are closed under taking transpose.

Theorem 11.4. If $A \in \text{Sp}(2n, F)$, then $A^T \in \text{Sp}(2n, F)$.

Proof. Taking transpose on both sides of the equation $A^T J A = J$, we have

$$A^T J^T A = J^T.$$

Since $J^T = -J$, we have

$$A^T J A = -J.$$

Multiplying both sides by -1 , we have

$$A^T J A = J.$$

Thus $A^T \in \text{Sp}(2n, F)$. □

12 Exercises

12.1 Useful Exercises

Exercise 12.1. Suppose A is an invertible real square matrix with singular values $\sigma_1, \dots, \sigma_n$. Find the singular values of A^{-1} .

Exercise 12.2 (Polar Decomposition). In this problem, you will prove the polar decomposition of an invertible complex matrix using the singular value decomposition.

1. Try to state without proof the singular value decomposition theorem for complex matrices similar to the one stated in the class for real matrices.

2. Use it to prove the following: Let A be an $n \times n$ invertible complex matrix. Prove that there exist unitary matrices $U \in \mathbf{U}(n)$ and P is a positive definite hermitian matrix such that $A = UP$. This is called the polar decomposition of A .
3. Show that the polar decomposition is unique. (This shows that $\mathbf{GL}(n, \mathbb{C})$ is homeomorphic to $\mathbf{U}(n) \times \mathcal{H}_n^+$ where $\mathcal{H}_n^+$ is the set of all positive definite hermitian matrices, a convex cone in the vector space of all hermitian matrices.)

Exercise 12.3. Artin Chapter 8, Exercise 6.20

Prove the circulant, the matrix below, is normal.

$$C = \begin{bmatrix} c_0 & c_1 & c_2 & \cdots & c_n \\ c_n & c_0 & c_1 & \cdots & c_{n-1} \\ c_{n-1} & c_n & c_0 & \cdots & c_{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_1 & c_2 & c_3 & \cdots & c_0 \end{bmatrix}$$

(How to diagonalize it? Hint: write C as a polynomial in the shift matrix. see Artin Chapter 8, Exercise 6.19)

Exercise 12.4. Prove that for any square submatrix of a unitary matrix, the modulus of any of its complex eigenvalues does not exceed 1.

Exercise 12.5. 1. If A and B are normal matrices, is AB necessarily normal? What if we additionally assume $AB = BA$?

2. Determine if the matrix $A = \begin{pmatrix} \sqrt{-1} & -\sqrt{-1} \\ -\sqrt{-1} & \sqrt{-1} \end{pmatrix}$ is normal, Hermitian, or unitary.

Exercise 12.6 (Characterization of Normal Matrices in terms of Singular Values). Let A be an $n \times n$ complex matrix. Let $\sigma_i(A)$ denote the i -th singular value of A (such that $\sigma_1(A) \geq \cdots \geq \sigma_n(A) \geq 0$), and let $\lambda_1, \dots, \lambda_n$ be the complex eigenvalues of A (counted with algebraic multiplicity). Prove that $\sum_{i=1}^n \sigma_i(A)^2 \geq \sum_{i=1}^n |\lambda_i|^2$, with equality if and only if A is a normal matrix.

(Hint: Recall that any complex matrix can be conjugate to upper triangular matrix by an invertible matrix, by choosing eigenvectors and induction. Prove that actually a unitary matrix can do this job, then compare the Frobenius norms of both sides.)

Exercise 12.7. Lang Algebra Chapter XV, 1.

Here we choose σ to be complex conjugation.

1. Let E be a finite dimensional space over the complex numbers, and let

$$h : E \times E \rightarrow \mathbb{C}$$

be a hermitian form. Write

$$h(x, y) = g(x, y) + if(x, y)$$

where g, f are real valued. Show that g, f are $\mathbb{R}$ -bilinear, g is symmetric, f is alternating.

2. Let E be finite dimensional over $\mathbf{C}$. Let $g : E \times E \rightarrow \mathbf{C}$ be $\mathbb{R}$ -bilinear. Assume that for all $x \in E$, the map $y \mapsto g(x, y)$ is $\mathbf{C}$ -linear, and that the $\mathbb{R}$ -bilinear form

$$f(x, y) = g(x, y) - g(y, x)$$

is real-valued on $E \times E$. Show that there exists a hermitian form h on E and a symmetric $\mathbf{C}$ -bilinear form ψ on E such that $2ig = h + \psi$. Show that h and ψ are uniquely determined.

Exercise 12.8. Let $X = A + \sqrt{-1}B$ be a complex square matrix, where $A, B \in M_n(\mathbb{R})$. Prove that X is a unitary matrix if and only if

$$\begin{pmatrix} A & -B \\ B & A \end{pmatrix}$$

is an orthogonal matrix.

12.2 Optional problems

Exercise 12.9. Prove the additive inequality of singular values. Let A, B be two $m \times n$ real matrices. Prove that

$$\sigma_{k+l-1}(A+B) \leq \sigma_k(A) + \sigma_l(B)$$

for $k+l-1 \leq \min(m, n)$.

Exercise 12.10. Let (V, ω) be a symplectic space if ω is a non-degenerate skew-symmetric form on the F -vector space V . An F -linear transformation T is called a symplectic transformation if $\omega(T(v), T(w)) = \omega(v, w)$. A basis $(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n)$ is called a symplectic basis if $\omega(\alpha_i, \beta_j) = \delta_{ij}$ and $\omega(\alpha_i, \alpha_j) = \omega(\beta_i, \beta_j) = 0$. When $F = \mathbf{C}$, prove that for any symplectic transformation T of a symplectic space, there exists a symplectic basis such that the matrix of T under this basis has the form

$$\begin{bmatrix} B_n & 0_n \\ 0_n & B_n^T \end{bmatrix}$$

where B_n is an n -dimensional Jordan normal form.

Exercise 12.11. Prove the following inequality between singular values and eigenvalues. Let A be an $n \times n$ complex matrix with eigenvalues $\lambda_1, \dots, \lambda_n$ (counted with algebraic multiplicity) and singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0$. Then for any $k = 1, \dots, n$,

$$|\lambda_1 \lambda_2 \dots \lambda_k| \leq \sigma_1 \sigma_2 \dots \sigma_k.$$

Exercise 12.12 (Challenge Problem). Let A be a real square matrix of order n , and let the eigenvalues of $A^T A$ be $\lambda_1^2, \dots, \lambda_n^2$, where $0 \leq \lambda_i \leq 1$ for $i = 1, 2, \dots, n$. Prove that:

$$\det(I_n - A) \geq (1 - \lambda_1)(1 - \lambda_2) \dots (1 - \lambda_n)$$

13 Orthogonal representation of $SU(2)$

We will use the spectral theorem for unitary matrices to study the special unitary group of degree 2 over $\mathbf{C}$:

$$SU(2) = \{A \in M_2(\mathbf{C}) : AA^* = I_2, \det A = 1\}.$$

Then we can study the topology of $SU(2)$ via its explicit form.

Proposition 13.1. Any matrix $A \in \text{SU}(2)$ can be written as

$$A = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix},$$

where $\alpha, \beta \in \mathbb{C}$ satisfy $|\alpha|^2 + |\beta|^2 = 1$.

Proof. First it is straightforward to verify that any matrix of the above form is in $\text{SU}(2)$. Conversely, let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SU}(2)$. Then the conditions $AA^* = I_2$ and $\det A = 1$ imply that $A^{-1} = A^*$. Thus

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}.$$

Comparing the entries gives us the desired form. □

So the topology of $\text{SU}(2)$ is the same as that of the unit sphere S^3 in $\mathbb{R}^4$.

The conjugacy classes of $\text{SU}(2)$ can be described as follows:

Proposition 13.2. The conjugacy classes of $\text{SU}(2)$ are determined by the trace. More precisely, two matrices $A, B \in \text{SU}(2)$ are conjugate if and only if $\text{tr}(A) = \text{tr}(B)$.

Proof. First the eigenvalues of any matrix $A \in \text{SU}(2)$ are of the form $e^{\sqrt{-1}\theta}, e^{-\sqrt{-1}\theta}$ for some $\theta \in \mathbb{R}$. Thus the trace of A is $2\cos\theta$. If two matrices $A, B \in \text{SU}(2)$ have the same trace, then they have the same eigenvalues. By the spectral theorem for unitary matrices, they are conjugate by a matrix $P \in U(2)$. And we can always rescale P such that $\det P = 1$, so that $P \in \text{SU}(2)$. If two matrices $A, B \in \text{SU}(2)$ are conjugate, then they have the same eigenvalues, so they have the same trace. □

The trace zero matrices in $\text{SU}(2)$ are of the form

$$\begin{pmatrix} x\sqrt{-1} & y + \sqrt{-1}z \\ -y + \sqrt{-1}z & -x\sqrt{-1} \end{pmatrix}, \text{ and } x^2 + y^2 + z^2 = 1.$$

Next we will show that there is a natural homomorphism from $\text{SU}(2)$ to the special orthogonal group $\text{SO}(3, \mathbb{R})$. Consider the Lie algebra of $\text{SU}(2)$: Let

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) : X^* = -X, \text{tr}(X) = 0\}.$$

Then $\mathfrak{su}(2)$ is a 3-dimensional real vector space with basis

$$\sqrt{-1}\sigma_1 = \begin{pmatrix} 0 & \sqrt{-1} \\ \sqrt{-1} & 0 \end{pmatrix}, \quad \sqrt{-1}\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \sqrt{-1}\sigma_3 = \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix},$$

where $\sigma_1, \sigma_2, \sigma_3$ are the Pauli matrices. Define an inner product on $\mathfrak{su}(2)$ by

$$\langle X, Y \rangle = -\frac{1}{2}\text{tr}(XY).$$

Then $\{i\sigma_1, i\sigma_2, i\sigma_3\}$ is an orthonormal basis of $\mathfrak{su}(2)$ with respect to this inner product.

Proposition 13.3. For any $A \in \text{SU}(2)$, the map $\varphi_A : \mathfrak{su}(2) \rightarrow \mathfrak{su}(2)$ defined by

$$\varphi_A(X) = AXA^{-1}$$

is an orthogonal transformation on the inner product space $\mathfrak{su}(2)$.

Proof. First we need to verify that $\varphi_A(X) \in \mathfrak{su}(2)$ for any $X \in \mathfrak{su}(2)$. Indeed,

$$(AXA^{-1})^* = (A^{-1})^* X^* A^* = -AXA^{-1},$$

and

$$\mathrm{tr}(AXA^{-1}) = \mathrm{tr}(X) = 0.$$

Next we verify that φ_A preserves the inner product. Indeed,

$$\begin{aligned} \langle \varphi_A(X), \varphi_A(Y) \rangle &= -\frac{1}{2} \mathrm{tr}(AXA^{-1}AY A^{-1}) \\ &= -\frac{1}{2} \mathrm{tr}(XY) \\ &= \langle X, Y \rangle. \end{aligned}$$

□

The main theorem we want to show is the following:

Theorem 13.1. *The map $\Phi : \mathrm{SU}(2) \rightarrow \mathrm{SO}(3, \mathbb{R})$ induced by*

$$\Phi(A) = \varphi_A$$

is a surjective group homomorphism with kernel $\{\pm I_2\}$.

It is easier to use quaternions to prove this theorem. We introduce the the following matrices:

$$\mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{i} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 & \sqrt{-1} \\ \sqrt{-1} & 0 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix}.$$

Then any matrix in $\mathfrak{su}(2)$ can be written as

$$X = x\mathbf{i} + y\mathbf{j} + z\mathbf{k},$$

for some $x, y, z \in \mathbb{R}$. Thus we can identify $\mathfrak{su}(2)$ with $\mathbb{R}^3$ via the map

$$(x, y, z) \mapsto x\mathbf{i} + y\mathbf{j} + z\mathbf{k}.$$

Under this identification, the inner product on $\mathfrak{su}(2)$ corresponds to the standard inner product on $\mathbb{R}^3$. The multiplication rules of $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -\mathbf{1}, \quad \mathbf{ij} = \mathbf{k}, \quad \mathbf{jk} = \mathbf{i}, \quad \mathbf{ki} = \mathbf{j},$$

The group $\mathrm{SU}(2)$ is isomorphic to the group of unit quaternions

$$\{a\mathbf{1} + b\mathbf{i} + c\mathbf{j} + d\mathbf{k} : a, b, c, d \in \mathbb{R}, a^2 + b^2 + c^2 + d^2 = 1\}.$$

Proof of Theorem 13.1. First we check that the image of Φ is in $\mathrm{SO}(3, \mathbb{R})$. Since φ_A is an orthogonal transformation on $\mathfrak{su}(2)$, we only need to check that $\det(\varphi_A) = 1$. Note that $\mathrm{SU}(2)$ is connected and the determinant is a continuous map to $\{\pm 1\}$, so the determinant must be constantly 1.

In order to prove surjectivity, we first verify the group operation of $\mathrm{SU}(2)$ on the unit sphere of $\mathfrak{su}(2)$ is transitive. This is because of the conjugacy class description of $\mathrm{SU}(2)$: any two trace zero matrices in $\mathrm{SU}(2)$ are conjugate.

Next we show that for any fixed axis, we have a surjective homomorphism from the stabilizer subgroup to $\text{SO}(2, \mathbb{R})$. Without loss of generality, we consider the axis $\mathbf{i}$. Then we can consider the subgroup

$$H = \{A_\theta \in \text{SU}(2) : A_\theta = \cos(\theta)\mathbf{1} + \sin(\theta)\mathbf{i}\}.$$

Then $A_\theta^{-1} = \cos(\theta)\mathbf{1} - \sin(\theta)\mathbf{i}$. So φ_{A_θ} fixes $\mathbf{i}$:

$$\begin{aligned}\varphi_{A_\theta}(\mathbf{i}) &= A_\theta \mathbf{i} A_\theta^{-1} \\ &= (\cos(\theta)\mathbf{1} + \sin(\theta)\mathbf{i})(\cos(\theta)\mathbf{1} - \sin(\theta)\mathbf{i}) \\ &= \mathbf{i}.\end{aligned}$$

Next we compute the action of φ_{A_θ} on the orthogonal complement of $\mathbf{i}$:

$$\begin{aligned}\varphi_{A_\theta}(\mathbf{j}) &= A_\theta \mathbf{j} A_\theta^{-1} \\ &= (\cos(\theta)\mathbf{1} + \sin(\theta)\mathbf{i})(\cos(\theta)\mathbf{1} - \sin(\theta)\mathbf{i}) \\ &= \cos(2\theta)\mathbf{j} + \sin(2\theta)\mathbf{k},\end{aligned}$$

and

$$\begin{aligned}\varphi_{A_\theta}(\mathbf{k}) &= A_\theta \mathbf{k} A_\theta^{-1} \\ &= (\cos(\theta)\mathbf{1} + \sin(\theta)\mathbf{i})(\cos(\theta)\mathbf{1} - \sin(\theta)\mathbf{i}) \\ &= -\sin(2\theta)\mathbf{j} + \cos(2\theta)\mathbf{k}.\end{aligned}$$

Thus the map φ_{A_θ} restricted on the orthogonal complement of $\mathbf{i}$ is a rotation by angle 2θ . So we have a surjective homomorphism from H to $\text{SO}(2, \mathbb{R})$.

Next we use a standard trick in group operations. Let $\mathbf{v}$ be any unit vector in $\mathfrak{su}(2)$. Then we can find some $P \in \text{SU}(2)$ such that $\varphi_P(\mathbf{i}) = \mathbf{v}$. Then the group PHP^{-1} is in the stabilizer subgroup of $\mathbf{v}$. And we have a surjective homomorphism from PHP^{-1} to $\text{SO}(2, \mathbb{R})$ by sending $PHP^{-1} \ni A \mapsto \varphi_P \circ \varphi_A \circ \varphi_{P^{-1}} \in \text{SO}(2, \mathbb{R})$.

Finally any element in $\text{SO}(3, \mathbb{R})$ fixed some $\mathbf{v}$ in the unit sphere of $\mathfrak{su}(2)$, we conclude that $\Phi : \text{SU}(2) \rightarrow \text{SO}(3, \mathbb{R})$ is surjective.

The kernel of Φ is $\{\pm I_2\}$ since these are the only two matrices act trivially on $\mathfrak{su}(2)$. Or you can check it from the explicit centers from the quaternions.

□

One important consequence of Theorem 13.1 is that $\text{SU}(2)$ is simply connected, and the special orthogonal group $\text{SO}(3, \mathbb{R})$ has fundamental group $\mathbb{Z}/2\mathbb{Z}$. This can be reflected by the famous experiment of Dirac's belt, or the plate trick.

14 Examples of Lie groups and Lie algebras

Lie groups and Lie algebras are important tools to study continuous symmetries. For example, the orthogonal group $\text{O}(3, \mathbb{R})$ is a Lie group which describes the continuous symmetries of the 3-dimensional space preserving the standard inner product. Continuous symmetry of a physic system usually corresponds to conservation laws. This is the famous Noether's theorem. If we consider all the symmetries of the 3-dimensional space preserving the distance, then we get a bigger Lie group containing both $\text{O}(3, \mathbb{R})$ and translation group $\mathbb{R}^3$ as subgroups. The total dimension of the symmetries is 6 and they correspond to three angular momenta and three linear momenta in physics. If we combine the symmetry in time axis, then we also have another

dimension corresponding to energy. The definition and general properties involves differential geometry, and we will only give some examples here instead of proving the full theory for general cases.

A *Lie group* is a group G which is also a smooth manifold such that the group operations (multiplication and inversion) are smooth maps. A *Lie algebra* is a vector space $\mathfrak{g}$ over a field $\mathbb{R}$ equipped with a bilinear map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ called the Lie bracket satisfying the following properties:

- (1) $[x, x] = 0$ for any $x \in \mathfrak{g}$ (anticommutativity);
- (2) $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$ for any $x, y, z \in \mathfrak{g}$ (Jacobi identity).

The definition of Lie algebra is motivated by the commutator operation on square matrices. If we consider the general linear group $GL(n, \mathbb{R})$, then its Lie algebra is the general linear Lie algebra $\mathfrak{gl}(n, \mathbb{R}) = M_n(\mathbb{R})$ with the Lie bracket defined by $[A, B] = AB - BA$ for any $A, B \in M_n(\mathbb{R})$. These two objects are related by the **exponential map**:

$$\exp : \mathfrak{gl}(n, \mathbb{R}) \rightarrow GL(n, \mathbb{R}), \quad \exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

The exponential map is a local diffeomorphism around the identity element of $GL(n, \mathbb{R})$ and the zero element of $\mathfrak{gl}(n, \mathbb{R})$. This is because the following differential equation:

$$\frac{d}{dt} \exp(tA) = A \exp(tA) = \exp(tA)A,$$

with initial condition $\exp(0) = I_n$ has a unique solution. So the differential of $\exp$ at 0 is the identity map:

$$d(\exp)_0(A) = A,$$

for any $A \in \mathfrak{gl}(n, \mathbb{R})$.

The exponential map also changes the addition in the Lie algebra to the multiplication in the Lie group. More precisely, we have the **Baker-Campbell-Hausdorff formula**:

$$\exp(tA) \exp(tB) = \exp \left(tA + tB + \frac{t^2}{2}[A, B] + \frac{t^3}{12}([A, [A, B]] + [B, [B, A]]) + \cdots \right),$$

for any $A, B \in \mathfrak{gl}(n, \mathbb{R})$.

When $A = B$, we have

$$\exp(tA) \exp(sA) = \exp((t+s)A).$$

So the bracket operation in the Lie algebra measures the noncommutativity of the multiplication in the Lie group.

The equality

$$\exp(tA) \exp(sA) = \exp((t+s)A)$$

can also be viewed as a group homomorphism from the additive group $\mathbb{R}$ to the Lie group $GL(n, \mathbb{R})$. So the exponential map gives us a **one-parameter subgroup** of the Lie group $GL(n, \mathbb{R})$ generated by $A \in \mathfrak{gl}(n, \mathbb{R})$.

Consider the subgroup called **special linear group** $SL(n, \mathbb{R})$ is defined by

$$SL(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) : \det A = 1\}.$$

Its Lie algebra is the **special linear Lie algebra**

$$\mathfrak{sl}(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : \text{tr}(A) = 0\}.$$

This can be read from the exponential map as follows:

$$\det(\exp(A)) = \exp(\text{tr}(A)).$$

So $\exp(A) \in \text{SL}(n, \mathbb{R})$ if and only if $\text{tr}(A) = 0$.

Similarly, the **orthogonal group** $\text{O}(n, \mathbb{R})$ is defined by

$$\text{O}(n, \mathbb{R}) = \{A \in \text{GL}(n, \mathbb{R}) : A^T A = \text{I}_n\}.$$

Its Lie algebra is the **orthogonal Lie algebra**

$$\mathfrak{o}(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : A^T = -A\}.$$

This can be read from the exponential map as follows:

$$(\exp(tA))^T \exp(tA) = \exp(tA^T) \exp(tA) = \exp(t(A^T + A)) + O(t^2).$$

So $\exp(A) \in \text{O}(n, \mathbb{R})$ if and only if $A^T = -A$.

We can also embed the **complex linear group** $\text{GL}(n, \mathbb{C})$ into $\text{GL}(2n, \mathbb{R})$ via the map

$$A + B\sqrt{-1} \mapsto \begin{pmatrix} A & -B \\ B & A \end{pmatrix},$$

for any $A, B \in M_n(\mathbb{R})$. Then the Lie algebra of $\text{GL}(n, \mathbb{C})$ can be identified with

$$\mathfrak{gl}(n, \mathbb{C}) = \{A + B\sqrt{-1} : A, B \in M_n(\mathbb{R})\}$$

with the bracket operation defined by

$$[X, Y] = XY - YX,$$

for any $X, Y \in \mathfrak{gl}(n, \mathbb{C})$.

The **unitary group** $\text{U}(n)$ is defined by

$$\text{U}(n) = \{A \in \text{GL}(n, \mathbb{C}) : AA^* = \text{I}_n\}.$$

Its Lie algebra is the **unitary Lie algebra**

$$\mathfrak{u}(n) = \{A \in M_n(\mathbb{C}) : A^* = -A\}.$$

The **special unitary group** $\text{SU}(n)$ is defined by

$$\text{SU}(n) = \{A \in \text{GL}(n, \mathbb{C}) : AA^* = \text{I}_n, \det A = 1\}.$$

Its Lie algebra is the **special unitary Lie algebra**

$$\mathfrak{su}(n) = \{A \in M_n(\mathbb{C}) : A^* = -A, \text{tr}(A) = 0\}.$$

A challenge problem is to compute the Lie algebra of the **symplectic group** $\text{Sp}(2n, \mathbb{R})$ defined by

$$\text{Sp}(2n, \mathbb{R}) = \{A \in \text{GL}(2n, \mathbb{R}) : A^T J A = J\},$$

where

$$J = \begin{pmatrix} O_n & I_n \\ -I_n & O_n \end{pmatrix}.$$

Its Lie algebra is

$$\mathfrak{sp}(2n, \mathbb{R}) = \{A \in M_{2n}(\mathbb{R}) : A^T J + JA = 0\}.$$

A very nontrivial and powerful result in Lie group theory is the following, which we will not prove here.

Theorem 14.1. *For any Lie group G , its closed subgroup H is also a Lie group.*

In other words, if you have any continuous equation defining a subgroup of $\mathrm{GL}(n, \mathbb{R})$, then the solution set is also a Lie group.

The study of Lie group is usually reduced to the study of Lie algebra via the exponential map. The following theorem shows that any finite dimensional Lie algebra can be integrated to a Lie group.

Theorem 14.2 (Lie's Third Theorem). *For any finite dimensional Lie algebra $\mathfrak{g}$ over $\mathbb{R}$, there exists a Lie group G such that the Lie algebra of G is isomorphic to $\mathfrak{g}$.*

The dimension of the Lie group is the same as the dimension of its Lie algebra as a vector space, or the number of parameters needed to describe the Lie group locally around the identity element.

Recall the spectral theorems for orthogonal and unitary matrices, we can now relate them by Lie groups and Lie algebras. For example, the spectral theorem for orthogonal matrices have 2×2 rotation blocks corresponding to complex eigenvalues and also 1×1 blocks corresponding to real eigenvalues ± 1 . The corresponding blocks in the Lie algebra (or skew-symmetric matrices) are 2×2 skew-symmetric matrices and also 1×1 zero blocks. The exponential map sends the 2×2 skew-symmetric blocks to the 2×2 rotation blocks, and sends the 1×1 zero blocks to the 1×1 identity blocks. Under the exponential map, each 2×2 skew-symmetric block generates a circle subgroup which is isomorphic to $\mathrm{SO}(2, \mathbb{R})$, and this is a one-parameter subgroup of $\mathrm{O}(n, \mathbb{R})$.

Example 14.1. *Two Lie groups may share the same Lie algebra. For example, the Lie algebras of $\mathrm{SO}(3, \mathbb{R})$ and $\mathrm{SU}(2)$ are both isomorphic to $\mathfrak{so}(3, \mathbb{R})$. We can find this using explicit bases:*

$$\begin{aligned} \mathfrak{so}(3, \mathbb{R}) &= \left\{ \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}, \\ \mathfrak{su}(2) &= \left\{ a \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} + b \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & \sqrt{-1} \\ \sqrt{-1} & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}. \end{aligned}$$

The isomorphism can be given by sending

$$\begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix} \mapsto \frac{1}{2\sqrt{-1}} \left(a \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} + b \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & \sqrt{-1} \\ \sqrt{-1} & 0 \end{pmatrix} \right).$$

15 Exercices

15.1 Mandatory part

Exercise 15.1. *Is it true that the conjugacy classes of unitary group $\mathrm{U}(2)$ are determined by the trace and determinant? Prove your answer.*

Exercise 15.2.

Definition 15.1 (Orthogonal Group of Signature (p, q)). Let p, q be non-negative integers. The orthogonal group of signature (p, q) , denoted by $O_{p,q}$, is defined as the group of all linear transformations on $\mathbb{R}^{p+q}$ that preserve the bilinear form

$$\langle x, y \rangle = x_1y_1 + x_2y_2 + \cdots + x_py_p - x_{p+1}y_{p+1} - \cdots - x_{p+q}y_{p+q},$$

for all $x, y \in \mathbb{R}^{p+q}$.

Let W be the space of real trace-zero 2×2 matrices $W = \{A \in M_{2 \times 2}(\mathbb{R}) \mid \text{trace}(A) = 0\}$. W has a basis $\mathbf{B} = (w_1, w_2, w_3)$, where

$$w_1 = \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}, w_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, w_3 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

1. Show that the symmetric bilinear form defined by $\langle A, A' \rangle = \text{trace}(AA')$ has signature $(2, 1)$. (Hint: use basis $\mathbf{B}$)
2. Prove that $P \star A = PAP^{-1}$ defines a linear group operation of $\text{SL}(2, \mathbb{R})$ on the space W .
3. Use this operation to define a group homomorphism $\varphi : \text{SL}(2, \mathbb{R}) \rightarrow O_{2,1}$.
4. Prove the kernel of this homomorphism is $\{\pm I\}$.

This construction actually shows that $\text{SL}(2, \mathbb{R})$ is a double cover of $\text{SO}_{2,1}^+$, the connected component of $O_{2,1}$ containing the identity matrix. Or equivalently, $\text{PSL}(2, \mathbb{R}) = \text{SL}(2, \mathbb{R})/\{\pm I\}$ is isomorphic to the spin group $\text{SO}^+(2, 1)$. It is an interesting question to show that the orthogonal group $O_{2,1}$ has four connected components and identify the geometry of each component.

A similar exercise is to relate $\text{SL}(2, \mathbb{C})$ to the orthogonal group $O(3, 1)$. see Artin Algebra Chapter 9, 4.8

Exercise 15.3. Let A be the set of all $n \times n$ upper triangular matrices with real entries and all diagonal entries equal to 1. Find the Lie algebra of A and compute its dimension.

Exercise 15.4. Show that the intersection of symplectic group $\text{Sp}(2n, \mathbb{R})$ and orthogonal group $O(2n, \mathbb{R})$ in $\text{GL}(2n, \mathbb{R})$ is isomorphic to the unitary group $U(n)$. Here we use the embedding of these groups into $\text{GL}(2n, \mathbb{R})$ as Section Examples of Lie groups and Lie algebras.

Exercise 15.5. Prove the Jacobi identity of Lie algebra $\mathfrak{gl}(n, \mathbb{R}) = M_n(\mathbb{R})$ using the properties of the matrix commutator. Here

$$[A, B] = AB - BA.$$

15.2 Optional exercises

Exercise 15.6. Prove that the Lie algebra of the symplectic group $\text{Sp}(2n, \mathbb{R})$ is

$$\mathfrak{sp}(2n, \mathbb{R}) = \{A \in M_{2n}(\mathbb{R}) : A^T J + JA = 0\},$$

where

$$J = \begin{pmatrix} O_n & I_n \\ -I_n & O_n \end{pmatrix}.$$

Exercise 15.7. Prove the second and the third order term in the Baker-Campbell-Hausdorff formula when X, Y are elements of $M_n(\mathbb{R})$. That is, prove

$$\exp(tX)\exp(tY) = \exp\left(t(X+Y) + \frac{1}{2}t^2[X, Y] + \frac{1}{12}t^3([X, [X, Y]] + [Y, [Y, X]]) + \cdots\right)$$

You may use the following expansion of logarithm:

$$\log(I + A) = A - \frac{1}{2}A^2 + \frac{1}{3}A^3 - \dots$$

16 Group representations: basic concepts

Group representation theory studies the ways in which a group can act on vector spaces via linear transformations. It is a powerful way to study symmetry and structure in mathematics. It is much more than a tool in abstract group theory; it provides deep insights into their nature. Actually, groups are usually appearing as their representations. Representations themselves are crucial and central objects in mathematics. This section introduces the basic definitions and concepts in group representation theory.

16.1 Linear operation, Matrix Representations and Conjugacy

Definition 16.1. The *general linear group* over a field F is defined as:

$$\mathrm{GL}(n, F) = \{A \in M_n(F) \mid \det A \neq 0\}.$$

Definition 16.2. Let V be a finite-dimensional vector space over F . The *general linear group of V* is:

$$\mathrm{GL}(V) = \{f: V \rightarrow V \mid f \text{ is an invertible linear transformation}\}.$$

Let $\dim V = n$ and fix a basis $B = (e_1, \dots, e_n)$ of V . For any linear transformation $f: V \rightarrow V$, we have the matrix representation $R_B(f)$ such that

$$f(e_1, \dots, e_n) = (e_1, \dots, e_n) \cdot R_B(f).$$

If $B' = (v_1, \dots, v_n)$ is another basis with change-of-basis matrix P (i.e., $(v_1, \dots, v_n) = (e_1, \dots, e_n) \cdot P$), then

$$R_{B'}(f) = P^{-1}R_B(f)P.$$

Definition 16.3. A *representation* of a group G on a vector space V is a group homomorphism

$$\rho: G \rightarrow \mathrm{GL}(V).$$

Equivalently, it is a group action $G \times V \rightarrow V$, denoted by $(g, v) \mapsto g \cdot v$, satisfying:

$$\begin{aligned} g \cdot (v + w) &= g \cdot v + g \cdot w, \\ g \cdot (\lambda v) &= \lambda(g \cdot v), \end{aligned}$$

for all $g \in G$, $v, w \in V$, and $\lambda \in F$. Such an action is called a *linear group action*.

Given a representation $\rho: G \rightarrow \mathrm{GL}(V)$ and a basis B of V , we obtain a **matrix representation** or a group homomorphism:

$$R: G \rightarrow \mathrm{GL}(n, F), \quad g \mapsto R_B(\rho(g)).$$

Two matrix representations R and R' are said to be **conjugate** if there exists $P \in \mathrm{GL}(n, F)$ such that

$$PR(g)P^{-1} = R'(g) \quad \forall g \in G.$$

Definition 16.4. Two representations $\rho : G \rightarrow \text{GL}(V)$ and $\rho' : G \rightarrow \text{GL}(V')$ are **isomorphic** if there exists an isomorphism $f : V \rightarrow V'$ such that

$$f(g \cdot v) = g \cdot f(v) \quad \forall g \in G, v \in V.$$

From the discussion about change of basis, we see that

Proposition 16.1. Two representations are isomorphic if and only if their corresponding matrix representations are conjugate.

To summarize, we have three equivalent viewpoints of group representations:

- A group homomorphism $\rho : G \rightarrow \text{GL}(V)$;
- A linear group action of G on V ;
- A matrix representation $R : G \rightarrow \text{GL}(n, F)$ for some n .

The isomorphism classes of finite-dimensional representations correspond to conjugacy classes of matrix representations.

Definition 16.5 (Faithful Representation). A representation $\rho : G \rightarrow \text{GL}(V)$ is called **faithful** if it is injective as a group homomorphism, i.e., if $\ker \rho = \{e\}$. In terms of group actions, this means that if $g \cdot v = v$ for all $v \in V$, then $g = e$.

16.2 Unitary Representations

Assume $F = \mathbb{C}$. Let V be a complex inner product space, with positive definite Hermitian form $\langle \cdot, \cdot \rangle$. A representation $\rho : G \rightarrow \text{GL}(V)$ is called a **unitary representation** if $\text{Im } \rho \subseteq U(V)$, where $U(V)$ is the unitary group.

Theorem 16.1. Let G be a finite group and $\rho : G \rightarrow \text{GL}(V)$ a representation. Then there exists a positive definite Hermitian form $\langle \cdot, \cdot \rangle$ on V that is G -invariant, i.e.,

$$\langle g \cdot v, g \cdot w \rangle = \langle v, w \rangle \quad \forall g \in G, v, w \in V.$$

Hence, ρ is unitary with respect to this form.

Proof. Start with any positive definite Hermitian form $\langle \cdot, \cdot \rangle_0$ on V . Define a new form by averaging over the group:

$$\langle v, w \rangle_G = \frac{1}{|G|} \sum_{g \in G} \langle g \cdot v, g \cdot w \rangle_0.$$

One checks that $\langle \cdot, \cdot \rangle_G$ is still a positive definite Hermitian form and is G -invariant. $\square$

This is a fundamental result in representation theory of finite groups, allowing us to always work with unitary representations when the field is $\mathbb{C}$. The key idea is to average an arbitrary inner product over the group to obtain a G -invariant one. This technique is known as **averaging** and will be used frequently in representation theory. For example, if you work with real representations, you can also try to prove that every real representation of a finite group admits an invariant real inner product, or equivalently, any matrix representation is conjugate to one with image in the orthogonal group $O(n)$.

16.3 Invariant Subspaces and Orthogonal Decomposition

Definition 16.6. Let $\rho: G \rightarrow \text{GL}(V)$ be a representation. A subspace $W \subseteq V$ is called *G-invariant* if $g \cdot w \in W$ for all $g \in G$ and $w \in W$. This also means that the restriction of ρ to W defines a *subrepresentation*.

Theorem 16.2. Let G be a finite group and $\rho: G \rightarrow \text{GL}(V)$ a unitary representation. For any G -invariant subspace $W \subseteq V$, the orthogonal complement $W^\perp$ (with respect to a G -invariant inner product) is also G -invariant, and we have the decomposition

$$V = W \oplus W^\perp.$$

Proof. Choose a G -invariant inner product $\langle \cdot, \cdot \rangle$ on V . For $w \in W$ and $v \in W^\perp$, we have

$$\langle w, g \cdot v \rangle = \langle g^{-1} \cdot w, v \rangle = 0,$$

since $g^{-1} \cdot w \in W$. Hence $g \cdot v \in W^\perp$, so $W^\perp$ is G -invariant. $\square$

16.4 Irreducible Representations

Definition 16.7. A representation $\rho: G \rightarrow \text{GL}(V)$ is called *irreducible* if V has no nontrivial G -invariant subspaces (i.e., the only G -invariant subspaces are $\{0\}$ and V itself).

Example 16.1. Let $F = \mathbb{F}_p$ (the finite field with p elements), $G = \mathbb{F}_p$ (additive group), and $V = \mathbb{F}_p^2$. Define the representation $R: G \rightarrow \text{GL}(2, \mathbb{F}_p)$ by

$$t \mapsto \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Then the subspace $W = \left\{ \begin{pmatrix} x \\ 0 \end{pmatrix} \mid x \in \mathbb{F}_p \right\}$ is G -invariant. However, there is no complementary G -invariant subspace W' such that $V = W \oplus W'$. This shows that the theorem on orthogonal decomposition requires an inner product (and hence the field to be $\mathbb{C}$ and the representation to be unitary).

16.5 Semisimplicity

A fundamental result in the representation theory of finite groups over $\mathbb{C}$ is that every representation can be decomposed into irreducible pieces. This property is known as complete reducibility or semisimplicity.

Theorem 16.3 (Maschke's Theorem). Let G be a finite group and V a finite-dimensional representation of G over $\mathbb{C}$. Then V is isomorphic to a direct sum of irreducible representations. That is,

$$V \cong V_1^{\oplus m_1} \oplus V_2^{\oplus m_2} \oplus \cdots \oplus V_k^{\oplus m_k},$$

where the V_i are pairwise non-isomorphic irreducible representations of G .

Proof. We proceed by induction on $\dim V$. If V is irreducible, we are done. Otherwise, let $W \subset V$ be a nontrivial G -invariant subspace. By Theorem 3.2, we can find a G -invariant inner product on V , and then $V = W \oplus W^\perp$ with $W^\perp$ also G -invariant. By induction, both W and $W^\perp$ decompose into irreducibles, hence so does V . $\square$

This theorem shows that irreducible representations are the basic building blocks of all representations of finite groups over $\mathbb{C}$. The study of these irreducible representations—their classification, dimensions, and characters—is the central goal of representation theory.

17 Examples of Group Representations

Now that we have established the basic framework, let us examine some concrete examples of group representations to illustrate these concepts in action.

Example 17.1 (Trivial Representation). *The simplest example is the **trivial representation**:*

$$\rho: G \rightarrow \mathrm{GL}(1, \mathbb{C}) = \mathbb{C}^\times, \quad g \mapsto 1.$$

Here every group element acts as the identity transformation.

Example 17.2 (Dihedral Group). *The dihedral group D_n has a natural representation as orthogonal transformations of the plane:*

$$D_n \rightarrow O(2) \subset \mathrm{GL}(2, \mathbb{R}) \subset \mathrm{GL}(2, \mathbb{C}).$$

This representation is generated by a rotation a and a reflection b satisfying the relations $a^n = 1$, $b^2 = 1$, and $bab^{-1} = a^{-1}$.

Definition 17.1 (Group Algebra). *Given a group G and a field F , the **group algebra** $F[G]$ is defined as:*

$$F[G] = \left\{ \sum_{g \in G} a_g g \mid a_g \in F \right\}.$$

The group G acts on $F[G]$ by left multiplication:

$$h \cdot \left(\sum_{g \in G} a_g g \right) = \sum_{g \in G} a_g (hg).$$

This is called the **regular representation** of G .

Example 17.3 (Permutation Representation). *Let G act on a finite set $X = \{x_1, \dots, x_n\}$. Consider the vector space $F^X = \{\sum_{x \in X} a_x x \mid a_x \in F\}$. Then G acts on F^X by permuting the basis vectors:*

$$g \cdot \left(\sum_{x \in X} a_x x \right) = \sum_{x \in X} a_x (g \cdot x).$$

This yields a representation $G \rightarrow \mathrm{GL}(n, F)$.

For example, take $G = S_n$ acting on the standard basis $\{e_1, \dots, e_n\}$ of $\mathbb{C}^n$. Then the subspace $W = \mathrm{span}\{e_1 + \dots + e_n\}$ is G -invariant. With respect to the standard Hermitian form, its orthogonal complement

$$W^\perp = \left\{ \sum_{i=1}^n a_i e_i \mid \sum_{i=1}^n a_i = 0 \right\}$$

is also G -invariant, giving a decomposition $\mathbb{C}^n = W \oplus W^\perp$ as S_n -representations.

18 Constructions of Representations

Having seen several examples, we now turn to systematic methods for constructing new representations from existing ones. These constructions are fundamental for building more complex representations from simpler building blocks.

18.1 Direct Sums and Quotients

Definition 18.1 (Direct Sum). Let V and W be representations of G . Their **direct sum** $V \oplus W$ is the representation on the direct sum of vector spaces, with G acting componentwise:

$$g \cdot (v, w) = (g \cdot v, g \cdot w).$$

In matrix form, if V and W have matrix representations R_V and R_W , then the representation on $V \oplus W$ is given by block diagonal matrices:

$$g \mapsto \begin{bmatrix} R_V(g) & 0 \\ 0 & R_W(g) \end{bmatrix}.$$

Definition 18.2 (Quotient Representation). Let $W \subseteq V$ be a G -invariant subspace. The **quotient representation** on V/W is defined by

$$g \cdot (v + W) = (g \cdot v) + W.$$

In matrix form, after choosing a basis adapted to W , the representation on V takes the form:

$$g \mapsto \begin{bmatrix} R_W(g) & * \\ 0 & R_{V/W}(g) \end{bmatrix},$$

where R_W and $R_{V/W}$ are the matrix representations on W and V/W respectively.

18.2 Dual Representation

Another important construction is the dual representation, which corresponds to the contragredient action.

Definition 18.3 (Dual Representation). Let V be a representation of G . The **dual representation** $V^* = \text{Hom}(V, \mathbb{C})$ is defined by

$$(g \cdot f)(v) = f(g^{-1} \cdot v) \quad \forall f \in V^*, v \in V, g \in G.$$

In matrix terms, if we choose a basis $B = (v_1, \dots, v_n)$ of V and the dual basis $B^* = (f_1, \dots, f_n)$ of V^* (satisfying $f_i(v_j) = \delta_{ij}$), then the matrix representation for V^* is given by:

$$R_{V^*}(g) = (R_V(g)^{-1})^\top.$$

19 Exercices

19.1 Mandatory part

Exercise 19.1. Let V be an irreducible representation of a finite group G over $\mathbb{C}$. Assume V is not the trivial representation. Prove that for any $v \in V$, we have $\sum_{g \in G} g \cdot v = 0$. (For cyclic groups, this is known to be an identity of roots of unity.)

Exercise 19.2. (Artin Algebra Chapter 10, 3.5) Let x be a generator of a cyclic group G of order p . Sending

$$x \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

defines a matrix representation $G \rightarrow \text{GL}_2(\mathbb{F}_p)$. Prove that this representation is not the direct sum of irreducible representations.

Exercise 19.3. (Artin Algebra Chapter 10, 3.4) Let $\langle \cdot, \cdot \rangle$ be a nondegenerate skew-symmetric form on a vector space V , and let ρ be a representation of a finite group G on V . Prove that the averaging process produces a G -invariant skew-symmetric form on V , and show by example that the form obtained in this way need not be nondegenerate.

Exercise 19.4. (Artin algebra Chapter 10, 2.2) Consider the standard two-dimensional complex representation of the dihedral group D_n . For which n is this an irreducible complex representation?

Here the standard representation is given by the action of D_n as the group of symmetries of a regular n -gon in the plane, or equivalently, the representation defined by the matrices

$$r \mapsto \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix}, \quad s \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

where r is a rotation by $2\pi/n$ and s is a reflection.

Exercise 19.5. (Artin Algebra Chapter 10, 3.1) Let G be a cyclic group of order 3. The matrix

$$A = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$$

has order 3, so it defines a matrix representation of G on $\mathbb{C}^2$. Use the averaging process to produce a G -invariant form from the standard Hermitian product $\langle X, Y \rangle = X^*Y$ on $\mathbb{C}^2$.

Exercise 19.6. (Artin Algebra Chapter 10, 3.2) Let $\rho: G \rightarrow \text{GL}(V)$ be a representation of a finite group G on a real vector space V . Prove the following:

1. There exists a G -invariant, positive definite symmetric form $\langle \cdot, \cdot \rangle$ on V .
2. ρ is a direct sum of irreducible representations.
3. Every finite subgroup of $\text{GL}_n(\mathbb{R})$ is conjugate to a subgroup of $\text{O}(n)$.

Exercise 19.7. Show that the dual representation of an irreducible representation is also irreducible.

19.2 Optional exercises

Exercise 19.8. In this part, we prove the semisimplicity theorem (Maschke's theorem) for any representation of group G and field F such that the characteristic of F does not divide the order of G , using the averaging technique.

Let V be a representation of G over F , and let W be a G -invariant subspace of V . Prove that there exists a complementary G -invariant subspace W' of V such that $V = W \oplus W'$. (Hint: start with any complementary subspace and then use averaging to construct a projection onto W whose kernel is W' .)

Exercise 19.9. Show that the dual representation is isomorphic to the original representation for any finite group representation if and only if there exists a nondegenerate G -invariant bilinear form on the representation space.

20 Hom, tensor product and Schur's lemma

20.1 Hom space

Given two representations, we can form the space of linear maps between them, which itself carries a natural representation.

Definition 20.1 (Hom Representation). *Let V and W be representations of G . The space $\text{Hom}(V, W)$ of linear maps from V to W becomes a representation of G via:*

$$(g \cdot T)(v) = g \cdot T(g^{-1} \cdot v) \quad \forall T \in \text{Hom}(V, W), v \in V, g \in G.$$

20.2 Tensor Product

Tensor products provide a way to combine representations, which is particularly important in physics and the study of product groups.

Definition 20.2 (Tensor Product of Representations). *Let V and W be representations of G . Their **tensor product** $V \otimes W$ is the representation on the tensor product of vector spaces, with G acting diagonally:*

$$g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w).$$

If V has basis $\{v_1, \dots, v_n\}$ and W has basis $\{w_1, \dots, w_m\}$, then $\{v_i \otimes w_j\}$ is a basis of $V \otimes W$, and any element can be written as $\sum a_{ij} v_i \otimes w_j$. The action of G in coordinates is given by the Kronecker product of the matrices for V and W .

Lemma 20.1. *There is a natural isomorphism of vector spaces:*

$$\Gamma : V^* \otimes W \rightarrow \text{Hom}(V, W),$$

defined by $\Gamma(f \otimes w)(v) = f(v)w$. Moreover, if V and W are representations of G , then Γ is an isomorphism of representations when $\text{Hom}(V, W)$ is equipped with the action

$$(g \cdot T)(v) = g \cdot T(g^{-1} \cdot v).$$

Proof. For surjectivity, let $B = (v_1, \dots, v_n)$ be a basis of V and $C = (w_1, \dots, w_m)$ a basis of W . Given $T \in \text{Hom}(V, W)$ with matrix (a_{ij}) relative to B and C , let $(f_1, \dots, f_n)$ be the dual basis of B . Then

$$\Gamma \left(\sum_{i,j} a_{ij} f_j \otimes w_i \right) (v_k) = \sum_i a_{ik} w_i = T(v_k).$$

Thus Γ is surjective, and by dimension count it is an isomorphism. $\square$

20.3 Matrix form of tensor product and Hom space

Definition 20.3 (Kronecker Product). *Let $A = (a_{ij})$ be an $n \times n$ matrix and B an $m \times m$ matrix. The **Kronecker product** $A \otimes B$ is the $nm \times nm$ block matrix defined by replacing each entry a_{ij} of A with the block $a_{ij}B$:*

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{bmatrix}.$$

Proposition 20.1. *If $R_V(g)$ and $R_W(g)$ are the matrix representations of g on V and W respectively, then the matrix representation of g on $V \otimes W$ is given by the Kronecker product:*

$$R_{V \otimes W}(g) = R_V(g) \otimes R_W(g).$$

More precisely, if R_V is the matrix representation of G on V with respect to a basis $B : v_1, \dots, v_n$, and R_W is the matrix representation of G on W with respect to a basis $C : w_1, \dots, w_m$, then the matrix representation of G on $V \otimes W$ with respect to the basis $\{v_1 \otimes w_1, \dots, v_1 \otimes w_m, v_2 \otimes w_1, \dots, v_2 \otimes w_m, \dots, v_n \otimes w_m\}$ is given by the Kronecker product of the matrices for R_V and R_W .

Corollary 20.1. *Let R_V and R_W be the matrix representations of G on V and W respectively. Then the matrix representation of G on $\text{Hom}(V, W)$ with respect to the basis $\{f_i \otimes w_j\}$ (where $\{f_i\}$ is the dual basis of V and $\{w_j\}$ is a basis of W) is given by the Kronecker product:*

$$R_{\text{Hom}(V, W)}(g) = (R_V(g)^{-1})^T \otimes R_W(g).$$

20.4 G-Homomorphisms

To compare different representations and understand their structure, we need the notion of maps that respect the group action.

Definition 20.4 (G-Homomorphism). *Let V and W be representations of G . A linear map $f : V \rightarrow W$ is called a **G-homomorphism** (or **intertwining operator**) if*

$$f(g \cdot v) = g \cdot f(v) \quad \forall g \in G, v \in V.$$

We denote by $\text{Hom}_G(V, W)$ the space of all such maps.

Proposition 20.2. *Let $f : V \rightarrow W$ be a G-homomorphism. Then $\ker f$ is a G-invariant subspace of V , and $\text{im } f$ is a G-invariant subspace of W . Moreover, the induced map $\bar{f} : V/\ker f \rightarrow \text{im } f$ is an isomorphism of representations.*

20.5 Schur's Lemma

One of the most powerful tools in representation theory is Schur's Lemma, which describes the structure of G-homomorphisms between irreducible representations.

Theorem 20.1 (Schur's Lemma). *Let V and W be irreducible representations of G over $\mathbb{C}$. Then:*

1. *If V and W are not isomorphic, then $\text{Hom}_G(V, W) = 0$.*
2. *If $V = W$, then $\text{Hom}_G(V, V) = \mathbb{C} \cdot I_V$.*

Proof. Let $T \in \text{Hom}_G(V, W)$. Since V is irreducible and $\ker T$ is G-invariant, either $\ker T = V$ or $\ker T = 0$. Similarly, $\text{im } T$ is G-invariant in W , so either $\text{im } T = 0$ or $\text{im } T = W$.

If $T \neq 0$, then $\ker T = 0$ and $\text{im } T = W$, so T is an isomorphism. This proves (1).

For (2), let $T \in \text{Hom}_G(V, V)$. Since $\mathbb{C}$ is algebraically closed, T has an eigenvalue λ . Then $T - \lambda I_V \in \text{Hom}_G(V, V)$ is not invertible, hence must be zero by the argument above. Thus $T = \lambda I_V$. $\square$

21 Characters of Representations

Now that we have introduced various constructions of representations, we turn to one of the most powerful tools in representation theory: characters. Characters provide a compact way to encode information about representations and play a central role in their classification.

21.1 Definition and Basic Properties

Definition 21.1 (Character). *Let $\rho : G \rightarrow \text{GL}(V)$ be a representation of a finite group G over $\mathbb{C}$. The **character** of ρ is the function $\chi_\rho : G \rightarrow \mathbb{C}$ defined by*

$$\chi_\rho(g) = \text{tr}(\rho(g)),$$

where tr denotes the trace of a linear transformation. The character is called **irreducible** if ρ is irreducible.

Remark 21.1. *The character is a class function: $\chi_\rho(hgh^{-1}) = \chi_\rho(g)$ for all $g, h \in G$, since trace is invariant under conjugation. Moreover, $\chi_\rho(1) = \dim V$, because $\rho(1) = I_V$ and $\text{tr}(I_V) = \dim V$.*

Characters have remarkable algebraic properties that reflect the operations on representations:

Proposition 21.1 (Properties of Characters). *Let ρ, ρ_1, ρ_2 be representations of G over $\mathbb{C}$ with characters χ, χ_1, χ_2 respectively. Then:*

1. $\chi_{\rho_1 \oplus \rho_2} = \chi_1 + \chi_2$.
2. $\chi_{\rho^*} = \overline{\chi}$, where ρ^* is the dual representation and $\overline{\chi}$ denotes complex conjugation.
3. $\chi_{\rho_1 \otimes \rho_2} = \chi_1 \cdot \chi_2$ (pointwise product).

Proof. The first property follows because the trace of a block diagonal matrix is the sum of the traces of the blocks. For the second, recall that the matrix of $\rho^*(g)$ is $(R(g)^{-1})^\top$, whose trace is the complex conjugate of $\text{tr}(R(g))$. The third property is a consequence of the fact that $\text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B)$ for matrices A and B . $\square$

21.2 Tensor Product and Trace Formula

To understand the third property more concretely, let us examine the tensor product in coordinates.

Suppose $\rho_V : G \rightarrow \text{GL}(V)$ and $\rho_W : G \rightarrow \text{GL}(W)$ are representations with bases $\{v_1, \dots, v_n\}$ and $\{w_1, \dots, w_m\}$ respectively. Let $R_V(g) = (a_{ij})$ and $R_W(g) = (b_{kl})$ be the matrices of $\rho_V(g)$ and $\rho_W(g)$ relative to these bases. Then the action on $V \otimes W$ is given by:

$$g \cdot (v_j \otimes w_k) = \left(\sum_i a_{ij} v_i \right) \otimes \left(\sum_l b_{lk} w_l \right) = \sum_{i,l} a_{ij} b_{lk} (v_i \otimes w_l).$$

Thus the matrix of $\rho_V \otimes \rho_W(g)$ on $V \otimes W$ (with basis ordered lexicographically) is the Kronecker product $R_V(g) \otimes R_W(g)$, whose trace is indeed $\text{tr}(R_V(g)) \text{tr}(R_W(g))$.

Example 21.1 (Character of the Regular Representation). Let ρ_{reg} be the regular representation of G on $\mathbb{C}[G]$. Its character is given by:

$$\chi_{\text{reg}}(g) = \begin{cases} |G| & \text{if } g = 1, \\ 0 & \text{if } g \neq 1. \end{cases}$$

This is because $\rho_{\text{reg}}(g)$ permutes the basis $\{h \mid h \in G\}$ without fixed points when $g \neq 1$, so its trace is zero.

21.3 Class Functions and the Space of Characters

We now formalize the space in which characters live and study its structure.

Definition 21.2 (Class Functions). The space of **class functions** on G is the $\mathbb{C}$ -vector space:

$$\mathcal{C}(G) = \{f : G \rightarrow \mathbb{C} \mid f(hgh^{-1}) = f(g) \text{ for all } g, h \in G\}.$$

That is, functions constant on conjugacy classes of G .

Proposition 21.2. The dimension of $\mathcal{C}(G)$ equals the number of conjugacy classes in G .

Proof. A class function is determined by its values on the conjugacy classes. The indicator functions of conjugacy classes form a basis for $\mathcal{C}(G)$. More precisely, if $C_1, \dots, C_r$ are the conjugacy classes of G , then the functions δ_{C_i} defined by

$$\delta_{C_i}(g) = \begin{cases} 1 & \text{if } g \in C_i, \\ 0 & \text{otherwise} \end{cases}$$

form a basis of $\mathcal{C}(G)$. Thus, $\dim \mathcal{C}(G) = r$. □

Since characters are class functions, they lie in $\mathcal{C}(G)$. In fact, the irreducible characters form a particularly nice basis, as we will see.

22 Orthogonality Relations

One of the deepest results in representation theory is the orthogonality relations for characters. These relations allow us to compute multiplicities of irreducible representations and provide a powerful computational tool.

22.1 Inner Product on Class Functions

Definition 22.1 (Inner Product on Class Functions). For two class functions $\phi, \psi : G \rightarrow \mathbb{C}$, define their inner product by

$$\langle \phi, \psi \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\phi(g)} \psi(g).$$

This is a Hermitian inner product on $\mathcal{C}(G)$.

22.2 Orthogonality of Irreducible Characters

Theorem 22.1 (Orthogonality Relations for Characters). *Let $\chi_1, \dots, \chi_r$ be the irreducible characters of G (corresponding to pairwise non-isomorphic irreducible representations $\rho_1, \dots, \rho_r$). Then:*

1. $\langle \chi_i, \chi_j \rangle = \delta_{ij}$ (orthonormality).
2. The irreducible characters form an orthonormal basis of $\mathcal{C}(G)$.

This theorem has several important consequences.

Corollary 22.1. *Let ρ be a representation of G with character χ . Suppose ρ decomposes into irreducibles as $\rho \cong \bigoplus_{i=1}^r \rho_i^{\oplus m_i}$. Then:*

1. $m_i = \langle \chi, \chi_i \rangle$.
2. ρ is irreducible if and only if $\langle \chi, \chi \rangle = 1$.
3. $\langle \chi_{\text{reg}}, \chi_i \rangle = \dim \rho_i = \chi_i(1)$.

Proof. For (1), by linearity of characters, $\chi = \sum_{i=1}^r m_i \chi_i$. Taking inner product with χ_j and using orthonormality gives $m_j = \langle \chi, \chi_j \rangle$.

For (2), see homework.

For (3), we compute using the formula for χ_{reg} :

$$\langle \chi_{\text{reg}}, \chi_i \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_{\text{reg}}(g) \overline{\chi_i(g)} = \frac{1}{|G|} \overline{\chi_{\text{reg}}(1)} \chi_i(1) = \chi_i(1),$$

since $\chi_{\text{reg}}(1) = |G|$ and $\chi_i(1) \in \mathbb{R}^+$. □

22.3 Proof of Orthogonality

We now outline the proof of the orthogonality relations, which relies on a beautiful connection between the inner product of characters and the dimension of invariant subspaces.

Lemma 22.1. *Let V and W be representations of G . Then*

$$\langle \chi_V, \chi_W \rangle = \dim_{\mathbb{C}} \text{Hom}_G(V, W).$$

Proof. Recall that $\text{Hom}(V, W) \cong V^* \otimes W$ as representations, with character $\chi_{V^* \otimes W}(g) = \overline{\chi_V(g)} \chi_W(g)$. Moreover, $\text{Hom}_G(V, W)$ is exactly the subspace of G -invariants in $\text{Hom}(V, W)$. Hence,

$$\langle \chi_V, \chi_W \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = \frac{1}{|G|} \sum_{g \in G} \chi_{V^* \otimes W}(g).$$

We claim that for any representation U , the average $\frac{1}{|G|} \sum_{g \in G} \chi_U(g)$ equals $\dim U^G$, where $U^G = \{u \in U \mid g \cdot u = u \ \forall g \in G\}$ is the subspace of invariants. Indeed, consider the averaging operator $P = \frac{1}{|G|} \sum_{g \in G} \rho_U(g)$. This is a projection onto U^G , and its trace is $\dim U^G$. But $\text{tr}(P) = \frac{1}{|G|} \sum_{g \in G} \chi_U(g)$. Applying this to $U = \text{Hom}(V, W)$ gives the result. □

Now if V and W are irreducible, Schur's Lemma tells us that $\dim \text{Hom}_G(V, W)$ is 1 if $V \cong W$ and 0 otherwise. This immediately yields the orthogonality relations.

22.4 Decomposition of the Regular Representation

A particularly important application of character theory is the decomposition of the regular representation.

Theorem 22.2. *The regular representation decomposes as*

$$\rho_{\text{reg}} \cong \bigoplus_{i=1}^r \rho_i^{\oplus \dim \rho_i},$$

where $\rho_1, \dots, \rho_r$ are the irreducible representations of G .

Proof. By the corollary above, the multiplicity of ρ_i in ρ_{reg} is $\langle \chi_{\text{reg}}, \chi_i \rangle = \dim \rho_i$. □

This theorem has the striking corollary that $|G| = \sum_{i=1}^r (\dim \rho_i)^2$, relating the group order to the dimensions of its irreducible representations.

23 Exercises

23.1 Mandatory part

In this homework, we always work over the complex field $\mathbb{C}$ and finite dimensional representations of finite groups.

Exercise 23.1. *Let V be an irreducible representation of a finite group G over $\mathbb{C}$ and χ be the character of V . Prove that V is irreducible if and only if $\langle \chi, \chi \rangle = 1$. Use this to reprove that the dual representation of an irreducible representation is also irreducible.*

Exercise 23.2. *Let G_1 and G_2 be two groups and V_i is an representation of G_i over $\mathbb{C}$ for $i = 1, 2$. Define the representation $V_1 \otimes V_2$ of $G_1 \times G_2$ by*

$$(g_1, g_2) \cdot (v_1 \otimes v_2) = (g_1 \cdot v_1) \otimes (g_2 \cdot v_2)$$

for $g_i \in G_i$ and $v_i \in V_i$. *Prove that $V_1 \otimes V_2$ is irreducible if and only if both V_1 and V_2 are irreducible. How do you express the character of $V_1 \otimes V_2$ in terms of the characters of V_1 and V_2 ?*

Exercise 23.3. *Let G be a finite group and operates on a finite set X . Let V be the vector space over $\mathbb{C}$ with basis X , i.e., $V = \mathbb{C}^X$. Then the G -action on X induces a linear representation of G on V . Show that number of orbits of G on X is equal to multiplicity of the trivial representation in the irreducible decomposition of V .*

Exercise 23.4. (Artin Algebra Chapter 10, 7.1) *Prove a converse to Schur's Lemma: Let ρ be a complex representation of a finite group G on a vector space V . If the only G -invariant linear operators on V are scalar multiplications (i.e., $\text{Hom}_G(V, V) = \mathbb{C} \cdot \text{Id}_V$), then ρ is irreducible.*

Exercise 23.5. *Let S_n be the symmetric group on $[n] = \{1, 2, \dots, n\}$. Prove the following combinatorial identity by characters*

$$\sum_{\sigma \in S_n} (\text{number of elements in } [n] \text{ fixed by } \sigma)^2 = 2n!$$

1. Let V be the representation of S_n on $\mathbb{C}^n$ induced by the action of S_n on standard basis $e_1, \dots, e_n$ of $\mathbb{C}^n$. Show that the character χ_V of V is given by

$$\chi_V(\sigma) = \text{number of elements in } [n] \text{ fixed by } \sigma$$

for $\sigma \in S_n$.

2. In class, we showed that the trivial representation of S_n is contained in the representation V given by subspace spanned by $e_1 + e_2 + \dots + e_n$. The G -invariant complement W is defined by $W = \{v = (v_1 \dots v_n)^T \in \mathbb{C}^n \mid \sum_{i=1}^n v_i = 0\}$. Show that W is irreducible by the following method. Let V' be a nonzero G -invariant subspace of W and $v \in V'$ be a non-zero vector. If any two component v_i and v_j of v are not equal, then use a permutation $\sigma \in S_n$ to prove that $e_i - e_j$ is also in V' . Then prove that V' must be all of W .
3. Use character of V to prove the combinatorial identity stated in the beginning.

23.2 Optional problems

Exercise 23.6. In this exercise, we work on representation of finite group G over field F whose characteristic does not divide the order of the group. Define the convolution of two functions on G by $\phi, \psi \in \text{Map}(G, F)$ by

$$(\phi * \psi)(g) = \frac{1}{|G|} \sum_{h \in G} \phi(h) \psi(h^{-1}g).$$

Prove the following:

1. The convolution is associative, i.e., $(\phi * \psi) * \theta = \phi * (\psi * \theta)$ for any $\phi, \psi, \theta \in \text{Map}(G, F)$.
2. The space of class functions $C(G, F)$ is closed under convolution.
3. If ϕ is a function on G such that $\phi * \psi = \psi * \phi$ for any $\psi \in \text{Map}(G, F)$, then ϕ is a class function.

24 Completeness of Irreducible Characters

Building on the orthogonality relations, we now prove that the irreducible characters form a basis for the space of class functions. This completeness result is fundamental for character theory.

Theorem 24.1 (Completeness of Irreducible Characters). *The irreducible characters $\chi_1, \dots, \chi_r$ form an orthonormal basis for $\mathcal{C}(G)$. In particular, $r = \dim \mathcal{C}(G)$ equals the number of conjugacy classes in G .*

Proof. We already know that the irreducible characters are orthonormal, hence linearly independent. To show they span $\mathcal{C}(G)$, let $\theta \in \mathcal{C}(G)$ be a class function orthogonal to all χ_i , i.e., $\langle \theta, \chi_i \rangle = 0$ for all i . For any representation $\rho: G \rightarrow \text{GL}(V)$, consider the operator

$$\rho_\theta = \frac{1}{|G|} \sum_{g \in G} \overline{\theta(g)} \rho(g) \in \text{Hom}_{\mathbb{C}}(V, V),$$

We claim that ρ_θ is a G -homomorphism in $\text{Hom}_G(V, V)$. Indeed, for any $h \in G$,

$$\rho(h) \rho_\theta \rho(h)^{-1} = \frac{1}{|G|} \sum_{g \in G} \overline{\theta(g)} \rho(hgh^{-1}) = \frac{1}{|G|} \sum_{g' \in G} \overline{\theta(h^{-1}g'h)} \rho(g') = \rho_\theta,$$

where we used the fact that θ is a class function. Thus ρ_θ commutes with the action of G . When V is irreducible, we apply Schur's lemma and ρ_θ is a scalar transformation $\rho_\theta = \lambda \cdot I_V$ for some $\lambda \in \mathbb{C}$. We can compute λ by taking the trace:

$$\lambda \cdot \dim V = \text{tr}(\rho_\theta) = \frac{1}{|G|} \sum_{g \in G} \overline{\theta(g)} \text{tr}(\rho(g)) = \frac{1}{|G|} \sum_{g \in G} \overline{\theta(g)} \chi(g) = \langle \theta, \chi \rangle = 0.$$

Thus $\rho_\theta = 0$ for any irreducible representation ρ . Since any representation decomposes into irreducibles, it follows that $\rho_\theta = 0$ for all representations ρ of G . In particular, consider the regular representation ρ_{reg} on $\mathbb{C}[G]$. The operator

$$T_\theta = \frac{1}{|G|} \sum_{g \in G} \theta(g) \rho_{\text{reg}}(g)$$

acts as zero on $\mathbb{C}[G]$. In particular, $T_\theta(e) = 0$, where e is the identity element of G (viewed as a basis vector of $\mathbb{C}[G]$). But

$$T_\theta(e) = \frac{1}{|G|} \sum_{g \in G} \overline{\theta(g)} \rho_{\text{reg}}(g)e = \frac{1}{|G|} \sum_{g \in G} \overline{\theta(g)} g = 0.$$

Since the elements $g \in G$ are linearly independent in $\mathbb{C}[G]$, we must have $\theta(g) = 0$ for all g , so $\theta = 0$. Therefore, the orthogonal complement of $\text{span}\{\chi_1, \dots, \chi_r\}$ in $\mathcal{C}(G)$ is zero, so the irreducible characters span $\mathcal{C}(G)$. $\square$

25 Character Tables

With the orthogonality and completeness theorems in hand, we can systematically study characters through character tables. A character table encodes the values of all irreducible characters on all conjugacy classes, providing a complete invariant for representations over $\mathbb{C}$.

25.1 Definition and Structure

Definition 25.1 (Character Table). *Let G be a finite group with conjugacy classes $C_1 = \{1\}, C_2, \dots, C_r$, and let $\chi_1, \dots, \chi_r$ be the irreducible characters of G . The **character table** of G is the $r \times r$ matrix with entries $\chi_i(C_j) = \chi_i(g)$ for any $g \in C_j$.*

The character table satisfies two orthogonality relations: the **row orthogonality** (from $\langle \chi_i, \chi_j \rangle = \delta_{ij}$) and the **column orthogonality** (which follows from the fact that the matrix is unitary up to scaling). Specifically, if we denote $h_j = |C_j|$, then

$$\sum_{k=1}^r h_k \chi_i(C_k) \overline{\chi_j(C_k)} = |G| \delta_{ij},$$

and the second orthogonality relation is

$$\sum_{i=1}^r \chi_i(C_k) \overline{\chi_i(C_\ell)} = \frac{|G|}{h_k} \delta_{k\ell}.$$

Proof of Column Orthogonality. Consider the matrix M whose (i, j) -entry is $\sqrt{h_j} \chi_i(C_j)$. The row orthogonality implies that $MM^* = |G| I_r$, where I_r is the $r \times r$ identity matrix. This means that M is unitary up to scaling by $\sqrt{|G|}$. Therefore, $M^*M = |G| I_r$ as well, which gives the column orthogonality relation. $\square$

25.2 Examples of Character Tables

Example 25.1 (Cyclic Group $C_n = \langle a \rangle$). The irreducible characters are one-dimensional: $\chi_k(a^m) = e^{2\pi i km/n}$ for $k = 0, 1, \dots, n-1$. The character table is:

C_n	1	a	$\dots$	a^{n-1}
χ_0	1	1	$\dots$	1
χ_1	1	ζ_n	$\dots$	ζ_n^{n-1}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
χ_{n-1}	1	ζ_n^{n-1}	$\dots$	$\zeta_n^{(n-1)^2}$

where $\zeta_n = e^{2\pi i/n}$.

Example 25.2 (Symmetric Group S_3). The group S_3 has three conjugacy classes: identity, transpositions, and 3-cycles. It has three irreducible representations: the trivial representation $\mathbf{1}$, the sign representation ε , and the 2-dimensional standard representation. Their characters are:

S_3	1	(12)	(123)
χ_1	1	1	1
χ_2	1	-1	1
χ_3	2	0	-1

Note that χ_3 is the character of the standard representation, which can be obtained as the permutation representation on $\mathbb{C}^3$ minus the trivial representation.

Example 25.3 (Dihedral Group D_n). The dihedral group $D_n = \langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle$ has n elements and its character table depends on whether n is even or odd. For odd n , there are $(n+3)/2$ conjugacy classes and irreducible representations. The characters can be computed from the 2-dimensional representation $D_n \rightarrow O(2)$ described earlier. For $n \geq 3$, the character of this representation is:

$$\chi(r^k) = 2 \cos\left(\frac{2\pi k}{n}\right), \quad \chi(sr^k) = 0.$$

Example 25.4 (Symmetric Group S_4). S_4 has 5 conjugacy classes: identity, transpositions, products of two disjoint transpositions, 3-cycles, and 4-cycles. It has 5 irreducible representations with dimensions 1, 1, 2, 3, 3. Their characters are:

S_4	1	(12)	(12)(34)	(123)	(1234)
χ_1	1	1	1	1	1
χ_2	1	-1	1	1	-1
χ_3	2	0	2	-1	0
χ_4	3	1	-1	0	-1
χ_5	3	-1	-1	0	1

Here χ_4 and χ_5 are the characters of the two 3-dimensional irreducible representations, which can be obtained from the permutation representation on $\mathbb{C}^4$ and its tensor product with the sign representation.

26 Symmetric and Alternating Squares

An important application of character theory is the decomposition of tensor products. In particular, for a representation V , the tensor product $V \otimes V$ decomposes into symmetric and alternating parts, which are themselves representations.

26.1 Definitions

Definition 26.1 (Symmetric and Alternating Squares). *Let V be a representation of G . Define the **symmetric square** $S^2(V)$ and the **alternating square** $\Lambda^2(V)$ (also denoted $A^2(V)$) as the subspaces of $V \otimes V$ spanned by:*

$$S^2(V) = \text{Span}\{v \otimes w + w \otimes v \mid v, w \in V\},$$

$$\Lambda^2(V) = \text{Span}\{v \otimes w - w \otimes v \mid v, w \in V\}.$$

These are G -invariant subspaces, and when $\text{char } F \neq 2$, we have the decomposition

$$V \otimes V = S^2(V) \oplus \Lambda^2(V).$$

If $\{v_1, \dots, v_n\}$ is a basis of V , then a basis of $S^2(V)$ is given by $\{v_i \otimes v_j + v_j \otimes v_i \mid i \leq j\}$, and a basis of $\Lambda^2(V)$ is given by $\{v_i \otimes v_j - v_j \otimes v_i \mid i < j\}$. The dimensions are $\dim S^2(V) = \frac{n(n+1)}{2}$ and $\dim \Lambda^2(V) = \frac{n(n-1)}{2}$.

26.2 Character Formulas

Theorem 26.1. *Let V be a representation of G with character χ . Then the characters of $S^2(V)$ and $\Lambda^2(V)$ are given by:*

$$\chi_{S^2(V)}(g) = \frac{1}{2} (\chi(g)^2 + \chi(g^2)),$$

$$\chi_{\Lambda^2(V)}(g) = \frac{1}{2} (\chi(g)^2 - \chi(g^2)).$$

Proof. Choose a basis of V in which $\rho(g)$ is diagonal, say with eigenvalues $\lambda_1, \dots, \lambda_n$. Then the eigenvalues of g on $V \otimes V$ are $\lambda_i \lambda_j$ for $i, j = 1, \dots, n$. The symmetric square $S^2(V)$ is the subspace spanned by symmetric tensors, and the eigenvalues on this subspace are $\lambda_i \lambda_j$ for $i \leq j$. Hence

$$\chi_{S^2(V)}(g) = \sum_{i \leq j} \lambda_i \lambda_j = \frac{1}{2} \left(\sum_{i,j} \lambda_i \lambda_j + \sum_i \lambda_i^2 \right) = \frac{1}{2} (\chi(g)^2 + \chi(g^2)).$$

Similarly, $\Lambda^2(V)$ has eigenvalues $\lambda_i \lambda_j$ for $i < j$, so

$$\chi_{\Lambda^2(V)}(g) = \sum_{i < j} \lambda_i \lambda_j = \frac{1}{2} \left(\sum_{i,j} \lambda_i \lambda_j - \sum_i \lambda_i^2 \right) = \frac{1}{2} (\chi(g)^2 - \chi(g^2)).$$

□

These formulas are extremely useful for decomposing tensor products and for finding new irreducible representations.

An interesting question to think about is to generalize this to higher tensor powers, such as $V^{\otimes k}$, and use the group operations of the symmetric group S_k to decompose it into irreducibles. This leads to the theory of Schur functors and Young symmetrizers, which are powerful tools in representation theory. Try to work out the decomposition and character formulas for the case of $V^{\otimes 3}$.

27 Kernel of characters and lifting of characters

Definition 27.1 (Kernel of a Character). Let χ be a character of a representation $\rho: G \rightarrow \text{GL}(V)$. The **kernel** of χ is defined as

$$\ker \chi = \{g \in G \mid \chi(g) = \dim V\}.$$

Proposition 27.1. The kernel of a character χ is the same as kernel of ρ , i.e.

$$\ker \chi = \ker \rho = \{g \in G \mid \rho(g) = I_V\}.$$

Proof. Let $g \in \ker \chi$. Then $\chi(g) = \dim V$, which means that the trace of $\rho(g)$ is equal to the dimension of V . Since the trace of a linear operator is equal to the sum of its eigenvalues, and the eigenvalues of $\rho(g)$ are all roots of unity (because G is finite), it follows that all eigenvalues of $\rho(g)$ are 1. Hence $\rho(g) = I_V$, and $g \in \ker \rho$. Conversely, if $g \in \ker \rho$, then $\rho(g) = I_V$, so $\chi(g) = \dim V$. Therefore, $\ker \chi = \ker \rho$. $\square$

In fact, the kernel of a character is a normal subgroup of G , and the representation ρ factors through the quotient group $G/\ker \chi$. This allows us to study representations of G by studying representations of its quotients.

Definition 27.2 (Lifting of representations and characters). Let ρ be a representation of a quotient group G/N , where N is a normal subgroup of G . The **lifting** of ρ to G is the representation $\tilde{\rho}$ of G defined by

$$\tilde{\rho}(g) = \rho(gN) \quad \text{for all } g \in G.$$

Similarly, if χ is a character of G/N , the **lifting** of χ to G is the character $\tilde{\chi}$ defined by

$$\tilde{\chi}(g) = \chi(gN) \quad \text{for all } g \in G.$$

An important property of lifting is that it preserves irreducibility: if ρ is an irreducible representation of G/N , then its lifting $\tilde{\rho}$ is an irreducible representation of G . This allows us to construct new irreducible representations of G from those of its quotients.

In fact, all the normal subgroups of G can be obtained as kernels of characters. This is a consequence of the following result.

Theorem 27.1. Let N be a normal subgroup of G . Then there exists a character χ of G such that $\ker \chi = N$, or equivalently, let $\chi_1, \dots, \chi_r$ be all the irreducible characters of G , then there exists a subset $I \subset \{1, 2, \dots, r\}$ such that

$$N = \bigcap_{i \in I} \ker \chi_i.$$

Proof. If $N = \{e\}$, we can take χ to be the character of any faithful representation of G , for example, the regular representation.

Since N is normal, we can consider the quotient group G/N . Let $\pi: G \rightarrow G/N$ be the natural projection. The group G/N has its own irreducible characters, say $\psi_1, \dots, \psi_s$. We can lift these characters to G by defining $\tilde{\psi}_i(g) = \psi_i(\pi(g))$. Each $\tilde{\psi}_i$ is a character of G , and its kernel contains N since $\tilde{\psi}_i(g) = \psi_i(\pi(g))$ depends only on the coset of g in G/N . Moreover, the intersection of the kernels of all $\tilde{\psi}_i$ is exactly N , because if $g \in G$ is such that $\tilde{\psi}_i(g) = \psi_i(\pi(g)) = \psi_i(e) = \dim \psi_i$ for all i , then $\pi(g) = e$ in G/N , which means $g \in N$. Thus, we can take χ to be sum of the $\tilde{\psi}_i$ to get a character of G with kernel exactly N . $\square$

So we can read all the normal subgroups of G from the character table.

28 Exercises on Character Theory

28.1 Mandatory part

In this homework, we always work over the complex field $\mathbb{C}$ and finite dimensional representations of finite groups.

Exercise 28.1. You can skip the first three questions if you already learned them in group theory last semester. Let $[G, G]$ be subgroup of G generated by all commutators $[g, h] = ghg^{-1}h^{-1}$ for $g, h \in G$.

1. Show that $[G, G]$ is a normal subgroup of G .
2. Show that $G/[G, G]$ is abelian.
3. Show that if H is abelian and $\phi: G \rightarrow H$ is a group homomorphism, then $[G, G] \subset \ker \phi$.
4. Prove that there is an one-to-one correspondence between the one-dimensional representations of G and the irreducible representations of $G/[G, G]$.
5. Show that the commutator subgroup $[G, G]$ is the intersection of kernels of all one-dimensional characters of G .

Exercise 28.2. From class, we know that the character χ_{reg} of regular representation satisfies $\chi_{\text{reg}}(g) = 0$ if $g \neq e$. There is an inverse of this proposition. Let χ be a character of G and satisfies $\chi(g) = 0$ if $g \neq e$. Prove that the corresponding representation is the direct sum of several copies of regular representation, i.e. $\rho \cong \rho_{\text{reg}} \oplus \rho_{\text{reg}} \cdots \oplus \rho_{\text{reg}}$ for some integer n .

Exercise 28.3. In this question, you will find the character table of A_4 the group of even permutations of S_4 . Alternating group A_4 is a subgroup of S_4 and has 4 conjugacy classes

$$\{(1)\}, \{(12)(34), (23)(14), (13)(24)\}, \{(123), (142), (134), (243)\}, \{(132), (124), (143), (234)\}.$$

1. Prove that $K = \{(1), (12)(34), (23)(14), (13)(24)\}$ is a normal subgroup of S_4 .
2. Prove that A_4/K is cyclic group of order 3. (You can use the fact that any group of prime order is a cyclic group, think about why this is true.)
3. Use lifting to find all the irreducible characters of A_4 .
4. (Optional) Compare this with character table of S_4 and describe all the irreducible representations of A_4 .

Exercise 28.4. Let G be a finite group and $g \in G$. Prove that g and g^{-1} are in the same conjugacy class if and only if $\chi(g)$ is in $\mathbb{R}$ for all characters χ .

Exercise 28.5. Below is a partial character table for G . One conjugacy class and one row are missing. Here e is the identity element. The numbers in the top row mean the numbers of elements in each conjugacy class.

	(1)	(1)	(2)	(2)	(3)
	e	u	v	w	x
χ_1	1	1	1	1	1
χ_2	1	1	1	1	-1
χ_3	1	-1	1	-1	i
χ_4	1	-1	1	-1	$-i$
χ_5	2	2	-1	-1	0

(3)

1. Complete the character table. (Hint: the number of elements in each conjugacy class divides the order of G .)
2. Determine the normal subgroups of G .
3. Find the orders of representative elements in each conjugacy class.

Exercise 28.6. Let $\mathbb{C}[G]$ be the underlying vector space for regular representation, i.e. the elements in $\mathbb{C}[G]$ are formal linear combinations of elements in G . Let V be a group representation of G . Prove that the map $F: \text{Hom}_G(\mathbb{C}[G], V) \rightarrow V$ defined by $F(f) = f(e)$ gives a linear space isomorphism. Use this isomorphism to give a new proof of the fact that the multiplicity of an irreducible representation W in regular representation is equal to $\dim W$.

28.2 Optional problems

Exercise 28.7. Write down the character table for group A_5 , the group of even permutations on 5 elements. (Hint: A_5 has 5 conjugacy classes and thus has 5 irreducible representations.)

29 Induced representations and restrictions

29.1 Restriction of representations

Definition 29.1. Let G be a finite group and $H \leq G$ a subgroup. If $\rho: G \rightarrow \text{GL}(V)$ is a representation of G , the **restriction** of ρ to H is the representation

$$\text{Res}_H^G V = (V, \rho|_H),$$

where we regard the same vector space V with the action of H given by the restriction $\rho|_H(h) = \rho(h)$ for $h \in H$.

The restriction simply "forgets" the action of $G \setminus H$, while the vector space, basis, and dimension remain unchanged. The character has an explicit relation: if χ_V is the character of V , then for all $h \in H$ we have

$$\chi_{\text{Res}_H^G V}(h) = \chi_V(h).$$

Example 29.1. Take the standard 2-dimensional representation of $G = S_3$ (permutations acting on the subspace where the sum of coordinates is zero) and let $H = \langle (123) \rangle \cong \mathbb{Z}/3\mathbb{Z}$ be the rotation subgroup. Restricting this representation to H gives two 1-dimensional representations (corresponding to two complex conjugate primitive third roots of unity). The reader can verify this directly by a change of basis.

29.2 Induced representations: Intuition and elementary definition

Given a representation of a subgroup H , can we construct a natural representation of G such that restricting it back to H allows us to "see" the original representation? This is the motivation behind induced representations: constructing representations of larger groups from representations of smaller groups.

Below is a completely elementary construction (without using the abstract tensor product of modules): Let W be a representation of H , denote $[G : H] = m$, and pick a set of left coset representatives

$$G = \bigsqcup_{i=1}^m g_i H.$$

Define the vector space

$$\mathrm{Ind}_H^G W := \bigoplus_{i=1}^m g_i W,$$

where each $g_i W$ is a formal copy of the same dimension as W , denoted by basis vectors $g_i \otimes w$ (here $\otimes$ is merely a label, not an abstract tensor of modules).

The action of G is defined by the following rule: for any $g \in G$ and basis vector $g_i \otimes w$, there exist unique j and $h \in H$ such that

$$gg_i = g_j h.$$

Then we define

$$g \cdot (g_j \otimes w) = g_i \otimes (h \cdot w),$$

and extend it linearly to the entire $\mathrm{Ind}_H^G W$. The reader can verify directly that this is a linear representation.

Remark 29.1. *One needs to check that this definition is independent of the choice of coset representatives; this is because replacing a representative will only introduce an element in H , the action of which is easily absorbed by $\rho(h)$ on w (the reader can work out the details as an exercise).*

Proposition 29.1. *The dimension is*

$$\dim \mathrm{Ind}_H^G W = [G : H] \cdot \dim W.$$

29.3 Description using tensor products

The above construction can be rephrased in a more abstract way using the tensor product of modules. The group algebra $\mathbb{C}[G]$ is a $\mathbb{C}[G]$ -module under left multiplication, and also a $\mathbb{C}[H]$ -module under right multiplication (since H is a subgroup of G). The representation W is a $\mathbb{C}[H]$ -module. We can then form the tensor product

$$\mathrm{Ind}_H^G W = \mathbb{C}[G] \otimes_{\mathbb{C}[H]} W,$$

where we identify $gh \otimes w$ with $g \otimes hw$ for any $g \in G$, $h \in H$, and $w \in W$. The action of G is given by left multiplication on the first factor: $g' \cdot (g \otimes w) = g'g \otimes w$. This construction is equivalent to the previous one, and it is often more convenient for theoretical purposes, such as proving properties of induced representations and understanding their relationship with restriction.

29.4 Matrix form of induced representations

If W has a basis $w_1, \dots, w_d$, then the basis of $\mathrm{Ind}_H^G W$ can be chosen as $\{g_i \otimes w_j\}_{1 \leq i \leq m, 1 \leq j \leq d}$. For a fixed $g \in G$, the transformation matrix is an $m \times m$ block matrix, where each block is $d \times d$, and each non-zero block is given by the matrix of some element in H acting on W . This explicit form is convenient for manual calculations and writing out the characters.

More explicitly, the (i, j) -block of the transformation matrix of g is given by $\rho(h)$ if $gg_j = g_i h$ for some $h \in H$, and is zero otherwise. This block structure reflects the way G permutes the cosets of H and how elements of H act on W . We define $\dot{\rho}(h) = 0$ for $h \in G \setminus H$ to unify the notation.

$$\mathrm{Ind}_H^G(\rho)(g) = (\dot{\rho}(g_i^{-1} g g_j))_{1 \leq i, j \leq m}.$$

Example 29.2. Take $G = S_3$ and $H = \{e, (12)\} \cong \mathbb{Z}/2\mathbb{Z}$. Let W be the sign representation of H (1-dimensional, where (12) acts as -1). Then $[G : H] = 3$, and $\text{Ind}_H^G W$ is a 3-dimensional representation. By writing down the coset representatives and computing the matrices of the generators of S_3 under this basis, one can decompose this induced representation into irreducible representations of S_3 (exercise).

29.5 Characters of induced representations

The character of an induced representation has a formula that is easy to compute: if ψ is the character of H , and we let $\chi = \text{Ind}_H^G \psi$, then for $g \in G$ we have

$$\chi(g) = \frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1}gx \in H}} \psi(x^{-1}gx). \quad (4)$$

This can also be written as a sum over the entire group G by extending values outside H to 0: let $\dot{\psi}$ equal ψ on H and 0 on $G \setminus H$, then

$$\chi(g) = \frac{1}{|H|} \sum_{x \in G} \dot{\psi}(x^{-1}gx).$$

The derivation of the formula can start from the function space description: $\chi(g)$, the trace of the transformation matrix, equals the sum of the traces contributed by the corresponding positions of all coset representatives g_i . Rearranging this yields the formula above.

29.6 Frobenius Reciprocity

Frobenius Reciprocity is the core relationship between induction and restriction; a simple and direct proof can be given using the inner product of class functions we've already learned.

Theorem 29.1 (Frobenius Reciprocity). *Let $H \leq G$, ψ be a class function of H (for example, the character of H), and χ be a class function of G , then*

$$\langle \chi, \text{Ind}_H^G \psi \rangle_G = \langle \text{Res}_H^G \chi, \psi \rangle_H.$$

Particularly, for representation V (with character χ_V) and W (with character χ_W) we have

$$\dim \text{Hom}_G(V, \text{Ind}_H^G W) = \dim \text{Hom}_H(\text{Res}_H^G V, W).$$

Proof. Using the definition of the inner product of class functions and substituting the induced character formula (4):

$$\begin{aligned} \langle \chi, \text{Ind}_H^G \psi \rangle_G &= \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} \left(\frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1}gx \in H}} \psi(x^{-1}gx) \right) \\ &= \frac{1}{|G||H|} \sum_{x \in G} \sum_{\substack{g \in G \\ x^{-1}gx \in H}} \overline{\chi(g)} \psi(x^{-1}gx). \end{aligned}$$

Let $y = x^{-1}gx \in H$, then $g = xyx^{-1}$. Substituting this into the summation gives

$$\langle \chi, \text{Ind}_H^G \psi \rangle_G = \frac{1}{|G||H|} \sum_{x \in G} \sum_{y \in H} \overline{\chi(xyx^{-1})} \psi(y).$$

Swapping the order of summation and using the fact that class functions are invariant under conjugation, we obtain

$$\langle \chi, \text{Ind}_H^G \psi \rangle_G = \frac{1}{|H|} \sum_{y \in H} \overline{\left(\frac{1}{|G|} \sum_{x \in G} \chi(xy x^{-1}) \right)} \psi(y).$$

Notice that since χ is a class function, we have $\chi(xy x^{-1}) = \chi(y)$ for any $y \in H$, hence $\frac{1}{|G|} \sum_{x \in G} \chi(xy x^{-1}) = \chi(y)$. Rearranging this gives

$$\langle \chi, \text{Ind}_H^G \psi \rangle_G = \frac{1}{|H|} \sum_{y \in H} \overline{\chi(y)} \psi(y) = \langle \text{Res}_H^G \chi, \psi \rangle_H.$$

This is exactly the relationship we want to prove. For the final equality regarding the Hom spaces, using the standard equivalence between representations and the inner product of class functions ($\langle \chi_V, \chi_W \rangle = \dim \text{Hom}_G(V, W)$) yields the result. $\square$

29.7 Several important corollaries and applications

- Take $H = \{e\}$. For the trivial representation $\mathbf{1}$ we have $\text{Ind}_{\{e\}}^G \mathbf{1} \cong \mathbb{C}[G]$ (the regular representation). Frobenius reciprocity specifies that the multiplicity of each irreducible representation ρ_i in the regular representation is $\dim \rho_i$, which implies the identity $|G| = \sum_i (\dim \rho_i)^2$.
- If W is an irreducible representation of H , Frobenius reciprocity says that the multiplicity of $\rho \in \text{Irr}(G)$ appearing in $\text{Ind}_H^G W$ is equal to the multiplicity of W appearing in $\text{Res}_H^G \rho$. This is extremely useful when constructing irreducible representations of a large group from subgroup representations.

29.8 Examples (detailed computation)

Example 29.3 (Two specific induced representations on S_3). Let $G = S_3$ and $H = \{e, (12)\}$. H has two 1-dimensional representations: the trivial representation $\mathbf{1}_H$ and the sign representation sgn_H (which takes the value -1 on (12)). Let's compute their induced representations to S_3 :

- $\text{Ind}_H^G \mathbf{1}_H \cong \mathbf{1}_G \oplus \text{std}$ (the sum of the trivial representation and the standard 2-dimensional representation).
- $\text{Ind}_H^G \text{sgn}_H \cong \text{sgn}_G \oplus \text{std}$ (the sum of the sign representation and the standard 2-dimensional representation).

Using the character formula (4) to compute values on each conjugacy class and comparing the inner products yields the above decompositions.

Example 29.4 (Dihedral groups). Let $G = D_n = \langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle$ and $H = \langle r \rangle \cong \mathbb{Z}/n\mathbb{Z}$. For the 1-dimensional representation of H given by $\chi_k(r) = e^{2\pi i k/n}$, inducing it to D_n usually results in a 2-dimensional representation (unless χ_k is equal to its conjugate, which typically depends on the relationship between k and $-k$). This provides most of the 2-dimensional irreducible representations of D_n .

29.9 Exercises

Optional problems

Exercise 29.1. Prove that the definition of an induced representation is independent of the choice of coset representatives.

Exercise 29.2. Prove that $\text{Ind}_{\{e\}}^G \mathbb{C} \cong \mathbb{C}[G]$ (the regular representation).

Exercise 29.3. Let $G = D_4$ and $H = \langle r \rangle \cong \mathbb{Z}/4\mathbb{Z}$. Compute the characters of the four 1-dimensional representations of H induced to G , and decompose them into irreducible representations.

Exercise 29.4. Use Frobenius reciprocity to prove: if ρ is an irreducible representation of G and $\langle \text{Res}_H^G \rho, \psi \rangle_H \neq 0$, then ρ appears in $\text{Ind}_H^G \psi$.

Exercise 29.5. Prove the transitivity of induction: if $K \leq H \leq G$, then $\text{Ind}_H^G \text{Ind}_K^H W \cong \text{Ind}_K^G W$.

Exercise 29.6. (A special case of Mackey's decomposition) Let H be an index 2 subgroup of G , and W be a representation of H . Prove that $\text{Res}_H^G \text{Ind}_H^G W \cong W \oplus (W^g)$, where g is any element in $G \setminus H$ and W^g is the representation of H defined by $(W^g)(h) = W(g^{-1}hg)$ for $h \in H$. Assume further that W is an irreducible representation of H . Prove that if $W \cong W^g$, then $\text{Ind}_H^G W$ is reducible; and if $W \not\cong W^g$, then $\text{Ind}_H^G W$ is irreducible.

30 Rings review

In this section, we review some basic concepts in ring theory and field extensions that are relevant to Galois theory.

30.1 Rings, ideals and quotient rings

Recall that a **ring** R is a set with two binary operations, addition and multiplication, such that $(R, +)$ is an abelian group, multiplication is associative and distributive over addition. We will assume all nonzero rings have a multiplicative identity $1 \neq 0$ and are **commutative** (i.e., $ab = ba$ for all $a, b \in R$) unless otherwise stated. An **integral domain** is a nonzero commutative ring with no zero divisors (i.e., $ab = 0 \implies a = 0$ or $b = 0$). A **field** is a commutative ring where every nonzero element is invertible. The basic examples of rings we will use include $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, polynomial rings $F[x]$ over a field F , and quotient rings R/I where I is an ideal of R .

Definition 30.1 (Ideal and Quotient Ring). A non-empty subset I of a ring R is an **ideal** if it is an additive subgroup and for all $r \in R, x \in I$, we have $rx \in I$. Given an ideal $I \subset R$, the **quotient ring** R/I is the set of cosets $\{r + I : r \in R\}$ with operations $(a + I) + (b + I) = (a + b) + I$ and $(a + I)(b + I) = ab + I$.

An ideal I in a commutative ring R is **prime** if R/I is an integral domain, and **maximal** if R/I is a field. By Zorn's lemma, every nonzero ring has a maximal ideal, and every maximal ideal is prime.

30.2 Polynomial rings

For a commutative ring R , the **polynomial ring** $R[x]$ consists of expressions $a_n x^n + \cdots + a_1 x + a_0$ with $a_i \in R$. The degree of a non-zero polynomial is the largest integer n such that $a_n \neq 0$. For zero polynomial, we define its degree to be $-\infty$ for convenience. The polynomial ring $R[x]$ is an integral domain if and only if R is an integral domain. If R is a field, then $R[x]$ has many nice properties, such as the division algorithm and unique factorization.

Theorem 30.1 (Division Algorithm). *Let F be a field and $f(x), g(x) \in F[x]$ with $g \neq 0$. There exist unique $q(x), r(x) \in F[x]$ such that*

$$f(x) = q(x)g(x) + r(x), \quad \text{where } \deg r < \deg g \text{ or } r = 0.$$

As a corollary, any ideal in $F[x]$ is generated by a single polynomial (a **principal ideal**), making $F[x]$ a **Principal Ideal Domain (PID)**. Then every ideal in $F[x]$ is of the form $(f(x)) = \{f(x)h(x) : h(x) \in F[x]\}$ for some polynomial $f(x)$. This also implies that the ring $F[x]$ is a **Unique Factorization Domain (UFD)**, meaning that every non-zero non-unit polynomial can be factored uniquely into irreducible polynomials. This also implies that every nonzero prime ideal in $F[x]$ is generated by an irreducible polynomial.

Another useful property of degree is that $\deg(fg) = \deg f + \deg g$ for non-zero polynomials f and g .

Proposition 30.1. *Let F be a field and $f(x) \in F[x]$ be a non-constant polynomial. Then the quotient ring $F[x]/(f(x))$ is a field if and only if $f(x)$ is irreducible over F .*

Example 30.1. *The polynomial $x^2 + 1$ is irreducible over $\mathbb{R}$ (it has no real roots). Therefore $\mathbb{R}[x]/(x^2 + 1)$ is a field. Denoting the image of x by i , every element has the form $a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$. Hence $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.*

30.3 Irreducible polynomials and criteria

Definition 30.2. *A non-constant polynomial $f(x) \in F[x]$ is **irreducible** over a field F if it cannot be written as $f(x) = g(x)h(x)$ where $g, h \in F[x]$ are non-constant polynomials.*

Theorem 30.2 (Rational Root Theorem). *Let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$. If $p/q \in \mathbb{Q}$ is a root of $f(x)$ with $\gcd(p, q) = 1$, then $p \mid a_0$ and $q \mid a_n$.*

Theorem 30.3 (Eisenstein's Criterion). *Let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$. If there exists a prime p such that:*

1. $p \nmid a_n$,
2. $p \mid a_i$ for all $0 \leq i < n$,
3. $p^2 \nmid a_0$,

then $f(x)$ is irreducible over $\mathbb{Q}$.

30.4 UFD and Gauss's Lemma

A **Unique Factorization Domain (UFD)** is an integral domain where every non-zero non-unit element can be written as a product of irreducible elements, and this factorization is unique up to order and multiplication by units.

Theorem 30.4. *If R is a UFD, then $R[x]$ is a UFD. In particular, $\mathbb{Z}[x]$ and $F[x_1, \dots, x_n]$ (where F is a field) are UFDs.*

Definition 30.3. *A polynomial $f(x) = a_n x^n + \dots + a_0 \in \mathbb{Z}[x]$ is **primitive** if the greatest common divisor of its coefficients is 1.*

Theorem 30.5 (Gauss's Lemma). *The product of two primitive polynomials in $\mathbb{Z}[x]$ is primitive.*

Corollary 30.1. *Let $f(x) \in \mathbb{Z}[x]$ be a primitive polynomial (i.e., the gcd of its coefficients is 1). Then $f(x)$ is irreducible over $\mathbb{Q}$ if and only if it is irreducible over $\mathbb{Z}$. More precisely, if $f(x)$ factors as $f(x) = g(x)h(x)$ with $g, h \in \mathbb{Q}[x]$ non-constant, then it already factors as a product of two non-constant polynomials in $\mathbb{Z}[x]$.*

31 Iterated quotients

A useful technique to analyze the structure of a ring is to take quotients by ideals iteratively. For example, we would like to know whether a prime integer p is still a prime element in the ring of Gaussian integers $\mathbb{Z}[i]$, or in other words, whether (p) remains a prime ideal in $\mathbb{Z}[i]$. This can be done by considering the quotient ring $\mathbb{Z}[i]/(p)$. For example,

$$\mathbb{Z}[i]/(5) \cong \mathbb{Z}[x]/(x^2 + 1)/(5) \cong \mathbb{Z}[x]/(5, x^2 + 1) \cong \mathbb{Z}[x]/(5)/(x^2 + 1) \cong \mathbb{F}_5[x]/(x^2 + 1)$$

Now, we only need to decide whether $x^2 + 1$ is irreducible over $\mathbb{F}_5$. Since the degree is less or equal to three, it suffices to check whether it has a root in $\mathbb{F}_5$. We can verify that $x^2 + 1$ has no root in $\mathbb{F}_5$, hence it is irreducible. Therefore $\mathbb{Z}[i]/(5)$ is a field, which implies that (5) is a prime ideal in $\mathbb{Z}[i]$. In general, we know that the multiplicative group $\mathbb{F}_p^\times$ is cyclic of order $p - 1$, so $x^2 + 1$ is reducible over $\mathbb{F}_p$ if and only if the order two element $-1 \in \mathbb{F}_p^\times$ is a square in $\mathbb{F}_p$, which happens if and only if 4 divides $p - 1$, or equivalently, $p \equiv 1 \pmod{4}$.

31.1 Review questions for practice

You don't need to submit the answers to these questions, but you can use them to test your understanding of the material in this section.

Exercise 31.1. *Determine whether the following polynomials are irreducible over $\mathbb{Q}$:*

1. $x^4 + 1$
2. $x^3 - 3x - 1$
3. $x^5 - 10x^3 + 25x^2 - 15x + 10$

Exercise 31.2. *Let p be a prime. The p -th cyclotomic polynomial is given by $\Phi_p(x) = x^{p-1} + x^{p-2} + \dots + x + 1 = \frac{x^p - 1}{x - 1}$. Prove that $\Phi_p(x)$ is irreducible over $\mathbb{Q}$. (Hint: Consider $\Phi_p(x + 1)$ and use Eisenstein's Criterion).*

Exercise 31.3. *Show that $\mathbb{Q}[x]/(x^3 - 2)$ is a field. Express $(x^2 + x + 1)^{-1}$ in the form $ax^2 + bx + c$ with $a, b, c \in \mathbb{Q}$ in this field.*

Exercise 31.4. *Prove that the ideal (x, y) in $\mathbb{Q}[x, y]$ is not a principal ideal. Conclude that $\mathbb{Q}[x, y]$ is a UFD but not a PID.*

32 Field extensions and degree of extensions

In this section, we review the basic theory of field extensions, algebraic and transcendental elements, degree of extensions, and the tower law. We also explore the implications of these concepts to classical compass and straightedge constructions.

32.1 Field extensions and algebraic elements

Definition 32.1. If K is a subfield of a field L , we say L is an **extension field** of K , and we denote this extension by L/K . We also emphasize the inclusion map by writing $\varphi: K \rightarrow L$. For example, $\mathbb{Q} \subset \mathbb{R}$ and $\mathbb{Q} \subset \mathbb{C}$ are field extensions. The natural map $\mathbb{Q} \rightarrow \mathbb{Q}[x]/(x^2 + 2)$ sending $q \in \mathbb{Q}$ to $q + (x^2 + 2)$ is also a field extension, and we can identify $\mathbb{Q}$ with its image in $\mathbb{Q}(\sqrt{2})$.

If L/K is an extension, we can view K as a subring of L . For any $\alpha \in L$, we can consider the smallest subfield of L containing both K and α , which is denoted by $K(\alpha)$ and called the **field generated by α over K** . It consists of all rational expressions in α with coefficients in K , i.e., elements of the form

$$\frac{f(\alpha)}{g(\alpha)}$$

where $f(x), g(x) \in K[x]$ and $g(\alpha) \neq 0$. The notion $K[\alpha]$ is for all polynomial expressions in α with coefficients in K , i.e., elements of the form $f(\alpha)$ where $f(x) \in K[x]$. We have $K[\alpha] \subseteq K(\alpha)$, and they are equal if and only if α is algebraic over K (to be defined below).

Given an extension L/K , (or $\varphi: K \rightarrow L$), and an element $\alpha \in L$, we can define a ring homomorphism $\Phi_\alpha: K[x] \rightarrow L$ by $f(x) \mapsto f(\alpha)$, or $\Phi_\alpha(\sum a_i x^i) \mapsto \sum \varphi(a_i) \alpha^i$. This homomorphism captures the algebraic relationship between α and K . This homomorphism is also called the **evaluation homomorphism** at α .

Definition 32.2. An element $\alpha \in L$ is **algebraic** over K if $\ker \Phi_\alpha \neq \{0\}$, meaning there exists a non-zero polynomial $f(x) \in K[x]$ such that $f(\alpha) = 0$. Otherwise, α is **transcendental** over K .

If α is algebraic over K , $\ker \phi_\alpha$ is a non-zero prime ideal in the PID $K[x]$, so it is generated by a unique monic irreducible polynomial $m_\alpha(x) \in K[x]$, called the **minimal polynomial** of α over K . By the first isomorphism theorem, $K(\alpha) \cong K[x]/(m_\alpha(x))$.

Example 32.1. $\sqrt{2}$ is algebraic over $\mathbb{Q}$ with minimal polynomial $m_{\sqrt{2}}(x) = x^2 - 2$. Therefore $\mathbb{Q}(\sqrt{2}) \cong \mathbb{Q}[x]/(x^2 - 2)$, and every element can be written uniquely as $a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$, i.e., $\{1, \sqrt{2}\}$ is a basis of $\mathbb{Q}(\sqrt{2})$ as a $\mathbb{Q}$ -vector space.

32.2 Degree of extensions and the tower law

Notice that if L/K is a field extension, L naturally has the structure of a vector space over K .

Definition 32.3. The **degree** of the extension L/K , denoted $[L : K]$, is the dimension of L as a vector space over K . If $[L : K]$ is finite, we call L/K a **finite extension**.

Proposition 32.1. Let $\alpha \in L$ be algebraic over K . Then $[K(\alpha) : K] = \deg m_\alpha(x)$.

Theorem 32.1 (Tower Law). Let $K \subseteq M \subseteq L$ be fields. Then L/K is a finite extension if and only if L/M and M/K are finite extensions. In this case,

$$[L : K] = [L : M][M : K].$$

Proof. Assume first that M/K and L/M are finite. Let

$$\{e_1, \dots, e_m\}$$

be a basis for M over K , and let

$$\{f_1, \dots, f_n\}$$

be a basis for L over M . We claim that the mn elements

$$e_i f_j \quad (1 \leq i \leq m, 1 \leq j \leq n)$$

form a basis for L over K .

To see that they span L over K , let $x \in L$. Since the f_j form an M -basis of L , we may write

$$x = \sum_{j=1}^n a_j f_j$$

with $a_j \in M$. Since the e_i form a K -basis of M , each coefficient a_j can be written as

$$a_j = \sum_{i=1}^m b_{ij} e_i \quad (b_{ij} \in K).$$

Substituting gives

$$x = \sum_{j=1}^n \sum_{i=1}^m b_{ij} e_i f_j,$$

so the elements $e_i f_j$ span L over K .

For linear independence, suppose

$$\sum_{j=1}^n \sum_{i=1}^m c_{ij} e_i f_j = 0 \quad (c_{ij} \in K).$$

Rearranging by the f_j gives

$$\sum_{j=1}^n \left(\sum_{i=1}^m c_{ij} e_i \right) f_j = 0.$$

Since the f_j are linearly independent over M , we must have

$$\sum_{i=1}^m c_{ij} e_i = 0$$

for every j . But the e_i are linearly independent over K , so each $c_{ij} = 0$. Thus the set $\{e_i f_j\}$ is linearly independent over K .

Therefore $\{e_i f_j\}$ is a K -basis of L , and hence

$$[L : K] = mn = [L : M][M : K].$$

Conversely, suppose L/K is finite. Then any subset of a finite-dimensional K -vector space is linearly dependent if it has too many elements, so every intermediate field is automatically finite-dimensional over K . In particular, M/K is finite. Also, if $\{u_1, \dots, u_r\}$ is a K -basis of L , then the same set spans L as an M -vector space because $K \subseteq M$. Hence L/M is finite as well. This proves the theorem. $\square$

Example 32.2. We have $[\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}] = 4$ because $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt[4]{2})$ with $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ and $[\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}(\sqrt{2})] = 2$.

Theorem 32.2 (Description of degree 2 extensions). *Let L/K be a field extension with $[L : K] = 2$ and the characteristic of K is not 2. Then there exists $\alpha \in L$ such that $L = K(\alpha)$ and $m_\alpha(x) = x^2 - a$ for some $a \in K$. In other words, every degree 2 extension is generated by the square root of some element in the base field.*

(We will give the precise definition of separable extensions later.) For characteristic two case, there are two different types of degree 2 extensions: the separable ones are generated by the root of a polynomial of the form $x^2 - x - a$ (Artin-Schreier) for some $a \in K$, and the inseparable ones are generated by the root of a polynomial of the form $x^2 - a$ for some $a \in K$.

Proof. The first part of the theorem can be proved by taking any $\alpha \in L \setminus K$ and considering its minimal polynomial $m_\alpha(x)$. Since $[L : K] = 2$, $m_\alpha(x)$ must have degree 2. If the characteristic of K is not 2, we can complete the square to write $m_\alpha(x)$ in the form $x^2 - a$ for some $a \in K$ and $\beta = \alpha - \frac{b}{2}$ for suitable $b \in K$. For the characteristic 2 case, the minimal polynomial can be transformed to either of the form $x^2 - x - a$ (separable) or $x^2 - a$ (inseparable), depending on whether the derivative of the minimal polynomial is zero or not. $\square$

32.3 Algebraic extensions and finite extensions

An extension L/K is called **algebraic** if every element of L is algebraic over K .

Proposition 32.2. *Every finite extension is an algebraic extension. Furthermore, an extension is finite if and only if it is generated by a finite number of algebraic elements.*

Corollary 32.1. *If α and β are algebraic over K , then $\alpha \pm \beta$, $\alpha\beta$, and α/β (for $\beta \neq 0$) are also algebraic over K .*

32.4 Compass and straightedge constructions

We apply the theory of field extensions to classical geometric problems. Let us define the set of **constructible numbers** as the coordinates of points that can be constructed from $(0, 0)$ and $(1, 0)$ using only an unmarked straightedge and a compass.

Theorem 32.3. *If α is a constructible number, then $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is a power of 2.*

This elegant theorem allows us to resolve three classic impossibilities of antiquity:

1. **Trisecting an angle:** Trisecting a 60° angle requires constructing $\cos 20^\circ$, which is a root of $8x^3 - 6x - 1 = 0$ (derived from the triple angle formula $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$). Since this polynomial is irreducible over $\mathbb{Q}$, $[\mathbb{Q}(\cos 20^\circ) : \mathbb{Q}] = 3 \neq 2^n$.
2. **Doubling a cube:** Constructing a cube with twice the volume of a unit cube requires constructing $\sqrt[3]{2}$, but $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3 \neq 2^n$.
3. **Squaring a circle:** This requires constructing $\sqrt{\pi}$. Since π is transcendental, $[\mathbb{Q}(\sqrt{\pi}) : \mathbb{Q}]$ is not finite.

33 Splitting fields

In this section, we study the formal construction of fields containing all roots of a given polynomial. These so-called splitting fields will allow us to define Galois extensions.

33.1 Definition and existence of splitting fields

Definition 33.1 (Splits). We say a polynomial $f(x) \in K[x]$ **splits** over an extension field L of K if $f(x)$ factors completely into linear factors in $L[x]$, i.e., there exist $\alpha_1, \dots, \alpha_n \in L$ such that

$$f(x) = c(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n)$$

for some $c \in K \setminus \{0\}$.

Definition 33.2. Let K be a field and $f(x) \in K[x]$ of degree $n > 0$. An extension field L of K is called a **splitting field** for $f(x)$ over K if:

1. $f(x)$ splits over L .
2. L is generated over K by the roots of $f(x)$, i.e., $L = K(\alpha_1, \dots, \alpha_n)$.

Theorem 33.1 (Existence of splitting fields). For any field K and any non-constant polynomial $f(x) \in K[x]$, there exists a splitting field for $f(x)$ over K .

Proof. We proceed by induction on $\deg f = n$. If $\deg f = 1$, K is already the splitting field. Suppose $n > 1$ and the result holds for polynomials of degree less than n . Let $p(x)$ be an irreducible factor of $f(x)$ in $K[x]$. Then $K_1 = K[x]/(p(x))$ is a field extension of K in which $p(x)$ (and hence $f(x)$) has a root, say α . In $K_1[x]$, we can write $f(x) = (x - \alpha)g(x)$ for some $g(x) \in K_1[x]$ of degree $n - 1$. By induction, $g(x)$ has a splitting field L over K_1 . It is easy to verify that L is a splitting field for $f(x)$ over K . $\square$

Example 33.1. Consider $f(x) = x^3 - 2 \in \mathbb{Q}[x]$. Its roots in $\mathbb{C}$ are $\sqrt[3]{2}$, $\sqrt[3]{2}\omega$, and $\sqrt[3]{2}\omega^2$, where $\omega = e^{2\pi i/3} = \frac{-1 + \sqrt{-3}}{2}$. The splitting field of $f(x)$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[3]{2}, \omega)$, which has degree 6 over $\mathbb{Q}$.

Example 33.2. Consider $f(x) = x^2 + x + 1 \in \mathbb{F}_2[x]$. One checks that $f(0) = f(1) = 1 \neq 0$, so f has no roots in $\mathbb{F}_2$ and is irreducible. The splitting field is $\mathbb{F}_2[x]/(x^2 + x + 1) = \mathbb{F}_4$, which has degree 2 over $\mathbb{F}_2$.

Corollary 33.1. If L is a splitting field of a polynomial $f(x) \in K[x]$ of degree n , then $[L : K]$ divides $n!$. In particular, $[L : K] \leq n!$.

Proof. Try to prove it using the iterated constructions of splitting fields. $\square$

33.2 Uniqueness of splitting fields

A fundamental property of splitting fields is their uniqueness up to isomorphism. The proof rests on understanding how a field homomorphism can be extended one simple extension at a time. We begin with the key building block—a *relative* version of the simple extension lemma—and then count the number of possible extensions.

Extending homomorphisms through a simple algebraic extension

Lemma 33.1 (Relative Simple Extension Lemma). Let $\sigma: K \rightarrow K'$ be a field homomorphism, let $p(x) \in K[x]$ be irreducible, and let α be a root of $p(x)$ in some extension of K . Write $\bar{p}(x) = \sigma(p)(x) \in K'[x]$ for the image of $p(x)$ under σ (apply σ to each coefficient).

Suppose L' is an extension of K' and $\beta \in L'$ is a root of $\bar{p}(x)$. Then σ extends to a (unique) field homomorphism

$$\tilde{\sigma}: K(\alpha) \longrightarrow L'$$

such that $\tilde{\sigma}(\alpha) = \beta$ and $\tilde{\sigma}|_K = \sigma$.

Proof. Every element of $K(\alpha)$ can be written uniquely as

$$a_0 + a_1\alpha + \cdots + a_{d-1}\alpha^{d-1}, \quad a_i \in K, \quad d = \deg p.$$

Define $\tilde{\sigma}$ by

$$\tilde{\sigma}\left(\sum_{i=0}^{d-1} a_i \alpha^i\right) = \sum_{i=0}^{d-1} \sigma(a_i) \beta^i.$$

Because $K(\alpha) \cong K[x]/(p(x))$ and $K'(\beta) \cong K'[x]/(\bar{p}(x))$ (here we use the fact that $\bar{p}(x)$ is irreducible in $K'[x]$, which follows from $p(x)$ being irreducible and σ being injective), the map $\tilde{\sigma}$ is a well-defined injective ring homomorphism $K(\alpha) \hookrightarrow L'$ extending σ with $\tilde{\sigma}(\alpha) = \beta$.

Uniqueness. Any homomorphism extending σ with $\alpha \mapsto \beta$ must send $\sum a_i \alpha^i$ to $\sum \sigma(a_i) \beta^i$ by ring-homomorphism properties, so $\tilde{\sigma}$ is the only such extension. $\square$

Remark 33.1 (Why $\bar{p}$ is irreducible). *Since σ is a field homomorphism it is injective. If $\bar{p}(x) = g(x)h(x)$ with $\deg g, \deg h \geq 1$ in $K'[x]$, pulling back via the isomorphism σ^{-1} (on the image of σ) would factor $p(x)$, contradicting its irreducibility. In general, if σ is only a homomorphism into K' (not necessarily surjective), the same argument works because a field homomorphism is always injective.*

Counting extensions

Lemma 33.1 produces *one* extension $\tilde{\sigma}$ per choice of root β of $\bar{p}(x)$ in L' . Different choices of β give different extensions (since $\tilde{\sigma}(\alpha) = \beta$). Therefore the number of extensions is exactly the number of distinct roots of $\bar{p}(x)$ in L' .

Proposition 33.1 (Number of extensions). *In the setting of Lemma 33.1, suppose L' contains a splitting field of $\bar{p}(x)$. Let r denote the number of distinct roots of $\bar{p}(x)$ in L' . Then the number of field homomorphisms $K(\alpha) \rightarrow L'$ extending σ is exactly r . Moreover:*

1. $r \leq \deg p = [K(\alpha) : K]$.
2. $r = [K(\alpha) : K]$ if and only if $\bar{p}(x)$ (equivalently $p(x)$) has no repeated roots in L' , i.e. $p(x)$ is **separable**.
3. If $p(x)$ is inseparable (which can only happen in characteristic $p > 0$), then r is a proper divisor of $[K(\alpha) : K]$.

Proof. As observed, each extension corresponds bijectively to a root β of $\bar{p}(x)$ in L' , so the count is r .

(1) Since $\bar{p}$ has degree $d = [K(\alpha) : K]$ and a polynomial of degree d has at most d roots, we have $r \leq d$.

(2) Equality $r = d$ holds exactly when $\bar{p}(x)$ has d distinct roots, which is the definition of separability. Because σ is injective, $p(x)$ and $\bar{p}(x)$ have the same structure of repeated roots (applying σ to a factorization over an algebraic closure is a bijection on roots with multiplicities), so $\bar{p}$ is separable if and only if p is.

(3) If $\text{char } K = p > 0$ and $p(x)$ is inseparable, then every root has multiplicity p^k for some $k \geq 1$, so r divides $d/p^k < d$. $\square$

Remark 33.2 (Towards the Galois group: why counting matters). *Proposition 33.1 is the embryo of many later counting results. When we iterate through a tower $K = K_0 \subset K_1 \subset \cdots \subset K_n = L$, multiplying the number of extensions at each step gives the inequality*

$$\#\{K\text{-homomorphisms } L \rightarrow \bar{K}\} \leq [L : K],$$

with equality precisely when every step is separable—i.e. L/K is a separable extension. This will be made precise in the Counting Embeddings Theorem (Theorem 35.1).

Let us preview how this bound leads to the full Galois theory. The automorphisms $\text{Aut}(L/K)$ are those K -homomorphisms $L \rightarrow \bar{K}$ that happen to land back in L . Therefore we always have a chain of inequalities

$$|\text{Aut}(L/K)| \leq |\text{Hom}_K(L, \bar{K})| \leq [L : K].$$

- **Separability controls the right-hand inequality.** It becomes an equality ($|\text{Hom}_K(L, \bar{K})| = [L : K]$) if and only if L/K is separable.
- **Normality controls the left-hand inequality.** When L is a splitting field over K , every K -embedding $L \hookrightarrow \bar{K}$ permutes the roots and therefore maps L into itself: $\text{Hom}_K(L, \bar{K}) = \text{Aut}(L/K)$. This “splitting” or “normal” property is exactly what closes the gap between Aut and Hom .

When both conditions hold—i.e. L is a splitting field of a separable polynomial—the chain collapses to

$$|\text{Aut}(L/K)| = [L : K].$$

This equality is the hallmark of a **Galois extension**, and it is already visible in Example 33.3 below.

Extension lemma for splitting fields

We now iterate the relative simple extension lemma to handle *all* roots of a polynomial at once.

Lemma 33.2 (Extension Lemma for splitting fields). *Let $\sigma : K \rightarrow K'$ be a field isomorphism, and let $f(x) \in K[x]$ be a polynomial of degree $n \geq 1$. Let L be a splitting field for $f(x)$ over K , and let L' be a splitting field for $\sigma(f)(x)$ over K' . Then:*

1. *There exists an isomorphism $\tau : L \xrightarrow{\sim} L'$ such that $\tau|_K = \sigma$.*
2. *The number of such isomorphisms is at most $[L : K]$ (equivalently at most $n!$); equality holds when $f(x)$ is separable.*

Proof. We argue by induction on $[L : K]$.

Base case: $[L : K] = 1$, i.e. $L = K$. Then f already splits over K , so $\sigma(f)$ already splits over K' and $L' = K'$. Take $\tau = \sigma$.

Inductive step: Assume $L \neq K$. Choose an irreducible factor $p(x) \mid f(x)$ in $K[x]$ with $d = \deg p \geq 2$, and fix a root $\alpha \in L$ of $p(x)$. The image $\bar{p}(x) = \sigma(p)(x)$ is irreducible in $K'[x]$. Since L' is a splitting field of $\sigma(f)$ and $\bar{p} \mid \sigma(f)$, the polynomial $\bar{p}$ has at least one root in L' .

Pick any root $\beta \in L'$ of $\bar{p}(x)$. By Lemma 33.1, σ extends to a unique isomorphism

$$\sigma_1 : K(\alpha) \xrightarrow{\sim} K'(\beta), \quad \sigma_1(\alpha) = \beta.$$

Now write

$$f(x) = (x - \alpha)g(x) \quad \text{in } K(\alpha)[x].$$

Then L is a splitting field of $g(x)$ over $K(\alpha)$, and L' is a splitting field of $\sigma_1(g)(x)$ over $K'(\beta)$. Since $[L : K(\alpha)] < [L : K]$, the induction hypothesis gives an isomorphism $\tau : L \xrightarrow{\sim} L'$ extending σ_1 . In particular $\tau|_K = \sigma$.

Counting. At the first step there are r choices for β (the number of distinct roots of $\bar{p}$ in L'), and for each choice, the induction hypothesis gives at most $[L : K(\alpha)]$ isomorphisms. Therefore the total count satisfies

$$\#\{\tau: L \xrightarrow{\sim} L' \text{ extending } \sigma\} \leq r \cdot [L : K(\alpha)] \leq d \cdot [L : K(\alpha)] = [K(\alpha) : K] \cdot [L : K(\alpha)] = [L : K],$$

where we used the tower law. If f is separable, then each irreducible factor is separable, so $r = d$ at every step, and the inductive counts achieve equality throughout, giving exactly $[L : K]$ isomorphisms. $\square$

Uniqueness of splitting fields

Theorem 33.2 (Uniqueness of splitting fields). *Any two splitting fields of a polynomial $f(x) \in K[x]$ over K are isomorphic by an isomorphism that fixes K pointwise.*

Proof. Apply Lemma 33.2 with $K' = K$ and $\sigma = I_K$. Since $\sigma(f) = f$, the lemma produces an isomorphism $\tau: L \xrightarrow{\sim} L'$ with $\tau|_K = I_K$. $\square$

Example 33.3. Take $f(x) = x^3 - 2 \in \mathbb{Q}[x]$. Inside $\mathbb{C}$, one splitting field is

$$L = \mathbb{Q}(\sqrt[3]{2}, \omega),$$

where $\omega = e^{2\pi i/3}$. If we choose a different cube root, for instance $\alpha' = \omega\sqrt[3]{2}$, then we obtain another field

$$L' = \mathbb{Q}(\alpha', \omega).$$

As subfields of $\mathbb{C}$, these are actually equal, since $\sqrt[3]{2} = \omega^2\alpha' \in L'$. The uniqueness theorem guarantees a $\mathbb{Q}$ -isomorphism $L \xrightarrow{\sim} L'$.

How many such $\mathbb{Q}$ -automorphisms of L are there? Since $f(x) = x^3 - 2$ is separable (we are in characteristic 0) and $[L : \mathbb{Q}] = 6$, the counting part of Lemma 33.2 gives exactly 6 automorphisms. Each automorphism is determined by the image of $\sqrt[3]{2}$ (any of the three roots of $x^3 - 2$) and the image of ω (either ω or ω^2), giving $3 \times 2 = 6$ choices. This is exactly the Galois group $\text{Aut}(L/\mathbb{Q}) \cong S_3$.

Example 33.4 (Inseparable case: fewer extensions). Let $K = \mathbb{F}_p(t)$ and consider $f(x) = x^p - t \in K[x]$. This polynomial is irreducible (by Eisenstein with the prime t), and in its splitting field $L = K(\alpha)$ where $\alpha^p = t$, it factors as $x^p - t = (x - \alpha)^p$. Here $[L : K] = p$, but there is only 1 root α (with multiplicity p), so there is only 1 K -automorphism of L , namely the identity. The strict inequality

$$\#\{K\text{-automorphisms of } L\} = 1 < p = [L : K]$$

reflects the inseparability of f .

Remark 33.3 (Road map: the two pillars of Galois theory). The examples above illustrate a pattern that will drive the rest of the course. The Extension Lemma (Lemma 33.2) gives us an upper bound: for a splitting field L of $f(x) \in K[x]$,

$$|\text{Aut}(L/K)| \leq [L : K],$$

with equality when f is separable. This is the **Counting Lemma** (to be generalized to arbitrary finite extensions in Theorem 35.1). It tells us how large the automorphism group can be.

In the Galois correspondence, we will meet a complementary result going in the opposite direction.

Artin’s Lemma (Theorem 40.1). If $H \leq \text{Aut}(L)$ is a finite group and $K = L^H$ is the fixed field, then $[L : K] = |H|$.

This is a lower bound: starting from a group of automorphisms, it guarantees that the extension degree is at least as large as the group. Together, the two results establish a tight “sandwich”:

Counting Lemma (upper bound): $|\text{Aut}(L/K)| \leq [L : K]$, equality $\Leftrightarrow$ normal + separable.

Artin’s Lemma (lower bound): $[L : L^H] \leq |H|$ is impossible; always $[L : L^H] = |H|$.

When both bounds are simultaneously sharp—that is, when L/K is normal and separable (equivalently, L is a splitting field of a separable polynomial)—the automorphism group “knows everything” about the extension:

$$|\text{Aut}(L/K)| = [L : K], \quad \text{Aut}(L/L^H) = H \quad \text{for every subgroup } H \leq \text{Aut}(L/K).$$

Extensions with this property are called **Galois extensions**, and the resulting bijection between subgroups and intermediate fields is the **Fundamental Theorem of Galois Theory**.

33.3 Algebraic closures

Definition 33.3. A field K is **algebraically closed** if every non-constant polynomial $f(x) \in K[x]$ has a root in K (or equivalently, splits completely in $K[x]$). An extension $\bar{K}$ of K is an **algebraic closure** of K if $\bar{K}$ is algebraically closed and algebraic over K .

The next two results show that every field has an algebraic closure and that it is essentially unique. We follow Artin’s elegant approach, which avoids transfinite induction and uses only the existence of maximal ideals (a consequence of Zorn’s lemma).

Existence

The key idea is to adjoin roots for *all* polynomials simultaneously.

Theorem 33.3 (Existence of algebraic closures). *Every field K has an algebraic closure.*

Proof. Step 1: One big extension in which every polynomial acquires a root.

For each monic irreducible polynomial $f \in K[x]$ of degree ≥ 1 , introduce a formal variable x_f . Consider the polynomial ring in all these variables:

$$R = K[\{x_f\}_{f \in \mathcal{F}}],$$

where $\mathcal{F}$ denotes the set of all monic irreducible polynomials in $K[x]$. Inside R , form the ideal

$$I = (f(x_f) : f \in \mathcal{F}),$$

generated by evaluating each f at its corresponding variable.

Claim. The ideal I is proper, i.e. $1 \notin I$.

Proof of claim. Suppose for contradiction that $1 \in I$. Then there exist finitely many irreducible polynomials $f_1, \dots, f_n \in \mathcal{F}$ and elements $g_1, \dots, g_n \in R$ such that

$$1 = g_1 \cdot f_1(x_{f_1}) + \dots + g_n \cdot f_n(x_{f_n}).$$

This relation involves only finitely many variables $x_{f_1}, \dots, x_{f_n}$. Let L be a splitting field of the product $f_1 \cdots f_n$ over K . In L , each f_i has a root α_i . Substituting $x_{f_i} \mapsto \alpha_i$ (and mapping any other occurring variable to 0, say) into the above relation gives

$$1 = g_1(\alpha_1, \dots) \cdot \underbrace{f_1(\alpha_1)}_{=0} + \cdots + g_n(\dots, \alpha_n) \cdot \underbrace{f_n(\alpha_n)}_{=0} = 0,$$

a contradiction. Therefore $I \neq R$.

Since I is a proper ideal, Zorn's lemma provides a maximal ideal $\mathfrak{m} \supset I$. The quotient

$$K_1 = R/\mathfrak{m}$$

is a field extension of K in which every $f \in \mathcal{F}$ has at least one root (namely the image of x_f).

Step 2: Iteration. In K_1 , there may be new irreducible polynomials that do not yet have roots. Apply the same construction to K_1 to obtain K_2 , and continue inductively:

$$K = K_0 \subset K_1 \subset K_2 \subset \cdots$$

Define

$$\bar{K} = \bigcup_{n=0}^{\infty} K_n.$$

$\bar{K}$ is a field: given any two elements $a, b \in \bar{K}$, both lie in some K_n , and we can perform field operations there.

$\bar{K}$ is algebraically closed: let $h(x) \in \bar{K}[x]$ be non-constant. Its finitely many coefficients all belong to some K_n , so $h(x) \in K_n[x]$. Any irreducible factor of h over K_n acquires a root in K_{n+1} by construction. Hence h has a root in $\bar{K}$.

$\bar{K}$ is algebraic over K : every element of K_1 is of the form $g(\alpha_1, \dots, \alpha_m)$ for some $g \in K[x_1, \dots, x_m]$ and roots α_i of irreducible polynomials over K , so it is algebraic over K . By induction, every element of K_n is algebraic over K_{n-1} , hence algebraic over K (since an algebraic tower of finite extensions is algebraic). Therefore every element of $\bar{K}$ is algebraic over K . $\square$

Remark 33.4. Artin's proof can be viewed as simultaneously adjoining “generic roots” of all polynomials and then verifying that no contradiction arises. The finiteness argument in Step 1 is the heart of the matter: any potential relation $1 \in I$ can only involve finitely many polynomials, and a splitting field for that finite set resolves the contradiction.

Uniqueness

Theorem 33.4 (Uniqueness of algebraic closures). *Any two algebraic closures of a field K are isomorphic over K . More precisely, if $\bar{K}_1$ and $\bar{K}_2$ are algebraic closures of K , there exists a K -isomorphism $\varphi: \bar{K}_1 \xrightarrow{\sim} \bar{K}_2$.*

The proof uses Zorn's lemma to extend homomorphisms step by step, applying the Relative Simple Extension Lemma (Lemma 33.1) at each stage.

Proof. Consider the set $\mathcal{S}$ of all pairs (M, σ) , where M is an intermediate field $K \subseteq M \subseteq \bar{K}_1$ and $\sigma: M \hookrightarrow \bar{K}_2$ is a K -embedding. We order $\mathcal{S}$ by saying $(M, \sigma) \leq (M', \sigma')$ if $M \subseteq M'$ and $\sigma'|_M = \sigma$.

$\mathcal{S}$ is non-empty: $(K, \text{id}_K) \in \mathcal{S}$ (viewing id_K as the natural inclusion $K \hookrightarrow \bar{K}_2$).

Every chain has an upper bound: if $\{(M_i, \sigma_i)\}_{i \in I}$ is a chain in $\mathcal{S}$, set $M = \bigcup_i M_i$ and define $\sigma: M \rightarrow \bar{K}_2$ by $\sigma(x) = \sigma_i(x)$ for any i with $x \in M_i$. This is well-defined by the chain condition, and $(M, \sigma) \in \mathcal{S}$.

By Zorn's lemma, $\mathcal{S}$ has a maximal element (M_0, σ_0) .

Claim: $M_0 = \bar{K}_1$. Suppose not, and pick $\alpha \in \bar{K}_1 \setminus M_0$. Let $p(x)$ be the minimal polynomial of α over M_0 , which is irreducible in $M_0[x]$. Since $\bar{K}_2$ is algebraically closed, $\sigma_0(p)(x)$ has a root $\beta \in \bar{K}_2$. By the Relative Simple Extension Lemma, σ_0 extends to a K -embedding $\tilde{\sigma}_0: M_0(\alpha) \hookrightarrow \bar{K}_2$ with $\tilde{\sigma}_0(\alpha) = \beta$. But then $(M_0(\alpha), \tilde{\sigma}_0) > (M_0, \sigma_0)$, contradicting maximality.

Hence $\sigma_0: \bar{K}_1 \hookrightarrow \bar{K}_2$ is a K -embedding.

Claim: σ_0 is surjective. Let $\gamma \in \bar{K}_2$. Since $\bar{K}_2/K$ is algebraic, γ is a root of some polynomial $h(x) \in K[x]$. The polynomial h also has a root in $\bar{K}_1$ (since $\bar{K}_1$ is algebraically closed), so $\sigma_0(h) = h$ (because σ_0 fixes K) splits completely in $\sigma_0(\bar{K}_1)$, and γ is among these roots. More precisely, $\sigma_0(\bar{K}_1)$ is algebraically closed (as the isomorphic image of an algebraically closed field) and contains K , and since γ is algebraic over K , we have $\gamma \in \sigma_0(\bar{K}_1)$.

Therefore σ_0 is a K -isomorphism $\bar{K}_1 \xrightarrow{\sim} \bar{K}_2$. $\square$

Remark 33.5. *The uniqueness isomorphism is not canonical: it depends on the choices made via Zorn's lemma. In general, an algebraic closure $\bar{K}$ has many K -automorphisms. For example, $\mathbb{Q}$ has uncountably many $\mathbb{Q}$ -automorphisms, even though $\mathbb{Q}$ itself is countable. The group $\text{Aut}(\bar{K}/K)$ is the **absolute Galois group** of K , and understanding its structure is a central problem in number theory.*

34 Exercises

34.1 Mandatory part

Exercise 34.1. Show that cyclotomic polynomials $\Phi_p(x) = \frac{x^p - 1}{x - 1}$ are irreducible over $\mathbb{Q}$ for all prime numbers p .

Exercise 34.2. Prove that the polynomial $x^4 + 3x + 3$ is irreducible polynomial over the field $\mathbb{Q}[\sqrt[3]{2}]$.

Exercise 34.3. Find the degree of field extension

1. $[\mathbb{Q}[\sqrt{p}, \sqrt{q}]: \mathbb{Q}]$ where p and q are two distinct prime numbers.
2. $[\mathbb{Q}[\sqrt[3]{2}, \sqrt{2}]: \mathbb{Q}]$

Exercise 34.4. Find the irreducible polynomial of $\sqrt[3]{2} + \sqrt{3}$ over $\mathbb{Q}$.

Exercise 34.5. Find whether (7) is a prime ideal in $\mathbb{Z}[\sqrt{-3}]$.

Exercise 34.6. Find the splitting field L of $x^8 - 2$ over $\mathbb{Q}$ (Write L in the form of $\mathbb{Q}(\alpha, \beta, \dots)$), and compute $[L: \mathbb{Q}]$.

Exercise 34.7. Let L be the splitting field of $x^3 - 1$ over $\mathbb{F}_2$. Compute $[L: \mathbb{F}_2]$.

Exercise 34.8. We call two extensions K_1 and K_2 of F isomorphic if there exists a field isomorphism $\varphi: K_1 \rightarrow K_2$ such that $\varphi|_F: F \rightarrow F$ is identity. Assume $\text{char}(F) \neq 2$. Classify the isomorphism classes of degree-two extensions (quadratic extensions) of F . (Find an identification with $F^\times / (F^\times)^2 - \{1\}$, and explain the structure of this group $F^\times / (F^\times)^2$ using prime factorization when $F = \mathbb{Q}$.)

34.2 Optional questions

Exercise 34.9. Let $p_1, \dots, p_n$ be distinct primes. Prove that $[\mathbb{Q}(\sqrt{p_1}, \dots, \sqrt{p_n}) : \mathbb{Q}] = 2^n$.

Exercise 34.10. Consider quadratic extensions K of F with $\text{char } F = 2$. Prove that either $K = F[\alpha]$ with $\alpha^2 \in F$ and $\alpha \notin F$ or $K = F[\alpha]$ with $\alpha^2 - \alpha \in F$ and $\alpha \notin F$. Can two cases of two different types above be isomorphic? Classify the isomorphism classes for quadratic extensions of F using certain quotient groups of $F^\times / (F^\times)^2$ and $(F, +) / \{a^2 - a \mid a \in F\}$ similar to question 34.8.

Exercise 34.11. Let p be a prime integer. Show that $x^p - x - a$ is irreducible over $\mathbb{F}_p$ for any $a \in \mathbb{F}_p^\times$. (Hint: If it has a root, then it splits by the shifting one root.)

35 Normal and separable extensions

In this section, we introduce the concepts of normal and separable extensions, which are crucial for understanding Galois theory. A Galois extension will simply be defined as an extension that is both normal and separable.

35.1 Normal extensions

Definition 35.1. An algebraic extension L/K is **normal** if every irreducible polynomial $p(x) \in K[x]$ that has a root in L actually splits completely into linear factors in $L[x]$.

An equivalent and often more useful characterization involves embeddings into an algebraic closure. Let $\bar{K}$ be an algebraic closure of K containing L .

Proposition 35.1. Let L/K be a finite extension. The following are equivalent:

1. L/K is normal.
2. L is the splitting field of some polynomial $f(x) \in K[x]$.
3. Every K -embedding $\sigma : L \hookrightarrow \bar{K}$ (i.e., a homomorphism fixing K pointwise) satisfies $\sigma(L) = L$. Thus, every embedding restricts to a K -automorphism of L .

Proof. (1) $\Rightarrow$ (2): Since L/K is finite, we may write

$$L = K(\alpha_1, \dots, \alpha_n).$$

For each i , let $m_{\alpha_i}(x)$ be the minimal polynomial of α_i over K . Because L/K is normal and $\alpha_i \in L$, each $m_{\alpha_i}(x)$ splits completely over L . Hence the product

$$f(x) = m_{\alpha_1}(x) \cdots m_{\alpha_n}(x)$$

splits over L . Conversely, L is generated by roots of these minimal polynomials, so L is the splitting field of $f(x)$ over K .

(2) $\Rightarrow$ (3): Suppose L is the splitting field of $f(x) \in K[x]$, and let $\sigma : L \hookrightarrow \bar{K}$ be a K -embedding. Since σ fixes K , it sends roots of f to roots of f . Therefore $\sigma(L)$ is generated by roots of f , so $\sigma(L) \subseteq L$. Because σ is injective and L/K is finite, we have

$$[\sigma(L) : K] = [L : K].$$

Since also $\sigma(L) \subseteq L$, it follows that $\sigma(L) = L$.

(3) $\Rightarrow$ (1): Let $p(x) \in K[x]$ be irreducible and suppose $\alpha \in L$ is a root of $p(x)$. Let $\beta \in \bar{K}$ be any other root of $p(x)$. Since p is irreducible, there is a K -isomorphism

$$K(\alpha) \longrightarrow K(\beta)$$

sending α to β . By the extension theorem for embeddings, this extends to a K -embedding

$$\sigma : L \hookrightarrow \bar{K}.$$

By assumption, $\sigma(L) = L$, so in particular

$$\beta = \sigma(\alpha) \in L.$$

Thus every root of $p(x)$ lies in L , so $p(x)$ splits completely in $L[x]$. Therefore L/K is normal. $\square$

Example 35.1. The extension $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is normal because it is the splitting field of $x^2 - 2$. However, $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is **not** normal. The irreducible polynomial $x^3 - 2 \in \mathbb{Q}[x]$ has a root $\sqrt[3]{2}$ in $\mathbb{Q}(\sqrt[3]{2})$, but its other roots $\sqrt[3]{2}\omega$ and $\sqrt[3]{2}\omega^2$ are non-real complex numbers, hence they do not lie in $\mathbb{Q}(\sqrt[3]{2})$.

The **normal closure** of a finite extension L/K is the smallest normal extension M/K containing L . If $L = K(\alpha_1, \dots, \alpha_n)$, then M is the splitting field of the product of the minimal polynomials $m_{\alpha_i}(x)$ over K .

35.2 Separable extensions

Definition 35.2. An irreducible polynomial $f(x) \in K[x]$ is **separable** if it has no multiple roots in its splitting field. An arbitrary polynomial is separable if all its irreducible factors are separable.

We can detect multiple roots using the formal derivative. For $f(x) = \sum a_i x^i$, define $f'(x) = \sum i a_i x^{i-1}$.

Proposition 35.2. A polynomial $f(x) \in K[x]$ has a multiple root in a splitting field if and only if $\gcd(f, f') \neq 1$ in $K[x]$. Because f is irreducible, this happens if and only if $f'(x) = 0$ identically.

In characteristic 0 (like $\mathbb{Q}, \mathbb{R}, \mathbb{C}$), $f'(x) = 0$ implies $f(x)$ is constant. Hence, **every irreducible polynomial over a field of characteristic 0 is separable**. In characteristic $p > 0$, an irreducible polynomial might have $f'(x) = 0$ if it is of the form $g(x^p)$, leading to inseparable polynomials.

Example 35.2. Let $K = \mathbb{F}_p(t)$, the field of rational functions in t over $\mathbb{F}_p$. The polynomial $f(x) = x^p - t \in K[x]$ is irreducible over K (by Eisenstein applied to the prime element t of $\mathbb{F}_p[t]$). Its formal derivative is $f'(x) = px^{p-1} = 0$ (since we are in characteristic p), so f is inseparable. In fact, if α is a root of f , then $x^p - t = (x - \alpha)^p$ in any extension field, so α is a root of multiplicity p .

Definition 35.3. An element $\alpha \in L$ is separable over K if its minimal polynomial over K is separable. An algebraic extension L/K is **separable** if every element in L is separable over K .

Fields where every algebraic extension is separable are called **perfect fields**. All fields of characteristic 0 and all finite fields are perfect.

We now collect the most useful equivalent characterizations of separability.

Proposition 35.3 (Characterizations of separable polynomials). *Let $f(x) \in K[x]$ be an irreducible polynomial. The following are equivalent:*

1. f is separable (i.e. has no multiple roots in a splitting field).
2. $\gcd(f, f') \neq f$, i.e. $f'(x) \neq 0$.
3. In characteristic $p > 0$: $f(x)$ is not of the form $g(x^p)$ for any $g \in K[x]$.
4. In characteristic 0: this condition is automatic (every irreducible polynomial is separable).

Proof. (1) $\Leftrightarrow$ (2): Since f is irreducible, $\gcd(f, f')$ is either 1 or (a scalar multiple of) f . It equals f only when $f \mid f'$, which forces $f' = 0$ since $\deg f' < \deg f$. But $f' = 0$ means every root has multiplicity > 1 , so f is inseparable.

(2) $\Leftrightarrow$ (3): Write $f(x) = \sum a_i x^i$. Then $f'(x) = \sum i a_i x^{i-1} = 0$ means $i a_i = 0$ for all i . In characteristic p , this means $a_i = 0$ whenever $p \nmid i$, i.e. $f(x) = g(x^p)$ where $g(y) = \sum_{p \mid i} a_i y^{i/p}$.

(4): In characteristic 0, $i a_i = 0$ with $i \geq 1$ forces $a_i = 0$, so $f' = 0$ only if f is constant. $\square$

Proposition 35.4 (Characterizations of separable extensions). *Let L/K be a finite extension. The following are equivalent:*

1. L/K is separable (every $\alpha \in L$ has separable minimal polynomial over K).
2. $L = K(\alpha_1, \dots, \alpha_n)$ where each α_i is separable over K .
3. $|\operatorname{Hom}_K(L, \bar{K})| = [L : K]$ (the separable degree equals the degree).

Proof. (1) $\Rightarrow$ (2) is immediate (choose generators of L over K ; they are all separable by assumption).

(2) $\Rightarrow$ (3): Use the tower $K \subset K(\alpha_1) \subset K(\alpha_1, \alpha_2) \subset \dots \subset L$. At each step, the element being adjoined has separable minimal polynomial over the current base, so the number of embeddings at that step equals the degree. By the tower formula for embeddings, the product gives $|\operatorname{Hom}_K(L, \bar{K})| = [L : K]$.

(3) $\Rightarrow$ (1): If some $\alpha \in L$ had inseparable minimal polynomial, then the step $K \subset K(\alpha)$ in any tower through α would contribute fewer embeddings than the degree, making $|\operatorname{Hom}_K(L, \bar{K})| < [L : K]$ by the tower formula—contradiction. $\square$

35.3 Counting embeddings: the role of $\operatorname{Hom}_K(L, \bar{K})$

Having defined normality and separability independently, we now introduce a single invariant that detects both properties at once: the set of K -embeddings of L into an algebraic closure.

Definition 35.4. *Let L/K be a finite extension and $\bar{K}$ an algebraic closure of K containing L . We write*

$$\operatorname{Hom}_K(L, \bar{K}) = \{ \sigma : L \hookrightarrow \bar{K} \mid \sigma|_K = \operatorname{Id}_K \}$$

*for the set of all K -embeddings of L into $\bar{K}$. Its cardinality $|\operatorname{Hom}_K(L, \bar{K})|$ is independent of the choice of $\bar{K}$ and is sometimes called the **separable degree** of L/K , denoted $[L : K]_s$.*

The key link to the Relative Simple Extension Lemma (Lemma 33.1) and Proposition 33.1 is the following multiplicativity formula.

Proposition 35.5 (Tower formula for embeddings). *Let $K \subseteq M \subseteq L$ be a tower of finite extensions. Then*

$$|\mathrm{Hom}_K(L, \bar{K})| = |\mathrm{Hom}_M(L, \bar{K})| \cdot |\mathrm{Hom}_K(M, \bar{K})|.$$

In other words, the separable degree is multiplicative in towers: $[L : K]_s = [L : M]_s \cdot [M : K]_s$.

Proof. Every K -embedding $\sigma: L \hookrightarrow \bar{K}$ restricts to a K -embedding $\sigma|_M: M \hookrightarrow \bar{K}$. We shall count σ in two stages: first choose its restriction $\tau = \sigma|_M \in \mathrm{Hom}_K(M, \bar{K})$, then count how many $\sigma \in \mathrm{Hom}_K(L, \bar{K})$ restrict to τ .

Fix a K -embedding $\tau: M \hookrightarrow \bar{K}$. An extension of τ to L is a K -embedding $\sigma: L \hookrightarrow \bar{K}$ with $\sigma|_M = \tau$. Since τ is injective and fixes K , such a σ is the same thing as a $\tau(M)$ -embedding $L \hookrightarrow \bar{K}$. The number of such extensions is $|\mathrm{Hom}_{\tau(M)}(L, \bar{K})|$, which equals $|\mathrm{Hom}_M(L, \bar{K})|$ because τ is an isomorphism $M \xrightarrow{\sim} \tau(M)$ and the structure of L over M is isomorphic to that of L over $\tau(M)$ (more precisely, the number of extensions depends only on the isomorphism class of L/M).

Since this count is the same for every choice of τ , we get

$$|\mathrm{Hom}_K(L, \bar{K})| = |\mathrm{Hom}_K(M, \bar{K})| \cdot |\mathrm{Hom}_M(L, \bar{K})|. \quad \square$$

Theorem 35.1 (Counting Embeddings Theorem). *Let L/K be a finite extension. Then:*

1. $|\mathrm{Hom}_K(L, \bar{K})| \leq [L : K]$, with equality if and only if L/K is separable.
2. If L/K is normal, then every element of $\mathrm{Hom}_K(L, \bar{K})$ maps L to itself, hence restricts to an automorphism:

$$\mathrm{Hom}_K(L, \bar{K}) = \mathrm{Aut}(L/K).$$

3. In general, $|\mathrm{Aut}(L/K)| \leq |\mathrm{Hom}_K(L, \bar{K})| \leq [L : K]$.

Proof. Part (1): We first treat the case $L = K(\alpha)$, a simple extension. Let $m_\alpha(x)$ be the minimal polynomial of α over K , with $d = \deg m_\alpha = [L : K]$.

Any $\sigma \in \mathrm{Hom}_K(L, \bar{K})$ is determined by $\sigma(\alpha)$, which must be a root of $m_\alpha(x)$ in $\bar{K}$. Conversely, each distinct root β of m_α gives a unique K -embedding σ with $\sigma(\alpha) = \beta$ (by the Relative Simple Extension Lemma). Therefore

$$|\mathrm{Hom}_K(K(\alpha), \bar{K})| = \text{number of distinct roots of } m_\alpha(x) \text{ in } \bar{K}.$$

This is at most $d = [K(\alpha) : K]$, with equality if and only if $m_\alpha(x)$ is separable.

For a general finite extension L/K , write $L = K(\alpha_1, \dots, \alpha_n)$ and use the tower

$$K \subset K(\alpha_1) \subset K(\alpha_1, \alpha_2) \subset \dots \subset L.$$

By Proposition 35.5 and the simple-extension case applied to each step,

$$|\mathrm{Hom}_K(L, \bar{K})| = \prod_{i=1}^n |\mathrm{Hom}_{K_{i-1}}(K_i, \bar{K})| \leq \prod_{i=1}^n [K_i : K_{i-1}] = [L : K],$$

where $K_0 = K$ and $K_i = K(\alpha_1, \dots, \alpha_i)$. Equality holds throughout if and only if each step is separable, i.e. each α_i is separable over K_{i-1} . This is equivalent to L/K being separable.

Part (2): Suppose L/K is normal and $\sigma \in \mathrm{Hom}_K(L, \bar{K})$. For any $\alpha \in L$, the element $\sigma(\alpha)$ is a root of the minimal polynomial of α over K . Since L/K is normal, all such roots lie in L . Hence $\sigma(L) \subseteq L$. Because σ is injective and L/K is finite, $\sigma(L) = L$, so $\sigma \in \mathrm{Aut}(L/K)$.

Part (3): Every K -automorphism of L is in particular a K -embedding into $\bar{K}$, so $\mathrm{Aut}(L/K) \subseteq \mathrm{Hom}_K(L, \bar{K})$ (after composing with the inclusion $L \hookrightarrow \bar{K}$). Combined with Part (1), this gives $|\mathrm{Aut}(L/K)| \leq |\mathrm{Hom}_K(L, \bar{K})| \leq [L : K]$. $\square$

The following summary table collects the diagnostic power of embedding counts. It is the single most useful criterion for checking normality, separability, and the Galois property.

Condition on L/K	Equivalent embedding criterion
separable	$ \operatorname{Hom}_K(L, \bar{K}) = [L : K]$
normal	$\operatorname{Hom}_K(L, \bar{K}) = \operatorname{Aut}(L/K)$
Galois (= normal + separable)	$ \operatorname{Aut}(L/K) = [L : K]$

Remark 35.1 (Separable closure and purely inseparable degree). *For a finite extension L/K in characteristic $p > 0$, let $L_s = \{\alpha \in L \mid \alpha \text{ is separable over } K\}$ be the **separable closure** of K in L . Then L_s/K is separable and L/L_s is **purely inseparable** (every element $\alpha \in L$ satisfies $\alpha^{p^e} \in L_s$ for some $e \geq 0$). One can show:*

$$|\operatorname{Hom}_K(L, \bar{K})| = [L_s : K] = [L : K]_s, \quad [L : K] = [L : K]_s \cdot [L : K]_i,$$

where $[L : K]_i = [L : L_s]$ is the **inseparable degree**. A purely inseparable extension has only one embedding into $\bar{K}$, namely the inclusion. Thus the “deficit” $[L : K] - |\operatorname{Hom}_K(L, \bar{K})|$ is entirely explained by inseparability.

Example 35.3 (Embedding counts for $\mathbb{Q}(\alpha)$, $\alpha^3 = 2$). *Let $L = \mathbb{Q}(\sqrt[3]{2})$. The minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is $m(x) = x^3 - 2$, which has three distinct roots in $\mathbb{C}$: $\sqrt[3]{2}$, $\sqrt[3]{2}\omega$, $\sqrt[3]{2}\omega^2$ where $\omega = e^{2\pi i/3}$. Therefore*

$$|\operatorname{Hom}_{\mathbb{Q}}(L, \mathbb{C})| = 3 = [L : \mathbb{Q}].$$

*So $L/\mathbb{Q}$ is **separable**. However, only one of these three embeddings maps L into L (the identity), since $\sqrt[3]{2}\omega \notin L \subset \mathbb{R}$. Therefore*

$$|\operatorname{Aut}(L/\mathbb{Q})| = 1 < 3 = |\operatorname{Hom}_{\mathbb{Q}}(L, \mathbb{C})|.$$

*By the table above, $L/\mathbb{Q}$ is **not normal** (and hence not Galois).*

Example 35.4 (Inseparable extension: $\mathbb{F}_p(t^{1/p})/\mathbb{F}_p(t)$). *Let $K = \mathbb{F}_p(t)$ and $L = K(\alpha)$ where $\alpha^p = t$. The minimal polynomial is $x^p - t = (x - \alpha)^p$, which has only one distinct root. Hence*

$$|\operatorname{Hom}_K(L, \bar{K})| = 1 < p = [L : K].$$

*By the table, L/K is **not separable**. Since every embedding automatically lands in L (there is only one embedding, the inclusion), we also have $\operatorname{Hom}_K(L, \bar{K}) = \operatorname{Aut}(L/K)$, so L/K is **normal**. This is an example of a normal but inseparable extension.*

36 Exercise

Exercise 36.1. *Let F be a field with finite characteristic p , and $f(x) \in F[x]$ be an irreducible polynomial. Prove that there exists a unique integer $e \geq 0$ and a unique irreducible separable polynomial $g(x) \in F[x]$ such that $f(x) = g(x^{p^e})$.*

Exercise 36.2. *Let F be a field. Let $f(x)$ and $g(x)$ be two coprime polynomials in $F[x]$.*

- Show that $f(x)y - g(x)$ is irreducible in $F[x, y]$, and also irreducible in both $F(y)[x]$ and $F(x)[y]$. (Hint: use Gauss’s lemma)*

2. Prove that $F(x)$ is a finite extension of $F(f(x)/g(x))$ and its degree is $\max\{\deg f, \deg g\}$.

Exercise 36.3. Let K/F be a field extension. Denote by $\text{Aut}_F(K)$ the automorphisms σ of K such that $\sigma|_F = \text{Id}_F$. Prove that

$$\text{Aut}_F(F(x)) = \left\{ x \mapsto \frac{ax+b}{cx+d} \mid \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} \neq 0 \right\}$$

and $\text{Aut}_F(F(x)) \cong \text{PGL}(2, F) = \text{GL}(2, F)/\{\lambda I_2 \mid \lambda \in F^\times\}$ as groups.

Exercise 36.4. Construct a splitting field for $x^5 - 2$ over $\mathbb{Q}$ by writing down a set of generators. What is its degree over $\mathbb{Q}$?

Exercise 36.5. Let $F \subset K \subset L$ be finite field extensions. Then L/F is separable if and only if K/F is separable and L/K is separable.

Exercise 36.6 (Application of uniqueness of splitting fields). Let p be a prime and consider $f(x) = x^{p^2} - x \in \mathbb{F}_p[x]$.

A useful fact about finite subgroups in $F^\times$ is that they are cyclic.

1. Show that the splitting field of f over $\mathbb{F}_p$ is a field with p^2 elements, denoted by $\mathbb{F}_{p^2}$.
2. Using the uniqueness of splitting fields, show that $\mathbb{F}_{p^2}$ is (up to isomorphism) the unique field with p^2 elements.
3. More generally, use the same argument to show that for any $n \geq 1$, $\mathbb{F}_{p^n}$ is the unique field (up to isomorphism) with p^n elements.

Exercise 36.7. Let L/K be an algebraic field extension. Show that all the separable elements in L over K form a subfield L_s of L containing K . (Optional: if $L \neq L_s$, then L is not separable over L_s .)

Exercise 36.8 (Degree bounds for splitting fields). Let $f(x) \in K[x]$ be a polynomial of degree n with splitting field L . Show that $[L : K]$ divides $n!$.

36.1 Optional

Exercise 36.9 (Perfect fields). Recall that a field K is **perfect** if every algebraic extension of K is separable.

1. Show that a field of characteristic 0 is perfect.
2. Show that a field K of characteristic $p > 0$ is perfect if and only if the Frobenius map $\varphi: K \rightarrow K, a \mapsto a^p$, is surjective (i.e. every element of K has a p -th root in K).
3. Deduce that every finite field is perfect.
4. Give an example of an imperfect field.

Exercise 36.10. Let $\overline{F}$ be the algebraic closure of F and F^{sep} be the subfield of $\overline{F}$ consisting of all elements α such that $F(\alpha)/F$ is separable. Prove that F^{sep} is the maximal separable extension of F in $\overline{F}$ in the sense that any separable extension of F in $\overline{F}$ can be embedded into F^{sep} , and F^{sep} has no finite separable extension of degree greater than 1. (Optional: show that these properties uniquely determine F^{sep}/F up to isomorphism. Is F^{sep}/F normal?)

37 Primitive Element Theorem

Theorem 37.1 (Primitive Element Theorem). *Every finite separable extension L/K is simple, meaning $L = K(\alpha)$ for some $\alpha \in L$. We call α a **primitive element**.*

Proof. We first treat the case $L = K(\alpha, \beta)$ with α, β separable over K , and then conclude by induction on the number of generators.

Let $\bar{K}$ be an algebraic closure of K . Since α and β are separable over K , their minimal polynomials have finitely many distinct roots in $\bar{K}$. Let

$$\alpha = \alpha_1, \alpha_2, \dots, \alpha_r \quad \text{and} \quad \beta = \beta_1, \beta_2, \dots, \beta_s$$

be the distinct conjugates of α and β over K .

Choose $c \in K$ such that

$$\alpha_i + c\beta_j \neq \alpha_1 + c\beta_1$$

for every pair $(i, j) \neq (1, 1)$. This is possible because for each such pair there is at most one bad choice of c , namely a solution of

$$\alpha_i + c\beta_j = \alpha_1 + c\beta_1.$$

Since there are only finitely many pairs and K is infinite in the standard proof, we can avoid all bad values. (For finite fields, every finite extension is automatically simple because finite multiplicative groups are cyclic, so the theorem is also true there.)

Set

$$\gamma = \alpha + c\beta.$$

Clearly $K(\gamma) \subseteq K(\alpha, \beta)$. We claim equality holds.

Let $h(x)$ be the minimal polynomial of β over $K(\gamma)$. Since $\beta \in K(\alpha, \beta)$, every root of $h(x)$ in $\bar{K}$ is among $\beta_1, \dots, \beta_s$. On the other hand, because

$$\alpha = \gamma - c\beta,$$

every embedding of $K(\alpha, \beta)$ over $K(\gamma)$ must send a pair (α, β) to another pair (α_i, β_j) satisfying

$$\alpha_i + c\beta_j = \gamma = \alpha_1 + c\beta_1.$$

By our choice of c , this forces $(i, j) = (1, 1)$. Hence β has no conjugate over $K(\gamma)$ other than itself, so $h(x) = x - \beta$ and therefore $\beta \in K(\gamma)$.

Then also

$$\alpha = \gamma - c\beta \in K(\gamma).$$

Thus $K(\alpha, \beta) = K(\gamma)$, proving the theorem when two generators are involved.

Now let $L = K(\alpha_1, \dots, \alpha_n)$ be a finite separable extension. By the two-generator case, $K(\alpha_1, \alpha_2) = K(\delta_2)$ for some δ_2 . Then

$$K(\alpha_1, \alpha_2, \alpha_3) = K(\delta_2, \alpha_3) = K(\delta_3)$$

for some δ_3 , and so on. Repeating this process, we finally obtain an element $\delta \in L$ such that

$$L = K(\delta).$$

Hence every finite separable extension is simple. □

38 Behavior in towers: inheritance and transitivity

When working with a tower of extensions $K \subset M \subset L$, a key question is whether normality and separability “pass” to the subextension M/K or the upper extension L/M (**inheritance**), and whether they “compose” (**transitivity**: if both M/K and L/M have a property, does L/K ?). The two properties behave quite differently.

Separability in towers

Proposition 38.1 (Separability is transitive and inherited). *Let $K \subset M \subset L$ be a tower of finite extensions.*

1. Inheritance downward: *If L/K is separable, then M/K is separable.*
2. Inheritance upward: *If L/K is separable, then L/M is separable.*
3. Transitivity: *If M/K and L/M are both separable, then L/K is separable.*

In short, for a tower of finite extensions, L/K is separable if and only if both M/K and L/M are separable.

Proof. (1) Every element of M is an element of L , hence separable over K .

(2) Let $\alpha \in L$. Its minimal polynomial $m_{\alpha,M}(x)$ over M divides $m_{\alpha,K}(x)$ (the minimal polynomial over K). Since $m_{\alpha,K}$ is separable (it has distinct roots), so is its factor $m_{\alpha,M}$.

(3) Let $\alpha \in L$. We need to show α is separable over K . Consider the tower $K \subset M \subset M(\alpha) \subseteq L$. Since L/M is separable, α is separable over M , so $m_{\alpha,M}(x)$ has distinct roots. Write $m_{\alpha,M}(x) = x^d + c_{d-1}x^{d-1} + \cdots + c_0$ with $c_i \in M$. Let $E = K(c_0, \dots, c_{d-1}) \subseteq M$. Since M/K is separable, E/K is separable (by part (1) applied to the tower $K \subseteq E \subseteq M$).

Now α is a root of $m_{\alpha,M}(x) \in E[x]$, and $m_{\alpha,M}$ has distinct roots. Hence α is separable over E . By the tower formula for separable degrees,

$$[E(\alpha) : K]_s = [E(\alpha) : E]_s \cdot [E : K]_s = [E(\alpha) : E] \cdot [E : K] = [E(\alpha) : K].$$

Since $K(\alpha) \subseteq E(\alpha)$, and $[K(\alpha) : K]_s \leq [E(\alpha) : K]_s = [E(\alpha) : K]$, while also $[K(\alpha) : K]_s \leq [K(\alpha) : K]$, the separable degree of α over K equals $[K(\alpha) : K]$, so α is separable over K . $\square$

Remark 38.1. *Property (3) can also be seen quickly from the multiplicativity of separable degrees: $[L : K]_s = [L : M]_s \cdot [M : K]_s$. If both steps are separable, then $[L : M]_s = [L : M]$ and $[M : K]_s = [M : K]$, so $[L : K]_s = [L : M] \cdot [M : K] = [L : K]$, hence L/K is separable.*

Normality in towers

The situation for normality is strikingly different: normality is inherited upward but *not* downward, and it is *not* transitive.

Proposition 38.2 (Normality: inheritance and failure of transitivity). *Let $K \subset M \subset L$ be a tower of finite extensions.*

1. Inheritance upward: *If L/K is normal, then L/M is normal.*
2. Failure of inheritance downward: *L/K normal does not imply M/K normal.*
3. Failure of transitivity: *M/K normal and L/M normal does not imply L/K normal.*

Proof. (1) L is the splitting field of some $f(x) \in K[x]$ over K . Then L is also the splitting field of $f(x)$ viewed as a polynomial in $M[x]$ over M . Hence L/M is normal.

(2) *Counterexample:* Let $K = \mathbb{Q}$, $L = \mathbb{Q}(\sqrt[4]{2}, i)$ (splitting field of $x^4 - 2$), and $M = \mathbb{Q}(\sqrt[4]{2})$. Then L/K is normal (it is a splitting field), but M/K is not: the polynomial $x^4 - 2 \in \mathbb{Q}[x]$ has a root $\sqrt[4]{2} \in M$, but its roots $\pm\sqrt[4]{2}, \pm i\sqrt[4]{2}$ do not all lie in $M \subset \mathbb{R}$.

(3) *Counterexample:* Let $K = \mathbb{Q}$, $M = \mathbb{Q}(\sqrt{2})$, $L = \mathbb{Q}(\sqrt[4]{2})$. Then:

- M/K is normal: M is the splitting field of $x^2 - 2$ over $\mathbb{Q}$.
- L/M is normal: $L = M(\sqrt[4]{2})$ and $\sqrt[4]{2}$ has minimal polynomial $x^2 - \sqrt{2}$ over M , which splits in L as $(x - \sqrt[4]{2})(x + \sqrt[4]{2})$.
- L/K is *not* normal: $x^4 - 2$ has root $\sqrt[4]{2} \in L$ but does not split in $L \subset \mathbb{R}$ (it has non-real roots $\pm i\sqrt[4]{2}$).

□

Remark 38.2. Part (2) can be rephrased as: a sub-extension of a normal extension need not be normal. By the Galois correspondence (which we will prove later), this corresponds to the fact that a subgroup of a normal subgroup need not be normal.

The following summary table collects these tower properties for easy reference. We include the Galois property for comparison, noting that “Galois = normal + separable.”

Property	Inherited by subext. M/K ?	Inherited by upper ext. L/M ?	Transitive?
Separable	Yes	Yes	Yes
Normal	No	Yes	No
Galois	No	Yes (separable; normal if splitting field)	No

Example 38.1 (The classic tower $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt[4]{2})$). This is the standard counterexample for transitivity of normality. To make it completely explicit:

$$[\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}] = 4, \quad \text{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z}.$$

The extension $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2})$ is Galois (splitting field of $x^2 - \sqrt{2}$), with $\text{Gal} \cong \mathbb{Z}/2\mathbb{Z}$. The extension $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is also Galois. Yet $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is not even normal: $x^4 - 2$ does not split in $\mathbb{Q}(\sqrt[4]{2}) \subset \mathbb{R}$.

To fix this, one passes to the normal closure $\mathbb{Q}(\sqrt[4]{2}, i)/\mathbb{Q}$, which is Galois of degree 8 with Galois group D_4 .

Example 38.2 (Tower where everything is Galois). In contrast, consider $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Each step is a degree-2 extension (hence normal and separable), and the composite $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ is also Galois (splitting field of $(x^2 - 2)(x^2 - 3)$) with $\text{Gal} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

More generally, any compositum of Galois extensions is Galois (because normality and separability are both preserved under compositum).

39 Galois extensions

Definition 39.1. An extension L/K is called a **Galois extension** if it is finite, normal, and separable.

For a finite extension L/K , let $\text{Aut}(L/K)$ be the group of field automorphisms of L that fix every element of K .

By Theorem 35.1, we always have $|\text{Aut}(L/K)| \leq |\text{Hom}_K(L, \bar{K})| \leq [L : K]$. The embedding criteria in the summary table give the following clean characterization.

Theorem 39.1. Let L/K be a finite extension. The following are equivalent:

1. L/K is Galois (i.e. normal and separable).
2. $|\text{Aut}(L/K)| = [L : K]$.
3. $|\text{Hom}_K(L, \bar{K})| = [L : K]$ and $\text{Hom}_K(L, \bar{K}) = \text{Aut}(L/K)$.

Proof. (1) $\Rightarrow$ (3): Separability gives $|\text{Hom}_K(L, \bar{K})| = [L : K]$ by Theorem 35.1(1). Normality gives $\text{Hom}_K(L, \bar{K}) = \text{Aut}(L/K)$ by Theorem 35.1(2). Together, (3) holds.

(3) $\Rightarrow$ (2): Immediate from $|\text{Aut}(L/K)| = |\text{Hom}_K(L, \bar{K})| = [L : K]$.

(2) $\Rightarrow$ (1): Since $|\text{Aut}(L/K)| \leq |\text{Hom}_K(L, \bar{K})| \leq [L : K]$ always holds, the hypothesis $|\text{Aut}(L/K)| = [L : K]$ forces both inequalities to be equalities. The first equality $|\text{Aut}(L/K)| = |\text{Hom}_K(L, \bar{K})|$ means every K -embedding $L \hookrightarrow \bar{K}$ maps L into L , which is the embedding characterization of normality. The second equality $|\text{Hom}_K(L, \bar{K})| = [L : K]$ is the characterization of separability. $\square$

When L/K is a Galois extension, we denote $\text{Aut}(L/K)$ by $\text{Gal}(L/K)$, called the **Galois group** of the extension L/K .

Example 39.1. The splitting field of $x^3 - 2$ over $\mathbb{Q}$ is $L = \mathbb{Q}(\sqrt[3]{2}, \omega)$ with $[L : \mathbb{Q}] = 6$. Since $\text{char}(\mathbb{Q}) = 0$, the extension is separable, so $|\text{Hom}_{\mathbb{Q}}(L, \mathbb{C})| = 6$. As a splitting field, it is also normal, so every $\mathbb{Q}$ -embedding lands in L , hence $|\text{Aut}(L/\mathbb{Q})| = 6 = [L : \mathbb{Q}]$. Thus $L/\mathbb{Q}$ is Galois. An automorphism σ is determined by $\sigma(\sqrt[3]{2})$ (any of the three cube roots of 2) and $\sigma(\omega)$ (either ω or ω^2). Mapping this to permutations of the three roots $\{\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2\}$ gives $\text{Gal}(L/\mathbb{Q}) \cong S_3$.

39.1 Extra Exercises

If you would like to practice more problems, here are some extra exercises.

Splitting fields

Exercise 39.1 (Splitting fields of low-degree polynomials). Find the splitting field L over $\mathbb{Q}$ and compute $[L : \mathbb{Q}]$ for each of the following polynomials:

1. $x^3 - 2$.
2. $x^4 - 2$.
3. $x^4 + 4$.
4. $x^3 - 3x - 1$. (Hint: the roots are $2\cos(2\pi k/9)$, $k = 1, 2, 4$.)

Exercise 39.2 (Splitting fields over finite fields). 1. Find the splitting field of $x^4 + x + 1$ over $\mathbb{F}_2$, and compute its degree over $\mathbb{F}_2$.

2. Find the splitting field of $x^3 - x - 1$ over $\mathbb{F}_3$, and compute its degree over $\mathbb{F}_3$.
3. Show that the splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$ is $\mathbb{F}_{p^n}$, and that $[\mathbb{F}_{p^n} : \mathbb{F}_p] = n$.

Exercise 39.3 (Degree bounds for splitting fields). Let $f(x) \in K[x]$ be a polynomial of degree n with splitting field L .

1. Show that $[L : K]$ divides $n!$.
2. Give an example where $[L : K] = n!$ (i.e. the upper bound is achieved).
3. Give an example where $[L : K] = n$ with f irreducible. (Hint: consider a cyclic extension.)
4. If f is irreducible of degree 3, show that $[L : K]$ is either 3 or 6. Determine which case occurs for $x^3 - 2$ over $\mathbb{Q}$ and for $x^3 - 3x - 1$ over $\mathbb{Q}$.

Exercise 39.4 (Splitting field of a product). Let $f(x), g(x) \in K[x]$. Show that the splitting field of $f(x)g(x)$ over K is the compositum of the splitting fields of f and g inside a common algebraic closure of K .

Normal extensions

Exercise 39.5 (Normality is not transitive: further examples). The classic counterexample $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt[4]{2})$ was discussed in Example 38.1. Construct two more counterexamples of towers $K \subset M \subset L$ where M/K and L/M are both normal but L/K is not:

1. A tower inside $\mathbb{C}$ with $[L : K] = 8$.
2. A tower using finite fields or function fields in characteristic $p > 0$.

Exercise 39.6 (Normal closure computation). For each of the following extensions, compute the normal closure M over K and determine $[M : K]$.

1. $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$.
2. $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$.
3. $\mathbb{Q}(\sqrt[5]{2})/\mathbb{Q}$. (Hint: for (c), the normal closure contains a primitive 5th root of unity.)

Exercise 39.7 (Characterization of normal extensions via minimal polynomials). Let L/K be a finite extension. Prove that the following are equivalent:

1. L/K is normal.
2. For every $\alpha \in L$, the minimal polynomial $m_{\alpha, K}(x)$ splits completely in $L[x]$.
3. L is the splitting field of a family of polynomials in $K[x]$.

(This generalizes the equivalence proved in the text to infinite families; the proof strategy is the same.)

Exercise 39.8 (Intersection and compositum of normal extensions). Let L_1/K and L_2/K be two normal extensions inside a common algebraic closure $\bar{K}$.

1. Show that $L_1 \cap L_2$ is normal over K .
2. Show that the compositum $L_1 L_2$ is normal over K .

3. Is the analogue true for separable extensions? For Galois extensions?

Exercise 39.9 (Normal extensions and K -conjugates). Let L/K be a finite extension and $\alpha \in L$. The K -conjugates of α are the roots of the minimal polynomial $m_{\alpha,K}(x)$ in $\bar{K}$.

1. Prove that L/K is normal if and only if L is closed under taking K -conjugates: for every $\alpha \in L$, all K -conjugates of α lie in L .
2. Use this to give a quick proof that $\mathbb{Q}(\sqrt{2} + \sqrt{3})/\mathbb{Q}$ is normal.

Separable extensions

Exercise 39.10 (Separability via the formal derivative). For each of the following polynomials, determine whether it is separable by computing $\gcd(f, f')$:

1. $f(x) = x^4 - 2 \in \mathbb{Q}[x]$.
2. $f(x) = x^4 + 4x^2 + 2 \in \mathbb{Q}[x]$.
3. $f(x) = x^3 + x + 1 \in \mathbb{F}_2[x]$.
4. $f(x) = x^4 + x \in \mathbb{F}_2[x]$.
5. $f(x) = x^p - t \in \mathbb{F}_p(t)[x]$.
6. $f(x) = x^{p^2} - t \in \mathbb{F}_p(t)[x]$.

Exercise 39.11 (Structure of inseparable polynomials). Let K be a field of characteristic $p > 0$.

1. Show that an irreducible polynomial $f(x) \in K[x]$ is inseparable if and only if $f(x) = g(x^p)$ for some polynomial $g \in K[x]$.
2. Show more generally that if $f(x) \in K[x]$ is irreducible and inseparable, then $f(x) = g(x^{p^e})$ for a unique maximal $e \geq 1$, where $g \in K[x]$ is irreducible and separable. (This e is called the **inseparability exponent**.)
3. Deduce that the degree of any inseparable irreducible polynomial is divisible by p .

Exercise 39.12 (Perfect fields). Recall that a field K is **perfect** if every algebraic extension of K is separable.

1. Show that a field of characteristic 0 is perfect.
2. Show that a field K of characteristic $p > 0$ is perfect if and only if the Frobenius map $\varphi: K \rightarrow K, a \mapsto a^p$, is surjective (i.e. every element of K has a p -th root in K).
3. Deduce that every finite field is perfect.
4. Give an example of an imperfect field.

Exercise 39.13 (Separability in towers and composita). The inheritance and transitivity of separability were proved in Proposition 38.1. This exercise extends those results.

1. Give an alternative proof of transitivity using the multiplicativity of separable degrees: $[L : K]_s = [L : M]_s \cdot [M : K]_s$.
2. Let L_1/K and L_2/K both be separable inside $\bar{K}$. Show that L_1L_2/K is separable.

3. Show that $L_1 \cap L_2$ is separable over K whenever L_1/K and L_2/K are separable.
4. Is the compositum of two inseparable extensions necessarily inseparable?

Exercise 39.14 (Separable closure). Let L/K be a finite extension and define

$$L_s = \{\alpha \in L \mid \alpha \text{ is separable over } K\}.$$

1. Prove that L_s is a subfield of L (so L_s/K is the maximal separable subextension of L/K).
2. Prove that L/L_s is purely inseparable, meaning that for every $\alpha \in L$, $\alpha^{p^e} \in L_s$ for some $e \geq 0$ (where $p = \text{char } K$).
3. Show that $[L_s : K] = |\text{Hom}_K(L, \bar{K})|$ and $[L : L_s] = [L : K]_i$ (the inseparable degree).

Exercise 39.15 (Purely inseparable extensions). Let K have characteristic $p > 0$.

1. Show that L/K is purely inseparable if and only if $|\text{Hom}_K(L, \bar{K})| = 1$.
2. Show that $[L : K]$ is a power of p whenever L/K is purely inseparable.
3. Give an example of a purely inseparable extension of degree p^2 that is not simple (i.e. cannot be written as $K(\alpha)$ for a single α). (Hint: consider $\mathbb{F}_p(s, t)$ and adjoin $s^{1/p}$ and $t^{1/p}$.)

Combining normality, separability, and Hom counting

The following exercises use the set $\text{Hom}_K(L, \bar{K})$ and the diagnostic table from the text to study the interplay between normality, separability, and the Galois property.

Exercise 39.16 (Diagnostic table in practice). For each of the following extensions, compute $[L : K]$, $|\text{Hom}_K(L, \bar{K})|$, and $|\text{Aut}(L/K)|$, and use the diagnostic table to determine which of the properties (normal, separable, Galois) hold.

1. $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$.
2. $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$.
3. $\mathbb{F}_{p^n}/\mathbb{F}_p$. (Hint: The Frobenius endomorphism $x \mapsto x^p$.)
4. $\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}$, where $\omega = e^{2\pi i/3}$.
5. $\mathbb{Q}(\zeta_7)/\mathbb{Q}$, where $\zeta_7 = e^{2\pi i/7}$.

Exercise 39.17. Show that an extension L/K of degree 2 is always normal. Is it always separable? (Consider fields of characteristic 2, e.g. $\mathbb{F}_2(t)(\sqrt{t})/\mathbb{F}_2(t)$.)

Exercise 39.18. Find a primitive element for the extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$, and verify your answer.

Exercise 39.19. Let $L = \mathbb{Q}(\sqrt[3]{2})$.

1. List all $\mathbb{Q}$ -embeddings $\sigma: L \hookrightarrow \mathbb{C}$. How many are there?
2. How many of these satisfy $\sigma(L) \subseteq L$? Conclude that $L/\mathbb{Q}$ is not normal.
3. Verify that $|\text{Hom}_{\mathbb{Q}}(L, \mathbb{C})| = [L : \mathbb{Q}]$, so $L/\mathbb{Q}$ is separable.

4. Compute the normal closure M of $L/\mathbb{Q}$ and verify that $|\operatorname{Aut}(M/\mathbb{Q})| = [M : \mathbb{Q}] = 6$.

Exercise 39.20 (Tower formula in practice). Let $L = \mathbb{Q}(\sqrt[4]{2})$.

1. Compute $|\operatorname{Hom}_{\mathbb{Q}}(L, \mathbb{C})|$ by listing all $\mathbb{Q}$ -embeddings of L into $\mathbb{C}$.
2. Verify the tower formula by computing $|\operatorname{Hom}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}), \mathbb{C})|$ and $|\operatorname{Hom}_{\mathbb{Q}(\sqrt{2})}(L, \mathbb{C})|$ separately and checking that their product equals $|\operatorname{Hom}_{\mathbb{Q}}(L, \mathbb{C})|$.
3. Compute $|\operatorname{Aut}(L/\mathbb{Q})|$. Is $L/\mathbb{Q}$ normal?

Exercise 39.21 (Hom counts detect inseparability). Let $K = \mathbb{F}_p(t)$ and $L = K(\alpha)$ where $\alpha^p = t$.

1. Compute $[L : K]$ and $|\operatorname{Hom}_K(L, \bar{K})|$.
2. Is L/K separable? Is L/K normal?
3. Now let $M = K(\beta)$ where $\beta^{p^2} = t$. Compute $[M : K]$, $[M : L]$, and $|\operatorname{Hom}_K(M, \bar{K})|$.

Exercise 39.22 (Normal but not separable). Show that the extension $L = \mathbb{F}_p(t^{1/p})$ over $K = \mathbb{F}_p(t)$ is normal but not separable. Compute $|\operatorname{Hom}_K(L, \bar{K})|$ and $|\operatorname{Aut}(L/K)|$, and verify that $\operatorname{Hom}_K(L, \bar{K}) = \operatorname{Aut}(L/K)$ (normality) but $|\operatorname{Hom}_K(L, \bar{K})| < [L : K]$ (not separable).

Exercise 39.23 (Separable but not normal). Let $f(x) = x^4 - 2 \in \mathbb{Q}[x]$ and let $L = \mathbb{Q}(\sqrt[4]{2})$.

1. Show that $L/\mathbb{Q}$ is separable but not normal.
2. Compute $|\operatorname{Hom}_{\mathbb{Q}}(L, \mathbb{C})|$ and $|\operatorname{Aut}(L/\mathbb{Q})|$. What is $|\operatorname{Hom}_{\mathbb{Q}}(L, \mathbb{C})| - |\operatorname{Aut}(L/\mathbb{Q})|$?
3. Find the splitting field M of f over $\mathbb{Q}$, and compute $|\operatorname{Hom}_{\mathbb{Q}}(M, \mathbb{C})| = [M : \mathbb{Q}]$.

Exercise 39.24 (The four cases). Give explicit examples to demonstrate that all four combinations of (normal / not normal) $\times$ (separable / not separable) can occur for finite field extensions:

1. Normal and separable (= Galois).
2. Normal and not separable.
3. Not normal and separable.
4. Not normal and not separable.

For each example, verify the claim by computing $|\operatorname{Hom}_K(L, \bar{K})|$ and $|\operatorname{Aut}(L/K)|$. (Hint: for (d), consider $\mathbb{F}_p(t)(\sqrt[p]{t} + \sqrt[p]{t})$ over $\mathbb{F}_p(t)$, or construct a suitable tower.)

Exercise 39.25 (Hom and the Galois group of a tower). Let $K \subset M \subset L$ with L/K Galois and $G = \operatorname{Gal}(L/K)$, $H = \operatorname{Gal}(L/M)$.

1. Show that $|\operatorname{Hom}_K(M, \bar{K})| = [G : H]$ by using the tower formula and the Galois property.
2. Show that M/K is normal if and only if $H \trianglelefteq G$, and in that case $\operatorname{Hom}_K(M, \bar{K})$ has a natural group structure isomorphic to G/H . (Hint: every K -embedding of M into $\bar{K}$ extends to an automorphism of L .)

Exercise 39.26 (Counting embeddings of compositum). Let $L_1 = \mathbb{Q}(\sqrt{2})$ and $L_2 = \mathbb{Q}(\sqrt{3})$, both contained in $\mathbb{R}$.

1. Compute $|\operatorname{Hom}_{\mathbb{Q}}(L_1, \mathbb{C})|$ and $|\operatorname{Hom}_{\mathbb{Q}}(L_2, \mathbb{C})|$.

2. Compute $|\operatorname{Hom}_{\mathbb{Q}}(L_1 L_2, \mathbb{C})|$ where $L_1 L_2 = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Is the product $|\operatorname{Hom}_{\mathbb{Q}}(L_1, \mathbb{C})| \cdot |\operatorname{Hom}_{\mathbb{Q}}(L_2, \mathbb{C})| = |\operatorname{Hom}_{\mathbb{Q}}(L_1 L_2, \mathbb{C})|$? Under what conditions on $L_1 \cap L_2$ would this be true?

Exercise 39.27 (Degree of splitting field via Hom counts). Let $f(x) \in K[x]$ be a separable polynomial of degree n with roots $\alpha_1, \dots, \alpha_n$ in $\bar{K}$, and let L be its splitting field over K .

1. Show that the map $\operatorname{Hom}_K(L, \bar{K}) \rightarrow S_n$ given by $\sigma \mapsto$ the permutation of $\{\alpha_1, \dots, \alpha_n\}$ induced by σ is an injective group homomorphism.
2. Deduce that $[L : K] = |\operatorname{Hom}_K(L, \bar{K})|$ divides $n!$. (This recovers the degree bound from the perspective of embedding counts.)
3. Use this to show $[L : K] = 6$ for $f(x) = x^3 - 2$ over $\mathbb{Q}$, and $[L : K] = 3$ for $f(x) = x^3 - 3x - 1$ over $\mathbb{Q}$.

Challenge problems

Exercise 39.28 (Artin's theorem on finite subgroups of $\operatorname{Aut}(L)$). Let L be a field and G a finite group of automorphisms of L . Let $K = L^G$ be the fixed field. Prove that $[L : K] = |G|$ and that L/K is Galois with $\operatorname{Gal}(L/K) = G$. (Hint: use the independence of characters, also known as Dedekind's lemma, to show $[L : K] \geq |G|$, and then use $|\operatorname{Aut}(L/K)| \leq [L : K]$ for the reverse inequality.)

Exercise 39.29 (Separable degree and tensor products). Let L/K be a finite separable extension.

1. Show that the $\bar{K}$ -algebra $L \otimes_K \bar{K}$ is isomorphic to $\bar{K}^{[L:K]}$ (a product of $[L : K]$ copies of $\bar{K}$).
2. What happens when L/K is purely inseparable of degree p ? Show that $L \otimes_K \bar{K} \cong \bar{K}[x]/(x^p)$, which is not reduced. Thus separability is equivalent to the tensor product being a reduced ring.

Exercise 39.30 (Finite fields: further questions). Let $q = p^n$.

1. Compute $|\operatorname{Hom}_{\mathbb{F}_q}(\mathbb{F}_{q^m}, \bar{\mathbb{F}}_p)|$ directly from the Hom-counting framework (Theorem 35.1), and verify $|\operatorname{Hom}_{\mathbb{F}_q}(\mathbb{F}_{q^m}, \bar{\mathbb{F}}_p)| = [\mathbb{F}_{q^m} : \mathbb{F}_q]_s = m$.
2. Let $N_q(d)$ denote the number of monic irreducible polynomials of degree d over $\mathbb{F}_q$. Verify the Möbius inversion formula from the counting-irreducible-polynomials example above for $q = 2$, $n \leq 6$ by listing all irreducible polynomials over $\mathbb{F}_2$ of degree ≤ 6 .
3. Show that for $n \geq 2$, there exist at least two non-isomorphic irreducible polynomials of degree n over $\mathbb{F}_q$.
4. Let α be a root of an irreducible polynomial $f(x) \in \mathbb{F}_q[x]$ of degree d . Show that the splitting field of f over $\mathbb{F}_q$ is $\mathbb{F}_{q^d}$ and $f(x) = \prod_{i=0}^{d-1} (x - \alpha^{q^i})$.

40 Galois correspondence

In this section, we establish the Galois correspondence between subgroups of the Galois group and intermediate fields of a Galois extension. This fundamental theorem translates difficult questions about field extensions into simpler questions about finite groups.

40.1 Fixed fields and Artin's Lemma

Definition 40.1. Let L be a field and H a group of automorphisms of L . The **fixed field** of H , denoted L^H , is the set of elements in L fixed by every automorphism in H :

$$L^H = \{x \in L \mid \sigma(x) = x \text{ for all } \sigma \in H\}.$$

It is straightforward to verify that L^H is a subfield of L .

Theorem 40.1 (Artin's Lemma). Let L be a field and $H \leq \text{Aut}(L)$ be a finite group of automorphisms of L . Then $[L : L^H] = |H|$ and L/L^H is a Galois extension with Galois group H .

Proof. Let $K = L^H$ and $n = |H|$. Since $H \subseteq \text{Aut}(L/K)$, we have $|H| \leq |\text{Aut}(L/K)| \leq [L : K]$. Thus, it suffices to show that $[L : K] \leq n$, which will imply all inequalities are equalities.

Suppose for contradiction that $[L : K] > n$. Then we can find $n+1$ elements $\alpha_1, \dots, \alpha_{n+1} \in L$ that are linearly independent over K . Let $H = \{\sigma_1, \dots, \sigma_n\}$ with $\sigma_1 = \text{Id}$. Consider the homogeneous system of n linear equations in $n+1$ variables $x_1, \dots, x_{n+1}$ over L :

$$\sum_{j=1}^{n+1} \sigma_i(\alpha_j)x_j = 0 \quad \text{for } i = 1, \dots, n.$$

Since there are more variables than equations, there exists a non-trivial solution in L^{n+1} . Choose a non-trivial solution $(x_1, \dots, x_{n+1})$ with the minimal number of non-zero entries. By reordering the α_j 's if necessary, we may assume $x_1, \dots, x_r$ are non-zero and $x_{r+1} = \dots = x_{n+1} = 0$. By scaling the solution, we can further assume $x_r = 1$.

If all $x_j \in K$, then the first equation ($i = 1$, where $\sigma_1 = \text{Id}$) gives $\sum_{j=1}^r \alpha_j x_j = 0$, contradicting the K -linear independence of $\alpha_1, \dots, \alpha_{n+1}$. Thus, some x_k is not in K ; without loss of generality, say $x_1 \notin K$. Since $x_1 \notin K = L^H$, there exists an automorphism $\tau \in H$ such that $\tau(x_1) \neq x_1$.

Applying τ to the system of equations, we get

$$0 = \tau \left(\sum_{j=1}^r \sigma_i(\alpha_j)x_j \right) = \sum_{j=1}^r (\tau\sigma_i)(\alpha_j)\tau(x_j).$$

As σ_i runs through H , the composition $\tau\sigma_i$ also runs through H , just permuting the equations. Thus, $(\tau(x_1), \dots, \tau(x_{r-1}), 1, 0, \dots, 0)$ is another solution to the original system. Subtracting this from our original solution yields

$$(x_1 - \tau(x_1), \dots, x_{r-1} - \tau(x_{r-1}), 0, \dots, 0).$$

This is a new solution with fewer non-zero entries (the r -th entry is $1 - 1 = 0$). Also, it is non-trivial because the first entry $x_1 - \tau(x_1) \neq 0$. This contradicts the minimality of our chosen solution.

Therefore, $[L : K] \leq n = |H|$, which completes the proof. $\square$

Conversely, if L/K is a finite Galois extension and $G = \text{Gal}(L/K)$, then $L^G = K$.

40.2 The Fundamental Theorem of Galois Theory

Theorem 40.2 (Fundamental Theorem of Galois Theory). *Let L/K be a finite Galois extension with Galois group $G = \text{Gal}(L/K)$. Let $\mathcal{F}$ be the set of intermediate fields M (so $K \subseteq M \subseteq L$) and $\mathcal{H}$ be the set of subgroups of G . The maps*

$$\begin{aligned}\mathcal{F} &\longleftrightarrow \mathcal{H} \\ M &\longmapsto \text{Gal}(L/M) \\ L^H &\longleftarrow H\end{aligned}$$

are inclusion-reversing bijections. Furthermore, this correspondence satisfies:

1. $[L : M] = |\text{Gal}(L/M)|$ and $[M : K] = [G : \text{Gal}(L/M)]$.
2. M/K is a normal extension if and only if $H = \text{Gal}(L/M)$ is a normal subgroup of G ($H \trianglelefteq G$).
3. If M/K is normal, then $\text{Gal}(M/K) \cong G/H$.

Proof. Let $\Phi(M) = \text{Gal}(L/M)$ for an intermediate field M , and let $\Psi(H) = L^H$ for a subgroup $H \leq G$.

First, both maps reverse inclusion. If $M_1 \subseteq M_2$, then any automorphism fixing M_2 also fixes M_1 , so

$$\Phi(M_2) \subseteq \Phi(M_1).$$

Similarly, if $H_1 \subseteq H_2$, then any element fixed by H_2 is fixed by H_1 , so

$$\Psi(H_2) \subseteq \Psi(H_1).$$

We next show that $\Psi \circ \Phi = \text{id}_{\mathcal{F}}$. Let M be an intermediate field and put $H = \Phi(M) = \text{Gal}(L/M)$. Since every element of H fixes M , we certainly have

$$M \subseteq L^H.$$

We claim that L/M is Galois. Separability is clear because every element of L is separable over K , hence also over the larger base field M . For normality, let $\iota : L \hookrightarrow \bar{K}$ be any M -embedding into a fixed algebraic closure $\bar{K}$ of K containing L . Since ι fixes M , it also fixes $K \subseteq M$, so ι is in particular a K -embedding. Because L/K is normal, the embedding characterization of normality shows that $\iota(L) = L$. Hence every M -embedding of L into $\bar{K}$ lands back in L , so L/M is normal. Therefore L/M is Galois, and by the theorem on automorphism groups of finite Galois extensions,

$$[L : M] = |\text{Gal}(L/M)| = |H|.$$

On the other hand, Artin's Lemma gives

$$[L : L^H] = |H|.$$

Therefore $[L : M] = [L : L^H]$. Since $M \subseteq L^H$, we must have $M = L^H$.

Now let $H \leq G$ and put $M = L^H$. By Artin's Lemma,

$$[L : M] = |H|.$$

Also $H \subseteq \text{Gal}(L/M)$ by definition of fixed field. But L/M is Galois again, so

$$|\text{Gal}(L/M)| = [L : M] = |H|.$$

Hence $H = \text{Gal}(L/M) = \Phi(\Psi(H))$, so $\Phi \circ \Psi = \text{id}_{\mathcal{H}}$. Therefore Φ and Ψ are inverse bijections between intermediate fields and subgroups.

The same argument also gives the degree formulas:

$$[L : M] = |\text{Gal}(L/M)|, \quad [M : K] = \frac{[L : K]}{[L : M]} = \frac{|G|}{|\text{Gal}(L/M)|} = [G : \text{Gal}(L/M)].$$

It remains to establish the normality criterion. Let $H = \text{Gal}(L/M)$.

If M/K is normal, then every K -embedding of M into an algebraic closure lands back in M . In particular, for every $\sigma \in G$, we have $\sigma(M) = M$. Therefore

$$\sigma H \sigma^{-1} = \text{Gal}(L/\sigma(M)) = \text{Gal}(L/M) = H,$$

so $H \trianglelefteq G$.

Conversely, suppose $H \trianglelefteq G$. Then for every $\sigma \in G$ and every $x \in M = L^H$, we have for all $\eta \in H$,

$$\eta(\sigma(x)) = \sigma((\sigma^{-1}\eta\sigma)(x)) = \sigma(x)$$

because $\sigma^{-1}\eta\sigma \in H$ and x is fixed by every element of H . Thus $\sigma(x) \in L^H = M$, so $\sigma(M) \subseteq M$. Applying the same argument to σ^{-1} gives $\sigma(M) = M$. Hence every element of G restricts to a K -automorphism of M , so every K -embedding of M into $\bar{K}$ lands in M . By the embedding characterization of normality, this shows that M/K is normal.

We therefore obtain a homomorphism

$$\rho : G \longrightarrow \text{Gal}(M/K), \quad \rho(\sigma) = \sigma|_M.$$

Its kernel consists exactly of those automorphisms fixing M pointwise, i.e. $\ker \rho = H$. Hence G/H injects into $\text{Gal}(M/K)$. But

$$|G/H| = [G : H] = [M : K].$$

Since always $|\text{Gal}(M/K)| \leq [M : K]$, we have

$$|\text{Gal}(M/K)| = [M : K] = |G/H|,$$

so the injection is actually an isomorphism. Therefore

$$\text{Gal}(M/K) \cong G/H.$$

□

Remark 40.1. *The theorem can be remembered in four parallel pieces. Intermediate fields correspond to subgroups in the opposite direction: larger fields correspond to smaller subgroups, and smaller fields correspond to larger subgroups. The degree $[L : M]$ is the order of the subgroup $\text{Gal}(L/M)$, while the degree $[M : K]$ is the index of that subgroup in G . Moreover, normal intermediate fields correspond exactly to normal subgroups, and in that case the Galois group of the smaller extension is the quotient group $G/\text{Gal}(L/M)$.*

40.3 Examples of Galois correspondence

Example 40.1 (The extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$). Let $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. This is the splitting field of $(x^2 - 2)(x^2 - 3)$ over $\mathbb{Q}$, so $L/\mathbb{Q}$ is Galois with degree 4. Any automorphism $\sigma \in \text{Gal}(L/\mathbb{Q})$

is determined by its action on $\sqrt{2}$ and $\sqrt{3}$: $\sigma(\sqrt{2}) = \pm\sqrt{2}$ and $\sigma(\sqrt{3}) = \pm\sqrt{3}$. Thus $G \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. (Why?)

The subgroup structure of G matches the intermediate fields. G has three non-trivial proper subgroups of order 2, generated by the non-identity elements. By the Fundamental Theorem, there are exactly four intermediate fields: $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, and $\mathbb{Q}(\sqrt{6})$. Write down the detailed arguments in homework.

Example 40.2 (The extension $\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}$). Let $L = \mathbb{Q}(\sqrt[3]{2}, \omega)$ be the splitting field of $x^3 - 2$ over $\mathbb{Q}$. As we saw, $[L : \mathbb{Q}] = 6$. Any $\sigma \in G = \text{Gal}(L/\mathbb{Q})$ permutes the three roots of $x^3 - 2$. In fact, $G \cong S_3$, the symmetric group on 3 letters. The non-trivial subgroups of S_3 are three subgroups of order 2 (each generated by a transposition) and one subgroup of order 3 (the alternating group A_3 , generated by a 3-cycle). The fields fixed by the three order-2 subgroups are $\mathbb{Q}(\sqrt[3]{2})$, $\mathbb{Q}(\sqrt[3]{2}\omega)$, and $\mathbb{Q}(\sqrt[3]{2}\omega^2)$. These are not normal over $\mathbb{Q}$, corresponding to the fact that order-2 subgroups are not normal in S_3 . The field fixed by A_3 is $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$, which is a normal extension of degree 2 over $\mathbb{Q}$, corresponding to the normal subgroup $A_3 \trianglelefteq S_3$.

Example 40.3 (The extension $\mathbb{Q}(\sqrt[4]{2}, i)/\mathbb{Q}$). Let $L = \mathbb{Q}(\sqrt[4]{2}, i)$ be the splitting field of $f(x) = x^4 - 2$ over $\mathbb{Q}$, whose roots are $\sqrt[4]{2}$, $i\sqrt[4]{2}$, $-\sqrt[4]{2}$, $-i\sqrt[4]{2}$. One computes $[L : \mathbb{Q}] = 8$.

Every $\sigma \in G = \text{Gal}(L/\mathbb{Q})$ is determined by $\sigma(\sqrt[4]{2})$ and $\sigma(i)$. Setting $\alpha = \sqrt[4]{2}$, we have $\sigma(\alpha) \in \{\alpha, i\alpha, -\alpha, -i\alpha\}$ and $\sigma(i) = \pm i$. Not all $4 \times 2 = 8$ combinations give well-defined automorphisms (we must respect $\alpha^4 = 2$ and $i^2 = -1$), but exactly 8 do, confirming $|G| = 8$.

Define

$$r: \alpha \mapsto i\alpha, \quad i \mapsto i, \quad s: \alpha \mapsto \alpha, \quad i \mapsto -i.$$

Then r has order 4, s has order 2, and $srs^{-1} = r^{-1}$ (since $s(r(\alpha)) = s(i\alpha) = -i\alpha = r^{-1}(\alpha)$). Therefore $G \cong D_4$, the dihedral group of order 8.

The subgroups of $D_4 = \langle r, s \mid r^4 = s^2 = 1, srs^{-1} = r^{-1} \rangle$ and the corresponding fixed fields are:

Subgroup H	$ H $	Fixed field L^H	$[L^H : \mathbb{Q}]$
$\{1\}$	1	$L = \mathbb{Q}(\sqrt[4]{2}, i)$	8
$\langle r^2 \rangle = \{1, r^2\}$	2	$\mathbb{Q}(\sqrt{2}, i)$	4
$\langle s \rangle = \{1, s\}$	2	$\mathbb{Q}(\sqrt[4]{2})$	4
$\langle rs \rangle = \{1, rs\}$	2	$\mathbb{Q}(i\sqrt[4]{2})$	4
$\langle r^2s \rangle = \{1, r^2s\}$	2	$\mathbb{Q}(\sqrt[4]{2}(1+i))$	4
$\langle r^2, s \rangle \cong V_4$	4	$\mathbb{Q}(\sqrt{2})$	2
$\langle r \rangle \cong \mathbb{Z}/4\mathbb{Z}$	4	$\mathbb{Q}(i)$	2
$\langle r^2, rs \rangle \cong V_4$	4	$\mathbb{Q}(i\sqrt{2})$	2
$G = D_4$	8	$\mathbb{Q}$	1

The three normal subgroups of D_4 are $\langle r^2 \rangle$, $\langle r \rangle$, and $\langle r^2, s \rangle$, corresponding to the three intermediate fields $\mathbb{Q}(\sqrt{2}, i)$, $\mathbb{Q}(i)$, and $\mathbb{Q}(\sqrt{2})$ that are normal (hence Galois) over $\mathbb{Q}$. The quotients are

$$G/\langle r \rangle \cong \mathbb{Z}/2\mathbb{Z}, \quad G/\langle r^2, s \rangle \cong \mathbb{Z}/2\mathbb{Z}, \quad G/\langle r^2 \rangle \cong V_4.$$

The remaining intermediate fields are not normal over $\mathbb{Q}$, consistent with their corresponding subgroups being non-normal.

Example 40.4 (Finite field extensions). Let $q = p^n$ and consider $\mathbb{F}_{q^m}/\mathbb{F}_q$. By Homework Exercise 36.6, this is a Galois extension with cyclic Galois group

$$\text{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q) = \langle \varphi_q \rangle \cong \mathbb{Z}/m\mathbb{Z},$$

where $\varphi_q: x \mapsto x^q$ is the Frobenius. The Galois correspondence identifies intermediate fields with subgroups of $\mathbb{Z}/m\mathbb{Z}$: for each $d \mid m$, the unique subfield of degree d over $\mathbb{F}_q$ is $\mathbb{F}_{q^d}$, corresponding to the subgroup $\langle \varphi_q^d \rangle$ of index d .

41 Exercises

Normal extensions and separable extensions

Exercise 41.1 (Recognizing normal extensions). Determine whether each of the following extensions is normal, and justify your answer.

1. $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$.
2. $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$.
3. $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2})$.
4. $\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}$, where $\omega = e^{2\pi i/3}$.
5. $\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}(\omega)$, where $\omega = e^{2\pi i/3}$.

Exercise 41.2 (Intersection and compositum of normal extensions). Let L_1/K and L_2/K be two normal extensions inside a common algebraic closure $\bar{K}$. (You can assume L_1 and L_2 are finite over K if you like, but it is not necessary.)

1. Show that $L_1 \cap L_2$ is normal over K .
2. Show that the subfield $L_1 L_2$ generated by L_1 and L_2 is normal over K .
3. Is the analogue true for separable extensions? For Galois extensions?

Exercise 41.3. For each of the following extensions, compute $[L : K]$, $|\text{Hom}_K(L, \bar{K})|$, and $|\text{Aut}_K(L)|$, and determine which of the properties (normal, separable, Galois) hold.

1. $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$.
2. $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$.
3. $\mathbb{F}_{p^n}/\mathbb{F}_p$. (Hint: The Frobenius endomorphism $x \mapsto x^p$.)
4. $\mathbb{Q}(\zeta_7)/\mathbb{Q}$, where $\zeta_7 = e^{2\pi i/7}$.

Galois correspondence: computation and structure

Exercise 41.4. Let L/K be a finite Galois extension. Prove that there are only finitely many intermediate fields M with $K \subseteq M \subseteq L$. If we only assume L/K is finite separable, does the same conclusion hold?

Exercise 41.5 (Galois correspondence for $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$). Let $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

1. Show that $\text{Gal}(L/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ by describing the four automorphisms explicitly.
2. List all subgroups of G and the corresponding intermediate fields.
3. Verify that every intermediate field is normal over $\mathbb{Q}$ (and explain why this follows from the fact that G is abelian).

Exercise 41.6 (Compositum and intersection). Let L/K be a finite Galois extension and let M_1, M_2 be intermediate fields. Let $M_1 M_2$ be subfield generated by M_1 and M_2 . Prove that:

1. $\text{Gal}(L/(M_1 M_2)) = \text{Gal}(L/M_1) \cap \text{Gal}(L/M_2)$.
2. $\text{Gal}(L/(M_1 \cap M_2))$ is generated by $\text{Gal}(L/M_1) \cup \text{Gal}(L/M_2)$.

Apply this to $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, $M_1 = \mathbb{Q}(\sqrt{2})$, $M_2 = \mathbb{Q}(\sqrt{3})$ and verify directly.

Exercise 41.7 (Fundamental theorem: degree formulas). Let L/K be a finite Galois extension with Galois group G , and let M be an intermediate field with corresponding subgroup $H = \text{Gal}(L/M)$. Show that: $[L : M] = |H|$ and $[M : K] = [G : H]$.

Optional

Exercise 41.8 (Galois correspondence for $x^4 - 2$). Compute the Galois group of $x^4 - 2$ over $\mathbb{Q}$. Find all intermediate fields. Which intermediate fields are normal over $\mathbb{Q}$? (You may use Example 40.3 as a guide, or work independently.)

Exercise 41.9 (Galois correspondence for $x^3 - 2$). Let L be the splitting field of $x^3 - 2$ over $\mathbb{Q}$.

1. Show that $\text{Gal}(L/\mathbb{Q}) \cong S_3$ by describing the action of each automorphism on the roots $\sqrt[3]{2}$, $\sqrt[3]{2}\omega$, $\sqrt[3]{2}\omega^2$.
2. Draw the subgroup lattice of S_3 and the corresponding intermediate fields.
3. Which intermediate fields are Galois over $\mathbb{Q}$? Verify using the normal subgroup criterion.

42 Exercises

Exercise 42.1 (Fundamental Theorem of Algebra). In this exercise, you will prove the Fundamental Theorem of Algebra using the Galois correspondence, Sylow theorems, and two elementary facts from analysis. The proof is broken into small steps; the necessary background facts are stated for you.

Theorem 42.1 (Fundamental Theorem of Algebra). Every non-constant polynomial $f(x) \in \mathbb{C}[x]$ has a root in $\mathbb{C}$. Equivalently, $\mathbb{C}$ is algebraically closed.

Facts from analysis (accepted without proof)

(A1) Every odd-degree polynomial in $\mathbb{R}[x]$ has a root in $\mathbb{R}$.

(This follows from the intermediate value theorem: if $\deg f$ is odd, then $f(x) \rightarrow +\infty$ and $f(x) \rightarrow -\infty$ as $x \rightarrow \pm\infty$, so f must cross the x -axis.)

(A2) Every complex number has a square root in $\mathbb{C}$.

(If $z = re^{i\theta}$ in polar form, then $\sqrt{r}e^{i\theta/2}$ is a square root of z .)

Facts from group theory (accepted without proof)

(G1) Sylow's theorem. If G is a finite group with $|G| = p^a \cdot m$ and $\gcd(p, m) = 1$, then G has a subgroup of order p^a .

(G2) Index- p subgroups of p -groups. If P is a non-trivial finite p -group (i.e. $|P| = p^k$ with $k \geq 1$), then P has a normal subgroup of index p .

(Sketch: the class equation shows $|Z(P)| > 1$; since $Z(P)$ is abelian of p -power order, Cauchy's theorem gives an element of order p in $Z(P)$, which generates a normal subgroup of order p . Quotient and induct.)

Work through the following steps to prove Theorem 42.1.

Step 1 (Reduction to $\mathbb{R}[x]$). Let $f(x) \in \mathbb{C}[x]$ be non-constant. Define $\bar{f}(x)$ by conjugating all coefficients of f .

- (a) Show that $g(x) := f(x)\bar{f}(x)$ has real coefficients.
- (b) Explain why, if g splits completely over $\mathbb{C}$, then f also has a root in $\mathbb{C}$.
- (c) Conclude: it suffices to show that every non-constant polynomial in $\mathbb{R}[x]$ splits over $\mathbb{C}$.

Step 2 (Setting up the Galois extension). Let $h(x) \in \mathbb{R}[x]$ be non-constant. Replace h by $h(x) \cdot (x^2 + 1)$ if necessary.

- (a) Show that a splitting field K of h over $\mathbb{R}$ contains a root of $x^2 + 1$, and hence $\mathbb{C} = \mathbb{R}(i) \subseteq K$.
- (b) Write $|G| = 2^a \cdot m$ where $G = \text{Gal}(K/\mathbb{R})$ and m is odd. Our goal is to show $K = \mathbb{C}$, i.e. $a = 1$ and $m = 1$.

Step 3 (Killing the odd part: $m = 1$).

- (a) By **(G1)**, G has a subgroup P of order 2^a . Using the Galois correspondence, show that the fixed field $E = K^P$ satisfies $[E : \mathbb{R}] = m$.
- (b) Let $\alpha \in E$. Explain why the minimal polynomial of α over $\mathbb{R}$ is irreducible of odd degree (dividing m).
- (c) Apply **(A1)** to conclude $\alpha \in \mathbb{R}$.
- (d) Since $\alpha \in E$ was arbitrary, conclude $E = \mathbb{R}$ and hence $m = 1$.

Step 4 (Killing the 2-part: $a = 1$). Now we know $|G| = 2^a$, i.e. G is a 2-group.

- (a) Let $H = \text{Gal}(K/\mathbb{C})$. Show that $[G : H] = [\mathbb{C} : \mathbb{R}] = 2$, hence $|H| = 2^{a-1}$.
- (b) Suppose for contradiction $H \neq 1$ (i.e. $a \geq 2$). Apply **(G2)** to H (with $p = 2$): conclude that H has a subgroup H_1 of index 2.
- (c) Show that the fixed field $L = K^{H_1}$ satisfies $[L : \mathbb{C}] = 2$.
- (d) Deduce that $L = \mathbb{C}(\alpha)$ where α satisfies an irreducible quadratic $x^2 + bx + c \in \mathbb{C}[x]$. Use the quadratic formula and **(A2)** to show $\alpha \in \mathbb{C}$. Derive a contradiction.
- (e) Conclude $H = 1$, hence $K = \mathbb{C}$, hence h splits over $\mathbb{C}$.

Step 5 (Corollaries).

(a) Prove: $\mathbb{C}$ is the unique (up to $\mathbb{R}$ -isomorphism) degree-2 extension of $\mathbb{R}$.

(Hint: if $[L : \mathbb{R}] = 2$, write $L = \mathbb{R}(\alpha)$ with $\alpha^2 + b\alpha + c = 0$, complete the square to find an element $\gamma \in L$ with $\gamma^2 = -1$.)

(b) Prove: the only finite extensions of $\mathbb{R}$ are isomorphic to $\mathbb{R}$ itself or $\mathbb{C}$.

Exercise 42.2 (Orbits under the Galois group and irreducible polynomials). Let K/F be a finite Galois extension with $G = \text{Gal}(K/F)$, and let $\alpha \in K$.

Step 1 (The orbit). Define the Galois orbit of α to be $\text{Orb}_G(\alpha) = \{\sigma(\alpha) : \sigma \in G\}$. The stabilizer $\text{Stab}_G(\alpha) = \{\sigma \in G : \sigma(\alpha) = \alpha\}$ is a subgroup of G .

(a) Show that $\text{Stab}_G(\alpha) = \text{Gal}(K/F(\alpha))$.

(b) Conclude by the orbit-stabilizer theorem that $|\text{Orb}_G(\alpha)| = [G : \text{Stab}_G(\alpha)] = [F(\alpha) : F]$.

Step 2 (From orbit to irreducible polynomial). Let $\text{Orb}_G(\alpha) = \{\alpha_1, \dots, \alpha_d\}$ where $d = [F(\alpha) : F]$.

(a) Form $f(x) = \prod_{j=1}^d (x - \alpha_j) \in K[x]$. Show that $f(x) \in F[x]$.

(b) Conclude that $f(x) = m_{\alpha, F}(x)$ is the minimal polynomial of α over F .

(c) Use this method to find the minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$.

Exercise 42.3 (The Galois group of $\mathbb{C}(x)/\mathbb{C}(x^n + x^{-n})$ is the dihedral group). Let $n \geq 2$ be an integer. We work with the field of rational functions $\mathbb{C}(x)$, and let $t = x^n + x^{-n}$.

(a) Let $\zeta = e^{2\pi i/n}$. Define $\rho: \mathbb{C}(x) \rightarrow \mathbb{C}(x)$ by $\rho(f(x)) = f(\zeta x)$. Show that ρ is a field automorphism of $\mathbb{C}(x)$ fixing $\mathbb{C}(t)$.

(b) Define $\sigma: \mathbb{C}(x) \rightarrow \mathbb{C}(x)$ by $\sigma(f(x)) = f(x^{-1})$. Show σ is a field automorphism of $\mathbb{C}(x)$ fixing $\mathbb{C}(t)$.

(c) Compute $\sigma\rho\sigma^{-1}(x)$. Show that $\sigma\rho\sigma^{-1} = \rho^{-1}$.

(d) Conclude that ρ and σ generate a subgroup $D \leq \text{Aut}_{\mathbb{C}(t)}(\mathbb{C}(x))$ satisfying $\rho^n = \text{Id}$, $\sigma^2 = \text{Id}$, and $\sigma\rho\sigma^{-1} = \rho^{-1}$, i.e. $D \cong D_n$ (the dihedral group of order $2n$).

(e) Use previous exercise to show $[\mathbb{C}(x) : \mathbb{C}(t)] = 2n$.

(f) Show that $\mathbb{C}(x)/\mathbb{C}(t)$ is Galois by showing that $\mathbb{C}(x)$ is a splitting field of $Q(y) = y^{2n} - ty^n + 1$ over $\mathbb{C}(t)$.

(g) Show that $\text{Gal}(\mathbb{C}(x)/\mathbb{C}(t)) \cong D_n$.

43 Galois group of polynomials

Definition 43.1. Let $f(x) \in K[x]$. The **Galois group of the polynomial** f , denoted $\text{Gal}(f)$, is the Galois group $\text{Gal}(L/K)$ where L is the splitting field of $f(x)$ over K .

Since any $\sigma \in \text{Gal}(f)$ maps roots of f to roots of f , we can view $\text{Gal}(f)$ as a subgroup of the symmetric group S_n , where n is the number of distinct roots. If $f(x)$ is irreducible and separable of degree n , then S_n acts transitively on the roots. One important tool to determine whether $\text{Gal}(f) \subseteq A_n$ is the **discriminant** of the polynomial.

Definition 43.2. Let $f(x) \in K[x]$ have roots $\alpha_1, \dots, \alpha_n$ in a splitting field. The **discriminant** of f is

$$\Delta(f) = \prod_{i < j} (\alpha_i - \alpha_j)^2 \in K.$$

The element $\delta = \prod_{i < j} (\alpha_i - \alpha_j)$ satisfies $\delta^2 = \Delta$. Even permutations of the roots fix δ , while odd permutations negate it.

Proposition 43.1. $\text{Gal}(f) \subseteq A_n$ if and only if $\Delta(f)$ is a square in K .

Proof. Let L be the splitting field of f over K and let $G = \text{Gal}(L/K)$. Define

$$\delta = \prod_{i < j} (\alpha_i - \alpha_j) \in L.$$

For any $\sigma \in G$, the action of σ on the roots gives

$$\sigma(\delta) = \prod_{i < j} (\sigma(\alpha_i) - \sigma(\alpha_j)) = \text{sgn}(\sigma) \cdot \delta,$$

where $\text{sgn}(\sigma) \in \{+1, -1\}$ is the sign of the permutation induced by σ on the roots. This is because the product $\prod_{i < j} (\alpha_i - \alpha_j)$ is an alternating expression in the roots: each transposition of two roots introduces exactly one sign change.

Now $\Delta = \delta^2 \in K$ (since Δ is a symmetric function of the roots and f has coefficients in K).

If $G \subseteq A_n$, then $\sigma(\delta) = \delta$ for all $\sigma \in G$, so $\delta \in L^G = K$. Hence $\Delta = \delta^2$ is a square in K .

Conversely, if $\Delta = c^2$ for some $c \in K$, then $\delta^2 = c^2$, so $\delta = \pm c \in K$. For any $\sigma \in G$, $\sigma(\delta) = \delta$ (since $\delta \in K$), so $\text{sgn}(\sigma) = 1$. Hence $G \subseteq A_n$. $\square$

Note that $\Delta(f) \in K$ because it is a symmetric polynomial in the roots (in fact, Δ can be expressed directly in terms of the coefficients of f ; see the formulas below).

Example 43.1 (Discriminant formulas for low degrees). For a quadratic $f(x) = x^2 + bx + c$ with roots α_1, α_2 :

$$\Delta(f) = (\alpha_1 - \alpha_2)^2 = b^2 - 4c.$$

For a depressed cubic $f(x) = x^3 + px + q$ (without x^2 term), the roots satisfy $\alpha_1 + \alpha_2 + \alpha_3 = 0$, and

$$\Delta(f) = -4p^3 - 27q^2.$$

These can be verified by direct expansion. A general cubic $f(x) = x^3 + ax^2 + bx + c$ can be reduced to a depressed cubic by the substitution $x \mapsto x - a/3$; its discriminant is

$$\Delta(f) = a^2b^2 - 4b^3 - 4a^3c + 18abc - 27c^2.$$

Example 43.2 (Galois group of cubics via discriminant). Consider $f(x) = x^3 - 3x - 1 \in \mathbb{Q}[x]$. This is irreducible over $\mathbb{Q}$ (it has no rational roots). Here $p = -3$, $q = -1$, so

$$\Delta(f) = -4(-3)^3 - 27(-1)^2 = 108 - 27 = 81 = 9^2.$$

Since Δ is a perfect square in $\mathbb{Q}$, Proposition 43.1 gives $\text{Gal}(f) \subseteq A_3$. Since f is irreducible of degree 3, its Galois group acts transitively on 3 roots, so $|\text{Gal}(f)| \geq 3$. The only transitive subgroup of $A_3 \cong \mathbb{Z}/3\mathbb{Z}$ is A_3 itself, so

$$\text{Gal}(x^3 - 3x - 1) \cong A_3 \cong \mathbb{Z}/3\mathbb{Z}.$$

In contrast, $f(x) = x^3 - 2$ has $\Delta = -4(0) - 27(4) = -108$, which is not a perfect square in $\mathbb{Q}$. Hence $\text{Gal}(x^3 - 2) \not\subseteq A_3$, and since the only transitive subgroup of S_3 not contained in A_3 is S_3 itself, we recover $\text{Gal}(x^3 - 2) \cong S_3$.

43.1 Dedekind's method modulo p and Frobenius lifting

The discriminant helps decide whether the Galois group is contained in A_n , but in practice we often want more detailed information about the cycle structure of elements of the Galois group. A remarkably effective tool is to reduce the polynomial modulo suitable primes and study how it factors there.

Let $f(x) \in \mathbb{Z}[x]$ be a monic polynomial of degree n , and let L be its splitting field over $\mathbb{Q}$. For a prime p , denote by $\bar{f}(x)$ the reduction of f modulo p .

Theorem 43.1 (Dedekind's method, informal form). Assume that p does not divide the discriminant $\Delta(f)$, so that $\bar{f}(x)$ has no repeated roots. Suppose that over $\mathbb{F}_p$ the factorization of $\bar{f}(x)$ is

$$\bar{f}(x) = g_1(x) \cdots g_r(x),$$

where the g_i are distinct irreducible polynomials of degrees

$$d_1, \dots, d_r.$$

Then the Galois group $\text{Gal}(f)$ contains an element whose cycle decomposition on the roots of f has cycle lengths

$$d_1, \dots, d_r.$$

In particular, the factorization pattern of $\bar{f}$ gives concrete information about the cycle types occurring in $\text{Gal}(f) \subseteq S_n$.

Remark 43.1. One often summarizes this by saying: for an unramified prime p (that is, $p \nmid \Delta(f)$), the factorization of f modulo p records the cycle type of a Frobenius element in the Galois group. Thus linear factors correspond to fixed points, irreducible quadratic factors correspond to 2-cycles, irreducible cubic factors correspond to 3-cycles, and so on.

The underlying Galois-theoretic reason is the Frobenius automorphism. Over a finite field $\mathbb{F}_q$, every finite extension carries the canonical automorphism

$$\varphi_q : x \mapsto x^q.$$

When we reduce number-theoretic data modulo a prime and then pass to residue fields, this Frobenius automorphism can be lifted back to an element of the global Galois group.

Remark 43.2 (Frobenius lifting). *Let $L/\mathbb{Q}$ be a Galois extension, and let $\mathfrak{p}$ be a prime of L lying above a rational prime p . Assume p is unramified in L . Then there is a unique element*

$$\text{Frob}_{\mathfrak{p}} \in \text{Gal}(L/\mathbb{Q})$$

such that for every algebraic integer $\alpha \in \mathcal{O}_L$,

$$\text{Frob}_{\mathfrak{p}}(\alpha) \equiv \alpha^p \pmod{\mathfrak{p}}.$$

*This element is called the **Frobenius element** at $\mathfrak{p}$. Different choices of primes $\mathfrak{p}$ above p give conjugate elements, so the corresponding conjugacy class in the Galois group depends only on p .*

For our purposes, the crucial consequence is that the action of this Frobenius element on the roots of f is reflected exactly by the factorization of $\bar{f}(x)$ over $\mathbb{F}_p$. Each irreducible factor of degree d corresponds to one Frobenius orbit of size d by the Frobenius-orbit factorization proposition above, hence to one d -cycle in the corresponding permutation of the roots.

Example 43.3. *Consider*

$$f(x) = x^3 - 2.$$

Its discriminant is -108 , so any prime $p \neq 2, 3$ is unramified. Reducing modulo 5, we get

$$\bar{f}(x) = x^3 - 2 \equiv x^3 + 3 \pmod{5}.$$

A quick check shows that $x = 3$ is a root modulo 5, so

$$\bar{f}(x) = (x - 3)(x^2 + 3x + 4)$$

over $\mathbb{F}_5$, and the quadratic factor is irreducible. Therefore the Galois group contains an element of cycle type $(1)(2)$, i.e. a transposition. Since $x^3 - 2$ is irreducible over $\mathbb{Q}$, its Galois group is a transitive subgroup of S_3 ; a transitive subgroup of S_3 containing a transposition must be all of S_3 . This recovers

$$\text{Gal}(x^3 - 2) \cong S_3.$$

In practice, Dedekind's method is often combined with the discriminant criterion and simple group-theoretic arguments. By testing a polynomial modulo a few carefully chosen primes, one can frequently pin down its Galois group completely.

Remark 43.3 (A typical strategy for detecting S_n). *Suppose $f(x) \in \mathbb{Q}[x]$ is irreducible of degree n . Then $\text{Gal}(f)$ is a transitive subgroup of S_n . To prove that $\text{Gal}(f) = S_n$, one often combines the following kinds of information:*

- 1. a factorization modulo some prime showing that the group contains a cycle of a certain length;*
- 2. a factorization modulo another prime showing that the group contains a transposition, or more generally an odd permutation;*
- 3. the discriminant criterion to rule out containment in A_n ;*
- 4. additional real-root information, which may identify complex conjugation with a specific permutation type.*

This is extremely effective in concrete examples, especially for polynomials of prime degree.

Example 43.4 (Dedekind's method for $x^5 - 4x + 2$). Let $f(x) = x^5 - 4x + 2 \in \mathbb{Q}[x]$, which is irreducible by Eisenstein at $p = 2$.

- Modulo 3: $\bar{f}(x) = x^5 + 2x + 2$ is irreducible over $\mathbb{F}_3$ (check that it has no roots, and is not a product of irreducibles of degree 2 and 3). So $\text{Gal}(f)$ contains a 5-cycle.
- Modulo 5: $\bar{f}(x) = x^5 + x + 2 \equiv (x + 2)(x^4 + 3x^3 + 4x^2 + 2x + 1)$ over $\mathbb{F}_5$, giving cycle type $(1, 4)$, so $\text{Gal}(f)$ contains a 4-cycle with a fixed point.
- Modulo 7: $\bar{f}(x) = x^5 + 3x + 2 \equiv (x^2 + 4x + 6)(x^3 + 3x^2 + 3x + 5)$ over $\mathbb{F}_7$, giving σ with cycle type $(2, 3)$, so $\text{Gal}(f)$ contains a transposition σ^3 .

A transitive subgroup of S_5 containing a 5-cycle and a transposition must be S_5 ; the exercises below ask you to develop the full argument using Dedekind's method.

44 Extra exercises for practice

Galois groups of polynomials

Exercise 44.1 (Discriminant computation). Compute the discriminant $\Delta(f)$ for each of the following polynomials and decide whether $\text{Gal}(f) \subseteq A_n$:

1. $f(x) = x^3 + 3x + 1 \in \mathbb{Q}[x]$.
2. $f(x) = x^3 - 4x + 1 \in \mathbb{Q}[x]$.
3. $f(x) = x^3 - 7 \in \mathbb{Q}[x]$.
4. $f(x) = x^4 - x - 1 \in \mathbb{Q}[x]$. (Hint: the discriminant of a quartic $x^4 + px + q$ is $-27p^4 + 256q^3$.)

Exercise 44.2 (Cubics: complete classification). Let $f(x) = x^3 + px + q \in \mathbb{Q}[x]$ be irreducible.

1. Show that $\text{Gal}(f)$ is either S_3 or $A_3 \cong \mathbb{Z}/3\mathbb{Z}$, and that it is A_3 if and only if $\Delta(f) = -4p^3 - 27q^2$ is a perfect square in $\mathbb{Q}$.
2. Determine the Galois group for: $x^3 - 3x - 1$, $x^3 + 3x + 1$, $x^3 - 2$, $x^3 + x + 1$.
3. For $f(x) = x^3 - 3x - 1$, verify that the splitting field is $\mathbb{Q}(2\cos(2\pi/9))$ and that all three roots are real, giving a concrete example of A_3 .

Exercise 44.3 (Galois group of irreducible cubic: real roots criterion). Let $f(x) \in \mathbb{Q}[x]$ be an irreducible cubic.

1. Show that if f has exactly one real root (and two complex conjugate roots), then complex conjugation is a transposition in $\text{Gal}(f) \subseteq S_3$, so $\text{Gal}(f) = S_3$.
2. Show that if f has three real roots, then $\text{Gal}(f)$ could be either A_3 or S_3 , and the discriminant distinguishes the two cases.
3. The case of three real roots with $\text{Gal}(f) = S_3$ is the casus irreducibilis: the roots cannot be expressed using only real radicals.

Exercise 44.4 (Dedekind's method in practice). Let $f(x) = x^4 - x - 1 \in \mathbb{Q}[x]$.

1. Verify that f is irreducible over $\mathbb{Q}$.

- Factor f modulo 2, 3, and 5, and determine the cycle type in S_4 for each prime.
- Using the factorization data, determine $\text{Gal}(f)$. (Hint: the factorization mod 2 gives cycle type (4); this means the Galois group contains a 4-cycle. What transitive subgroups of S_4 contain a 4-cycle?)

Exercise 44.5 (Dedekind's method: another example). Let $f(x) = x^5 - 4x + 2 \in \mathbb{Q}[x]$.

- Verify irreducibility over $\mathbb{Q}$ (e.g. by Eisenstein's criterion).
- Factor f modulo 3 and show it is irreducible over $\mathbb{F}_3$. Deduce that $\text{Gal}(f)$ contains a 5-cycle.
- Factor f modulo 7 and find the cycle type. (You should find (2, 3), hence an odd permutation.)
- Conclude that $\text{Gal}(f) = S_5$. (Hint: classify transitive subgroups of S_5 containing a 5-cycle, then use the odd permutation from the modulo 7 factorization.)

Exercise 44.6 (Quartic Galois groups). The Galois group of an irreducible separable quartic $f(x) \in K[x]$ is a transitive subgroup of S_4 , hence one of: S_4 , A_4 , D_4 , $V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, or $\mathbb{Z}/4\mathbb{Z}$.

- Show that $\text{Gal}(f) \subseteq A_4$ if and only if $\Delta(f)$ is a square in K .
- The **resolvent cubic** of $f(x) = x^4 + px^2 + qx + r$ is

$$g(x) = x^3 - px^2 - 4rx + (4pr - q^2).$$

Show that $\text{Gal}(f)$ is contained in D_4 if and only if $g(x)$ has a root in K .

- Determine the Galois groups of $x^4 - 2$, $x^4 + 1$, and $x^4 - x - 1$ over $\mathbb{Q}$.

Exercise 44.7 (Cyclotomic Galois groups). 1. Show that $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.

- Determine the Galois group of $\mathbb{Q}(\zeta_7)/\mathbb{Q}$ and draw the subgroup lattice.
- What are the intermediate fields of $\mathbb{Q}(\zeta_7)/\mathbb{Q}$? (Hint: $(\mathbb{Z}/7\mathbb{Z})^\times \cong \mathbb{Z}/6\mathbb{Z}$ has subgroups of orders 1, 2, 3, 6.)

Challenge problems

Exercise 44.8 (Natural irrationalities). Let L/K be Galois with group G , and let M/K be an arbitrary finite extension. Assume L and M are contained in a common algebraic closure. Prove that

$$\text{Gal}(LM/M) \cong \text{Gal}(L/(L \cap M)).$$

This is sometimes called the theorem of natural irrationalities. (Hint: show that LM/M is Galois and restriction gives the isomorphism.)

Exercise 44.9 (The inverse Galois problem for abelian groups over $\mathbb{Q}$). The Kronecker–Weber theorem states that every finite abelian extension of $\mathbb{Q}$ is contained in a cyclotomic field $\mathbb{Q}(\zeta_n)$.

- Using the fact that $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, show that for every finite abelian group A there exists a Galois extension $L/\mathbb{Q}$ with $\text{Gal}(L/\mathbb{Q}) \cong A$. (Hint: by the Chinese Remainder Theorem, $(\mathbb{Z}/n\mathbb{Z})^\times$ surjects onto any quotient of the appropriate form.)

2. Find an explicit n and an intermediate field L of $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ such that $\text{Gal}(L/\mathbb{Q}) \cong \mathbb{Z}/5\mathbb{Z}$.

Exercise 44.10 (Galois group of a generic polynomial). Let $K = \mathbb{Q}(s_1, \dots, s_n)$ where $s_1, \dots, s_n$ are algebraically independent over $\mathbb{Q}$, and let

$$f(x) = x^n - s_1 x^{n-1} + s_2 x^{n-2} - \dots + (-1)^n s_n \in K[x].$$

This is called the **generic polynomial** of degree n .

1. Show that the splitting field of f over K is $L = \mathbb{Q}(x_1, \dots, x_n)$ where $x_1, \dots, x_n$ are independent variables and s_i is the i -th elementary symmetric polynomial in $x_1, \dots, x_n$.
2. Conclude that $\text{Gal}(f) = S_n$.
3. Deduce that $K = \mathbb{Q}(s_1, \dots, s_n) = L^{S_n}$, recovering the fundamental theorem of symmetric polynomials from Galois theory.

45 Application to classical problems

One of the supreme triumphs of Galois theory is providing complete, elegant answers to classical problems that eluded mathematicians for centuries: determining which regular polygons can be constructed by compass and straightedge, and proving that there is no general formula for the roots of polynomials of degree 5 or higher.

45.1 Constructibility of regular polygons

Recall that a number is constructible if and only if it lies in an extension of $\mathbb{Q}$ formed by a sequence of quadratic extensions. To construct a regular n -gon, we need to construct the primitive n -th root of unity $\zeta_n = e^{2\pi i/n}$, or equivalently, the length $\cos(2\pi/n)$. This means $[\mathbb{Q}(\zeta_n) : \mathbb{Q}]$ must be a power of 2.

Theorem 45.1 (Gauss-Wantzel Theorem). *A regular n -gon is constructible with compass and straightedge if and only if $n = 2^k p_1 p_2 \dots p_m$, where $k \geq 0$ and the p_i are distinct Fermat primes (primes of the form $p = 2^{2^i} + 1$).*

Proof. Let $\zeta_n = e^{2\pi i/n}$. A regular n -gon is constructible if and only if the point ζ_n is constructible in the plane, and this is equivalent to constructibility of the real number $\cos(2\pi/n)$. In particular, a necessary condition is that the cyclotomic extension $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ have degree a power of 2. Since $\mathbb{Q}(\zeta_n)$ is the splitting field of $x^n - 1$ over $\mathbb{Q}$, we know

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \phi(n).$$

Therefore constructibility implies that $\phi(n)$ is a power of 2.

Write

$$n = 2^k q_1^{a_1} \dots q_r^{a_r}$$

with the q_i distinct odd primes. Then

$$\phi(n) = \phi(2^k) \prod_{i=1}^r \phi(q_i^{a_i}) = 2^{k-1} \prod_{i=1}^r q_i^{a_i-1} (q_i - 1)$$

(with the usual convention $\phi(1) = 1$ and $\phi(2) = 1$). If $\phi(n)$ is a power of 2, then every odd factor on the right must be absent. Hence each exponent $a_i - 1$ must be 0, so every odd prime appears

only to the first power. Also each number $q_i - 1$ must itself be a power of 2. Thus every odd prime divisor q_i has the form

$$q_i = 2^s + 1.$$

If s were not a power of 2, say $s = 2^u v$ with v odd and $v > 1$, then

$$q_i = 2^{2^u v} + 1 = (2^{2^u})^v + 1$$

would be divisible by $2^{2^u} + 1$, so q_i would not be prime. Therefore s must be a power of 2, and each odd prime divisor of n is a Fermat prime. This proves the necessity.

Conversely, suppose

$$n = 2^k p_1 \cdots p_m$$

where the p_i are distinct Fermat primes. Then

$$\phi(n) = \phi(2^k) \prod_{i=1}^m (p_i - 1),$$

and every factor on the right is a power of 2. Hence $\phi(n) = 2^r$ for some $r \geq 0$.

Now $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is a finite Galois extension with Galois group

$$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times,$$

which has order $\phi(n) = 2^r$. In particular, this Galois group is a finite Abelian 2-group. Every finite 2-group has a subgroup of index 2, and therefore one can choose a chain of subgroups

$$\{1\} = H_0 \subset H_1 \subset \cdots \subset H_r = (\mathbb{Z}/n\mathbb{Z})^\times$$

with $[H_i : H_{i-1}] = 2$ for each i . By the Fundamental Theorem of Galois Theory, this corresponds to a tower of intermediate fields

$$\mathbb{Q} = K_0 \subset K_1 \subset \cdots \subset K_r = \mathbb{Q}(\zeta_n)$$

such that $[K_i : K_{i-1}] = 2$ for every i . Thus $\mathbb{Q}(\zeta_n)$ is obtained from $\mathbb{Q}$ by a sequence of quadratic extensions.

Any element of such a tower is constructible, so in particular both ζ_n and its real part

$$\cos(2\pi/n) = \frac{\zeta_n + \zeta_n^{-1}}{2}$$

are constructible. Therefore the regular n -gon is constructible. This proves the sufficiency. $\square$

For example, the regular 17-gon is constructible because $17 = 2^{2^2} + 1$ is a Fermat prime.

45.2 Solvable groups

To define "solvable by radicals", we first need the group-theoretic notion of solvability.

Definition 45.1. A group G is **solvable** if it has a subnormal series

$$\{1\} = G_0 \triangleleft G_1 \triangleleft G_2 \triangleleft \cdots \triangleleft G_k = G$$

such that every quotient group G_{i+1}/G_i is an Abelian group.

Example 45.1. All Abelian groups are solvable. S_3 is solvable because $\{1\} \triangleleft A_3 \triangleleft S_3$, and both A_3 (cyclic of order 3) and S_3/A_3 (cyclic of order 2) are Abelian. S_4 is also solvable, with subnormal series

$$\{1\} \triangleleft V_4 \triangleleft A_4 \triangleleft S_4,$$

where $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$ is the Klein four-group. The successive quotients $V_4/\{1\} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, $A_4/V_4 \cong \mathbb{Z}/3\mathbb{Z}$, and $S_4/A_4 \cong \mathbb{Z}/2\mathbb{Z}$ are all Abelian.

However, for $n \geq 5$, A_n is a simple non-Abelian group.

Theorem 45.2. A_5 is simple (has no non-trivial normal subgroups). Consequently, S_n is not solvable for $n \geq 5$.

Proof. We first prove that A_5 is simple. Since

$$|A_5| = \frac{5!}{2} = 60,$$

it is enough to show that A_5 has no non-trivial proper normal subgroup.

The elements of A_5 have the following cycle types:

- the identity;
- 3-cycles;
- products of two disjoint transpositions;
- 5-cycles.

There are 20 elements of type (abc) , since the number of 3-cycles in S_5 is

$$\binom{5}{3} \cdot 2 = 20.$$

There are 15 elements of type $(ab)(cd)$, because this is the number of ways to partition four of the five symbols into two unordered pairs. Finally, there are 24 different 5-cycles in S_5 , and all of them lie in A_5 because a 5-cycle is even.

To determine the conjugacy classes in A_5 , note that the 20 three-cycles form a single conjugacy class in A_5 , and the 15 double transpositions also form a single conjugacy class in A_5 . The 24 five-cycles split into two conjugacy classes of size 12 each. Thus the conjugacy classes of A_5 have sizes

$$1, \quad 20, \quad 15, \quad 12, \quad 12.$$

Now let $N \trianglelefteq A_5$ be a normal subgroup. Then N must be a union of conjugacy classes of A_5 containing the identity. Hence $|N|$ must be one of the sums

$$1, \quad 1 + 12 = 13, \quad 1 + 15 = 16, \quad 1 + 20 = 21, \quad 1 + 12 + 12 = 25, \quad 1 + 12 + 15 = 28,$$

$$1 + 12 + 20 = 33, \quad 1 + 15 + 20 = 36, \quad 1 + 12 + 12 + 15 = 40, \quad 1 + 12 + 12 + 20 = 45, \quad 1 + 12 + 15 + 20 = 48,$$

or 60. By Lagrange's theorem, $|N|$ must divide 60. Among these numbers, only 1 and 60 divide 60. Therefore the only normal subgroups of A_5 are $\{1\}$ and A_5 itself, so A_5 is simple.

Since A_5 is non-Abelian and simple, it is not solvable. For every $n \geq 5$, the subgroup A_5 embeds into A_n by letting it permute five symbols and fix the remaining ones. If S_n were solvable, then every subgroup would also be solvable; in particular A_5 would be solvable, a contradiction. Hence S_n is not solvable for $n \geq 5$. $\square$

The following basic closure properties of solvable groups are used constantly.

Proposition 45.1 (Closure properties of solvable groups). *1. Every subgroup of a solvable group is solvable.*

2. Every quotient group of a solvable group is solvable.

3. If $N \trianglelefteq G$ and both N and G/N are solvable, then G is solvable.

Proof. (1) Let $\{1\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G$ be a subnormal series with Abelian quotients, and let $H \leq G$. Set $H_i = H \cap G_i$. Then $H_i \trianglelefteq H_{i+1}$ (since $H_i = H \cap G_i \trianglelefteq H \cap G_{i+1} = H_{i+1}$), and the map $H_{i+1}/H_i \rightarrow G_{i+1}/G_i$ given by $hH_i \mapsto hG_i$ is an injective homomorphism. Since G_{i+1}/G_i is Abelian, so is H_{i+1}/H_i .

(2) Let $\pi: G \rightarrow G/N$ be the quotient map. Set $\overline{G}_i = \pi(G_i) = G_i N/N$. Then $\overline{G}_i \trianglelefteq \overline{G}_{i+1}$ and $\overline{G}_{i+1}/\overline{G}_i$ is a quotient of G_{i+1}/G_i (by the third isomorphism theorem), hence Abelian.

(3) Let $\{1\} = N_0 \trianglelefteq \cdots \trianglelefteq N_r = N$ and $\{N\} = \overline{G}_0 \trianglelefteq \cdots \trianglelefteq \overline{G}_s = G/N$ be subnormal series with Abelian quotients. Lifting the latter via the quotient map $\pi: G \rightarrow G/N$, we obtain subgroups $G_j = \pi^{-1}(\overline{G}_j)$ with $N = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_s = G$ and $G_{j+1}/G_j \cong \overline{G}_{j+1}/\overline{G}_j$ Abelian. Concatenating gives

$$\{1\} = N_0 \trianglelefteq \cdots \trianglelefteq N_r = N = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_s = G,$$

a subnormal series for G with Abelian quotients. $\square$

An equivalent characterization via the derived series is often useful.

Definition 45.2. *The **derived series** of a group G is defined by $G^{(0)} = G$ and $G^{(i+1)} = [G^{(i)}, G^{(i)}]$ (the commutator subgroup). A group is solvable if and only if $G^{(k)} = \{1\}$ for some k .*

45.3 Solvability by radicals

An extension $K(\alpha)/K$ is a **simple radical extension** if $\alpha^m \in K$ for some integer $m \geq 1$. A **radical extension** L/K is a tower of simple radical extensions:

$$K = K_0 \subset K_1 \subset \cdots \subset K_r = L, \quad K_i = K_{i-1}(\alpha_i), \quad \alpha_i^{m_i} \in K_{i-1}.$$

A polynomial $f(x) \in K[x]$ is **solvable by radicals** if its splitting field is contained in some radical extension of K . This formally captures the idea of expressing the roots of f using $+$, $-$, $\times$, $/$, $\sqrt[m]{}$.

The bridge to group theory is that each time one adjoins an m -th root, the new Galois group becomes built from a comparatively simple Abelian piece. The following key lemma makes this precise.

Lemma 45.1 (Adjoining roots of unity). *Let K be a field of characteristic 0 containing a primitive m -th root of unity ζ_m , and let α be an element of an extension of K with $\alpha^m = a \in K$. Then $K(\alpha)/K$ is a Galois extension with Abelian Galois group (in fact, cyclic of order dividing m).*

Proof. Since $\zeta_m \in K$, the polynomial $x^m - a$ splits over $K(\alpha)$: its roots are $\alpha, \zeta_m \alpha, \zeta_m^2 \alpha, \dots, \zeta_m^{m-1} \alpha$. Hence $K(\alpha)$ is the splitting field of $x^m - a$ over K , so the extension is normal (and separable, since $\text{char}(K) = 0$), hence Galois.

Any $\sigma \in \text{Gal}(K(\alpha)/K)$ must send α to some root $\zeta_m^{i(\sigma)} \alpha$. The map $\sigma \mapsto \zeta_m^{i(\sigma)}$ is an injective group homomorphism

$$\text{Gal}(K(\alpha)/K) \hookrightarrow \mu_m,$$

where μ_m is the cyclic group of m -th roots of unity. Hence the Galois group is cyclic of order dividing m . $\square$

Lemma 45.2 (Radical extensions give solvable Galois groups). *Let $\text{char}(K) = 0$. If L/K is a radical extension, then the Galois closure $\tilde{L}/K$ has solvable Galois group.*

Proof. Write the radical tower as

$$K = L_0 \subset L_1 \subset \cdots \subset L_r = L, \quad L_i = L_{i-1}(\alpha_i), \quad \alpha_i^{m_i} = a_i \in L_{i-1}.$$

For each i , let $\widetilde{L_i}$ be the Galois closure of L_i/K . We prove by induction on i that $\text{Gal}(\widetilde{L_i}/K)$ is solvable.

For $i = 0$, we have $\widetilde{L_0} = K$, so the Galois group is trivial.

Now assume $i \geq 1$ and that $\text{Gal}(\widetilde{L_{i-1}}/K)$ is solvable. Set

$$F_i = \widetilde{L_{i-1}}(\zeta_{m_i}),$$

where ζ_{m_i} is a primitive m_i -th root of unity. Since $K(\zeta_{m_i})/K$ is cyclotomic, it is Galois with Abelian Galois group; hence F_i/K , being a compositum of Galois extensions over K , is Galois. Moreover, the restriction map gives a surjection

$$\text{Gal}(F_i/K) \twoheadrightarrow \text{Gal}(\widetilde{L_{i-1}}/K),$$

whose kernel is the Abelian group $\text{Gal}(F_i/\widetilde{L_{i-1}})$. By Proposition 45.1(3), $\text{Gal}(F_i/K)$ is solvable.

For each $\sigma \in \text{Gal}(F_i/K)$, let $N_{i,\sigma}$ be the splitting field over F_i of the polynomial

$$x^{m_i} - \sigma(a_i).$$

Because $\zeta_{m_i} \in F_i$, Lemma 45.1 shows that each extension $N_{i,\sigma}/F_i$ is Galois with cyclic Galois group of order dividing m_i . Let M_i be the compositum of all these fields $N_{i,\sigma}$ inside a fixed algebraic closure. Since a compositum of Galois extensions over the same base field is again Galois, M_i/F_i is Galois; moreover,

$$\text{Gal}(M_i/F_i) \hookrightarrow \prod_{\sigma \in \text{Gal}(F_i/K)} \text{Gal}(N_{i,\sigma}/F_i),$$

so $\text{Gal}(M_i/F_i)$ is solvable.

Now consider the polynomial

$$P_i(x) = \prod_{\sigma \in \text{Gal}(F_i/K)} (x^{m_i} - \sigma(a_i)).$$

Every $\tau \in \text{Gal}(F_i/K)$ merely permutes the factors of $P_i(x)$, so the coefficients of $P_i(x)$ are fixed by $\text{Gal}(F_i/K)$. Hence

$$P_i(x) \in F_i^{\text{Gal}(F_i/K)}[x] = K[x].$$

By construction, M_i is the splitting field of $P_i(x)$ over F_i , hence also over K . Therefore M_i/K is Galois.

Since α_i is a root of $x^{m_i} - a_i$, we have

$$L_i = L_{i-1}(\alpha_i) \subset F_i(\alpha_i) = N_{i,\text{id}} \subset M_i.$$

Because M_i/K is Galois and contains L_i , the Galois closure $\widetilde{L}_i$ is contained in M_i . The restriction map

$$\text{Gal}(M_i/K) \twoheadrightarrow \text{Gal}(\widetilde{L}_i/K)$$

is therefore surjective. On the other hand, since F_i/K is Galois and M_i/K is Galois, restriction also gives a surjection

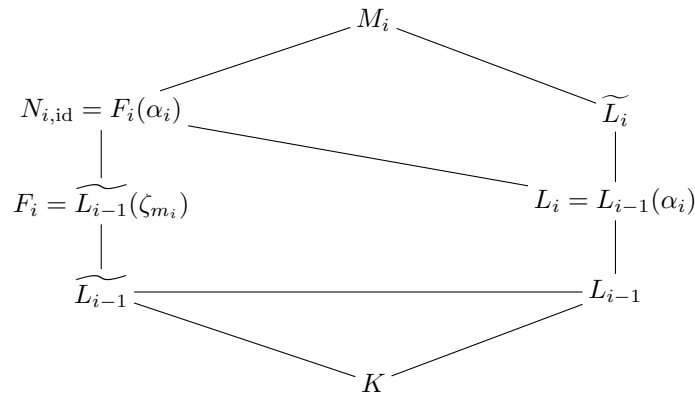
$$\text{Gal}(M_i/K) \twoheadrightarrow \text{Gal}(F_i/K)$$

with kernel $\text{Gal}(M_i/F_i)$. Both kernel and quotient are solvable, so Proposition 45.1(3) implies that $\text{Gal}(M_i/K)$ is solvable. Hence its quotient $\text{Gal}(\widetilde{L}_i/K)$ is solvable by Proposition 45.1(2).

This completes the induction. For $i = r$, we obtain that $\text{Gal}(\widetilde{L}/K)$ is solvable. $\square$

Remark 45.1 (One induction step as a Hasse diagram). *The point of the proof is that adjoining a single radical α_i is not enough: even after adjoining roots of unity, the field $F_i(\alpha_i)/K$ need not be Galois. To repair this, one adjoins all K -conjugate radical equations $x^{m_i} - \sigma(a_i)$ and takes the compositum of their splitting fields.*

At the i -th step, the fields are arranged as follows:



Here $N_{i,\text{id}}/F_i$ is cyclic Galois by Lemma 45.1, M_i/F_i is a compositum of such cyclic Galois extensions, and M_i/K is Galois because it is the splitting field over K of

$$P_i(x) = \prod_{\sigma \in \text{Gal}(F_i/K)} (x^{m_i} - \sigma(a_i)).$$

The Galois closure $\widetilde{L}_i$ sits inside M_i , which is why proving solvability for M_i/K is enough.

For the converse direction of the main theorem, we need the reverse implication: once the relevant roots of unity are present, cyclic Galois extensions are obtained by adjoining radicals.

Proposition 45.2 (Converse to Kummer theory). *Let E/F be a cyclic Galois extension of degree p where p is a prime integer. Assume $\text{char}(F) = 0$ and that F contains a primitive p -th root of unity ζ_p . Then there exists $\alpha \in E$ such that*

$$E = F(\alpha) \quad \text{and} \quad \alpha^p \in F.$$

Proof. Let $G = \text{Gal}(E/F) = \langle \sigma \rangle$, and view σ as an F -linear endomorphism of the F -vector space E . Since $\sigma^p = \text{id}$, the minimal polynomial of σ divides $x^p - 1$. Because F contains ζ_p and $\text{char}(F) = 0$, the polynomial $x^p - 1$ splits over F into distinct linear factors. Therefore σ is diagonalizable and all its eigenvalues are p -th roots of unity.

Since σ is diagonalizable, its order is the least common multiple of the orders of its eigenvalues. But σ has order p , so at least one eigenvalue λ has order exactly p . Choose $0 \neq \alpha \in E$ with $\sigma(\alpha) = \lambda\alpha$.

Then

$$\sigma(\alpha^p) = \sigma(\alpha)^p = (\lambda\alpha)^p = \lambda^p \alpha^p = \alpha^p,$$

so $\alpha^p \in E^G = F$. Moreover, the G -orbit of α is

$$\alpha, \sigma(\alpha), \sigma^2(\alpha), \dots, \sigma^{p-1}(\alpha) = \alpha, \lambda\alpha, \lambda^2\alpha, \dots, \lambda^{p-1}\alpha,$$

and these are distinct because λ has order p . Hence the minimal polynomial of α over F has degree p . Since $[E : F] = p$, we conclude that $E = F(\alpha)$, (here you need to use the result from exercise 42.2 that the degree of the minimal polynomial of an element is equal to the length of its orbit under Galois action). You can also use multiplicative property from degree of field extensions and p being prime to conclude that $[F(\alpha) : F]$ must be either 1 or p , and it cannot be 1 since $\alpha \notin F$. Therefore $[F(\alpha) : F] = p$, so $E = F(\alpha)$. and these are distinct because λ has order p . Therefore the minimal polynomial of α over F has degree at least p . On the other hand, its degree is at most $[E : F] = p$, so in fact

$$[F(\alpha) : F] = p.$$

Since $F(\alpha) \subseteq E$ and $[E : F] = p$, we conclude that $E = F(\alpha)$. □

Remark 45.2. Classically one constructs α by a Lagrange resolvent. If σ generates $\text{Gal}(E/F)$ and $\beta \in E$, define

$$R_\beta = \beta + \zeta_p^{-1}\sigma(\beta) + \zeta_p^{-2}\sigma^2(\beta) + \dots + \zeta_p^{-(p-1)}\sigma^{p-1}(\beta).$$

Then $\sigma(R_\beta) = \zeta_p R_\beta$. For a suitable choice of β , this resolvent is nonzero, so $R_\beta^p \in F$ and $E = F(R_\beta)$.

We can now state and prove the main theorem connecting solvability by radicals with solvability of the Galois group. The forward implication uses Lemma 45.2, whose proof explains exactly why one must adjoin not only roots of unity but also all K -conjugate radical equations in order to recover a Galois extension over the original base field.

Theorem 45.3 (Galois). *Let $f(x) \in K[x]$ where $\text{char}(K) = 0$. Then $f(x)$ is solvable by radicals if and only if its Galois group $\text{Gal}(f)$ is a solvable group.*

Proof. ($\Rightarrow$) If f is solvable by radicals, let L be the splitting field of f and choose a radical extension M/K with $L \subseteq M$. Let $\widetilde{M}/K$ be the Galois closure of M/K . By Lemma 45.2, the group $\text{Gal}(\widetilde{M}/K)$ is solvable. Since L/K is Galois and $L \subseteq \widetilde{M}$, restriction gives a surjective homomorphism

$$\text{Gal}(\widetilde{M}/K) \twoheadrightarrow \text{Gal}(L/K) = \text{Gal}(f).$$

Therefore $\text{Gal}(f)$ is solvable by Proposition 45.1(2).

($\Leftarrow$) Let L be the splitting field of f , and set $G = \text{Gal}(L/K)$. Since G is solvable, we may choose a descending subnormal series

$$G = G_0 \supseteq G_1 \supseteq \dots \supseteq G_r = \{1\}, \quad G_i \trianglelefteq G_{i-1},$$

such that each quotient G_{i-1}/G_i is cyclic of prime order p_i .

Set $F_i = L^{G_i}$. By the Galois correspondence,

$$K = F_0 \subset F_1 \subset \cdots \subset F_r = L,$$

and each extension F_i/F_{i-1} is Galois with cyclic Galois group G_{i-1}/G_i of order p_i .

Now adjoin all the necessary roots of unity at once:

$$K' = K(\zeta_{p_1}, \dots, \zeta_{p_r}).$$

This is already a radical extension of K , since each ζ_{p_i} satisfies $\zeta_{p_i}^{p_i} = 1 \in K$. Put $E_i = F_i K'$. Then

$$K' = E_0 \subset E_1 \subset \cdots \subset E_r = LK'.$$

Because F_i/F_{i-1} is Galois, each extension E_i/E_{i-1} is also Galois. Moreover, any automorphism of E_i fixing E_{i-1} preserves the normal subfield F_i , so restriction gives an injective homomorphism

$$\text{Gal}(E_i/E_{i-1}) \hookrightarrow \text{Gal}(F_i/F_{i-1}).$$

Hence $\text{Gal}(E_i/E_{i-1})$ is either trivial or cyclic of order p_i . In the nontrivial case, E_{i-1} contains ζ_{p_i} , so Proposition 45.2 shows that

$$E_i = E_{i-1}(\alpha_i) \quad \text{with} \quad \alpha_i^{p_i} \in E_{i-1}.$$

Thus each step E_i/E_{i-1} is either trivial or a simple radical extension.

Therefore LK'/K' is a radical extension. Since K'/K is radical, concatenating the two towers shows that LK'/K is radical. Because $L \subset LK'$, the polynomial f is solvable by radicals. $\square$

Example 45.2 (The abstract shape of the cubic formula). *Let L be the splitting field of an irreducible cubic $f(x) \in \mathbb{Q}[x]$ with $\text{Gal}(L/\mathbb{Q}) \cong S_3$; for example, $f(x) = x^3 - 2$. The subnormal series*

$$S_3 \supseteq A_3 \supseteq \{1\}$$

produces, via the Galois correspondence, a tower

$$\mathbb{Q} \subset F \subset L, \quad [F : \mathbb{Q}] = 2, \quad [L : F] = 3,$$

with L/F cyclic of degree 3. After adjoining ζ_3 if necessary, Proposition 45.2 yields an element α such that

$$L(\zeta_3) = F(\zeta_3, \alpha), \quad \alpha^3 \in F(\zeta_3).$$

Thus the splitting field is built from a quadratic radical step followed by a cubic radical step. This is the group-theoretic skeleton behind Cardano's formula. In the concrete example $f(x) = x^3 - 2$, one may take $F = \mathbb{Q}(\zeta_3)$ and $\alpha = \sqrt[3]{2}$.

Thus the algebraic problem of finding formulas by radicals is converted into a finite-group problem. Since $\text{Gal}(f)$ for degree ≤ 4 is a subgroup of S_4 , which is solvable, equations of degree up to 4 can always be solved by radicals (e.g., cubic and quartic formulas). To disprove the existence of a general radical formula for degree 5, it therefore suffices to exhibit one irreducible quintic whose Galois group is not solvable.

45.4 Insolvability of the general quintic

To prove that there is no general formula for the quintic (the **Abel–Ruffini Theorem**), one only needs to find a single degree 5 polynomial over $\mathbb{Q}$ whose Galois group is not solvable.

Proposition 45.3. *Let p be a prime. If $f(x) \in \mathbb{Q}[x]$ is an irreducible polynomial of degree p that has exactly two non-real roots in $\mathbb{C}$, then $\text{Gal}(f) \cong S_p$.*

Proof. Let L be the splitting field of f over $\mathbb{Q}$, and let $G = \text{Gal}(L/\mathbb{Q})$. By permuting the roots of f , we may regard G as a subgroup of S_p .

Because f is irreducible over $\mathbb{Q}$, the action of G on the set of roots is transitive. Equivalently, for any root α of f , the orbit of α under G consists of all p roots of f . By the orbit-stabilizer theorem, the orbit size p divides $|G|$. Since p is prime, Cauchy's theorem implies that G contains an element of order p . Viewed as a permutation of p roots, such an element must be a p -cycle.

Now use the hypothesis on the roots. Since f has exactly two non-real roots, they must be complex conjugates of one another, while the remaining $p-2$ roots are real. Complex conjugation is an automorphism of L fixing $\mathbb{Q}$, so it belongs to G . As a permutation of the roots, it swaps the two non-real roots and fixes every real root. Therefore G contains a transposition.

Finally, a transitive subgroup of S_p that contains a transposition and a p -cycle must be all of S_p . Indeed, after relabeling roots we may assume the p -cycle is $(1\ 2\ \dots\ p)$ and the transposition is $(i\ j)$. Conjugating $(i\ j)$ by powers of the p -cycle produces transpositions of the form

$$(i+k,\ j+k)$$

modulo p , and these generate all adjacent transpositions; hence they generate all of S_p . Therefore $G = S_p$. $\square$

Consider $f(x) = x^5 - 4x + 2$. By Eisenstein's criterion with $p = 2$ (check: $2 \nmid 1$, $2|0$, $2|0$, $2|0$, $2|-4$, $2|2$, and $4 \nmid 2$), it is irreducible over $\mathbb{Q}$. To determine the number of real roots, note that

$$f'(x) = 5x^4 - 4.$$

Hence the only critical points are

$$x = \pm a, \quad a := \left(\frac{4}{5}\right)^{1/4}.$$

Since $f'(x) > 0$ for $|x| > a$ and $f'(x) < 0$ for $|x| < a$, the function f is increasing on $(-\infty, -a)$, decreasing on $(-a, a)$, and increasing again on (a, ∞) . Thus $x = -a$ is a local maximum and $x = a$ is a local minimum.

Now compute the signs at the critical points. Because $a^4 = 4/5$, we have $a^5 = (4/5)a$, so

$$f(a) = a^5 - 4a + 2 = \frac{4}{5}a - 4a + 2 = 2 - \frac{16}{5}a < 0$$

since $a > (1/2)^{1/4} > 1/2$. Similarly,

$$f(-a) = -a^5 + 4a + 2 = -\frac{4}{5}a + 4a + 2 = 2 + \frac{16}{5}a > 0.$$

Because $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$, the monotonicity just established shows that:

- there is one real root in $(-\infty, -a)$;

- there is one real root in $(-a, a)$;
- there is one real root in (a, ∞) .

Therefore f has exactly three real roots. Since $\deg f = 5$, the remaining two roots must be non-real, and because the coefficients are real they form a complex conjugate pair. By the proposition above, $\text{Gal}(f) \cong S_5$. Because S_5 is not solvable, $x^5 - 4x + 2 = 0$ cannot be solved by radicals.

Remark 45.3. *This example is also a good illustration of how different Galois-theoretic tools reinforce one another. Earlier, using Dedekind's method modulo primes, one can already gather evidence that $\text{Gal}(x^5 - 4x + 2)$ should be large: for suitable primes p , the reduction modulo p exhibits cycle types such as a 5-cycle and a transposition. The argument given here is cleaner and more conceptual: irreducibility supplies transitivity, and the existence of exactly two non-real roots identifies complex conjugation with a transposition. Together with the prime-degree hypothesis, this forces the whole group to be S_5 .*

Remark 45.4 (Finite-field evidence for $x^5 - 4x + 2$). *The earlier Dedekind discussion can be applied concretely here. Modulo 3 we obtain*

$$x^5 - 4x + 2 \equiv x^5 - x - 1 \pmod{3},$$

and this polynomial is irreducible over $\mathbb{F}_3$. Hence $\text{Gal}(f)$ contains a 5-cycle. Modulo 5 we have

$$x^5 - 4x + 2 \equiv x^5 + x + 2 \equiv (x - 4)(x^4 + 4x^3 + x^2 + 4x + 2) \pmod{5},$$

so the factorization type is $(1, 4)$. Therefore the Galois group contains an element of cycle type $(1)(4)$. These two pieces of finite-field data already show that the group is far from small. What they do not immediately provide is a transposition. That is exactly where the real-root argument above becomes decisive: the two non-real roots produce complex conjugation, hence a transposition, and the proposition then gives $\text{Gal}(f) = S_5$.

45.5 Exercises

Constructibility

Exercise 45.1 (Constructibility of regular polygons). *Determine whether each of the following regular polygons is constructible with compass and straightedge, and justify your answer:*

1. 9-gon.
2. 15-gon.
3. 20-gon.
4. 257-gon.
5. 100-gon.

Exercise 45.2. *Show that the regular p^2 -gon is not constructible for any odd prime p .*

Exercise 45.3. *Let $n \geq 3$. Show that the angle $2\pi/n$ is constructible if and only if the regular n -gon is constructible.*

Exercise 45.4. *Prove that the angle $\pi/9$ (i.e. 20°) is not constructible. Deduce that it is impossible to trisect 60° with compass and straightedge.*

Solvable groups

Exercise 45.5. Show that every group of order p^n (p prime, $n \geq 1$) is solvable.

(Hint: use the fact that p -groups have nontrivial centers.)

Exercise 45.6 (Burnside's theorem, statement only). Every group of order $p^a q^b$ (with p, q prime) is solvable. Without proving this, verify it directly for $|G| = 12$ by listing the possible groups.

Exercise 45.7. Construct an explicit subnormal series for the dihedral group D_n to show that it is solvable. What is the derived length (the length of the shortest such series)?

Exercise 45.8. Show that A_4 is solvable but not nilpotent. In particular, exhibit its derived series.

Exercise 45.9 (Derived series). Compute the derived series of S_4 . At which step does it terminate?

Solvability by radicals

Exercise 45.10 (A cyclic cubic and converse to Kummer). Let L be the splitting field of $f(x) = x^3 - 3x - 1$ over $\mathbb{Q}$. By Example 43.2, we know that $\text{Gal}(L/\mathbb{Q}) \cong A_3 \cong \mathbb{Z}/3\mathbb{Z}$.

1. Show that $L/\mathbb{Q}$ is cyclic of degree 3.
2. Show that $L \neq \mathbb{Q}(\sqrt[3]{a})$ for every $a \in \mathbb{Q}$.
3. Let $M = L(\zeta_3)$. Use Proposition 45.2 to show that $M = \mathbb{Q}(\zeta_3, \alpha)$ for some α with $\alpha^3 \in \mathbb{Q}(\zeta_3)$.
4. Explain why this illustrates the necessity of adjoining roots of unity in the converse direction of Theorem 45.3.

Exercise 45.11. Show that the Galois group of $x^5 - 2$ over $\mathbb{Q}$ is the Frobenius group $F_{20} = \mathbb{Z}/5\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$ (a solvable group of order 20). Is this polynomial solvable by radicals?

Exercise 45.12. Let $f(x) = x^4 - 2$. Find the splitting field L of f over $\mathbb{Q}$ and show that $\text{Gal}(L/\mathbb{Q}) \cong D_4$. Exhibit an explicit radical tower that produces L .

Exercise 45.13. Show that every polynomial of degree ≤ 4 over a field of characteristic 0 is solvable by radicals. (Hint: every transitive subgroup of S_4 is solvable.)

Exercise 45.14. Find a polynomial of degree 6 over $\mathbb{Q}$ that is solvable by radicals but whose Galois group is nonabelian.

Exercise 45.15 (An irreducible quintic with $\text{Gal} = D_5$). Consider $f(x) = x^5 + x^4 - 4x^3 - 3x^2 + 3x + 1$, the minimal polynomial of $2 \cos(2\pi/11)$.

1. Show that f is irreducible over $\mathbb{Q}$.
2. Show that all roots of f are real.
3. Deduce that $\text{Gal}(f) \subseteq A_5$.
4. Show that $\text{Gal}(f) \cong \mathbb{Z}/5\mathbb{Z}$ (hence f is solvable by radicals).

Exercise 45.16. Let $f(x) = x^5 - 4x + 2$ and let p be a prime with $p \nmid \Delta(f)$. Using a computer or by hand, factor f modulo primes $p = 3, 5, 7, 11, 13$ and list the cycle types you observe. Verify that the set of cycle types is consistent with $\text{Gal}(f) \cong S_5$.

Extra problems

Exercise 45.17. Let G be a finite group. Show that G is solvable if and only if every composition factor of G is cyclic of prime order. (Use the Jordan–Hölder theorem.)

Exercise 45.18 (Natural irrationalities). Let $f(x) \in \mathbb{Q}[x]$ be irreducible of degree n , and let L be its splitting field. Let $E/\mathbb{Q}$ be a radical extension containing L . Show that if $\gcd(n, [E : L]) = 1$, then L is itself a radical extension of $\mathbb{Q}$.

(This is a strengthening of the idea that the “extra” roots of unity one must adjoin can sometimes be avoided.)

46 Optional topic: an explicit algebraic closure of a finite field

Fix a prime number p . To avoid set-theoretic ambiguity, we first choose a fixed algebraic closure Ω of $\mathbb{F}_p$. For each $n \geq 1$, let

$$\mathbb{F}_{p^n}(\Omega) := \{x \in \Omega : x^{p^n} = x\}.$$

As we will see below, this is a subfield of Ω with exactly p^n elements, hence isomorphic to the abstract finite field usually denoted $\mathbb{F}_{p^n}$. In this way, every $\mathbb{F}_{p^n}$ is realized as an actual subfield of the same ambient field Ω , so it now makes literal sense to form the union

$$\overline{\mathbb{F}}_p := \bigcup_{n \geq 1} \mathbb{F}_{p^n}.$$

More precisely, with the above convention one should read this as

$$\overline{\mathbb{F}}_p := \bigcup_{n \geq 1} \mathbb{F}_{p^n}(\Omega) \subseteq \Omega.$$

We will show that this is an algebraic closure of $\mathbb{F}_p$.

Proposition 46.1. *The field $\overline{\mathbb{F}}_p$ is algebraic over $\mathbb{F}_p$.*

Proof. Every element of $\overline{\mathbb{F}}_p$ lies in some finite extension $\mathbb{F}_{p^n}/\mathbb{F}_p$, hence is algebraic over $\mathbb{F}_p$. Therefore the whole union is algebraic over $\mathbb{F}_p$. $\square$

To prove that $\overline{\mathbb{F}}_p$ is algebraically closed, we use the polynomial

$$x^{p^n} - x.$$

Its derivative is -1 , so it has no repeated roots. Over any extension of $\mathbb{F}_p$, its roots are exactly the elements fixed by the n -th power of the Frobenius map $x \mapsto x^p$.

Theorem 46.1. *For each $n \geq 1$, the set of roots of $x^{p^n} - x$ in an algebraic closure of $\mathbb{F}_p$ is precisely the field $\mathbb{F}_{p^n}$.*

Proof. Every element $a \in \mathbb{F}_{p^n}$ satisfies $a^{p^n} = a$, so $\mathbb{F}_{p^n}$ is contained in the root set of $x^{p^n} - x$. Conversely, the polynomial $x^{p^n} - x$ has degree p^n and no repeated roots, so it has at most p^n roots. Since $\mathbb{F}_{p^n}$ already contributes p^n distinct roots, these are all the roots. $\square$

In particular, once we work inside the fixed ambient field Ω , the inclusions among these finite fields are canonical: if $m \mid n$, then every root of $x^{p^m} - x$ is also a root of $x^{p^n} - x$, so

$$\mathbb{F}_{p^m}(\Omega) \subseteq \mathbb{F}_{p^n}(\Omega).$$

Thus the family $\{\mathbb{F}_{p^n}(\Omega)\}_{n \geq 1}$ forms a directed system of subfields of Ω , and the union above is a genuine directed union of fields.

This already shows that the finite fields fit together in a very rigid way. In fact, if $m \mid n$, then $\mathbb{F}_{p^m}$ is exactly the unique subfield of $\mathbb{F}_{p^n}$ with p^m elements. Moreover, as we saw in the section on Galois extensions, the extension $\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois with cyclic Galois group generated by Frobenius:

$$\text{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) = \langle \varphi \rangle, \quad \varphi(x) = x^p,$$

and this group has order n .

Theorem 46.2. *The field $\overline{\mathbb{F}}_p = \bigcup_{n \geq 1} \mathbb{F}_{p^n}(\Omega)$ is algebraically closed.*

Proof. Let $f(x) \in \overline{\mathbb{F}}_p[x]$ be non-constant. Since there are only finitely many coefficients, all coefficients of f lie in some finite field $\mathbb{F}_{p^m}$. Let $d = \deg f$.

Now consider the finite extension $\mathbb{F}_{p^{md}}/\mathbb{F}_{p^m}$. By the finite field theorem proved earlier, the multiplicative group $\mathbb{F}_{p^{md}}^\times$ is cyclic of order $p^{md} - 1$. Hence every nonzero element of $\mathbb{F}_{p^{md}}$ satisfies $x^{p^{md}-1} = 1$, and therefore every element of $\mathbb{F}_{p^{md}}$ is a root of $x^{p^{md}} - x$. So $x^{p^{md}} - x$ splits completely over $\mathbb{F}_{p^{md}}$.

But f is a polynomial over $\mathbb{F}_{p^m}$ of degree d , and every irreducible factor of f over $\mathbb{F}_{p^m}$ has degree at most d . If $g(x)$ is such an irreducible factor of degree r , then the quotient field

$$\mathbb{F}_{p^m}[x]/(g(x))$$

has size p^{mr} , hence is isomorphic to $\mathbb{F}_{p^{mr}}$. Since $r \mid md$, we may view $\mathbb{F}_{p^{mr}}$ as a subfield of $\mathbb{F}_{p^{md}}$. Therefore every irreducible factor of f splits in $\mathbb{F}_{p^{md}}$, and so f itself splits in $\mathbb{F}_{p^{md}} \subseteq \overline{\mathbb{F}}_p$.

Thus every non-constant polynomial over $\overline{\mathbb{F}}_p$ has a root (indeed, splits completely) in $\overline{\mathbb{F}}_p$. Hence $\overline{\mathbb{F}}_p$ is algebraically closed. $\square$

Remark 46.1. *If one prefers not to choose a background algebraic closure in advance, one can phrase the same construction in slightly more abstract language. For each divisibility relation $m \mid n$, there is an $\mathbb{F}_p$ -embedding $\mathbb{F}_{p^m} \hookrightarrow \mathbb{F}_{p^n}$, unique up to a unique $\mathbb{F}_p$ -automorphism of the target. One may then form the direct limit*

$$\varinjlim_n \mathbb{F}_{p^n}$$

over the divisibility poset. The result is a field, and after choosing an embedding into an algebraic closure Ω it identifies with the concrete union $\bigcup_n \mathbb{F}_{p^n}(\Omega)$ above. For purposes of computation, however, the concrete realization inside a fixed Ω is the easiest model to keep in mind.

Remark 46.2. *This example is worth comparing with the case of $\mathbb{R} \subset \mathbb{C}$. For $\mathbb{R}$, the algebraic closure is a single quadratic extension. For $\mathbb{F}_p$, the algebraic closure is an infinite union of finite cyclic Galois extensions. In both cases, the algebraic closure is described very concretely, and the Galois group plays a decisive role. By contrast, for $\mathbb{Q}$ or for the p -adic field $\mathbb{Q}_p$, the algebraic closure exists and is fundamental, but it is much less concrete. One can still say a great deal about finite Galois extensions of $\mathbb{Q}_p$ using Hensel's lemma, ramification theory, and local class field theory, but there is no comparably simple description of the entire algebraic closure as in the finite-field case.*

46.1 Optional topic: the Newton–Puisseux viewpoint

There is another important family of “explicit” algebraic closures, this time for function fields rather than finite fields. Consider the field of formal Laurent series

$$\mathbb{C}((t)) = \left\{ \sum_{k \geq k_0} a_k t^k : a_k \in \mathbb{C}, k_0 \in \mathbb{Z} \right\}.$$

For each $n \geq 1$, we may adjoin an n -th root of t and obtain the extension

$$\mathbb{C}((t^{1/n})) = \left\{ \sum_{k \geq k_0} a_k t^{k/n} : a_k \in \mathbb{C}, k_0 \in \mathbb{Z} \right\}.$$

To make the notation precise, one may fix a background algebraic closure Ξ of $\mathbb{C}((t))$ and choose, for each n , an element $t^{1/n} \in \Xi$ satisfying $(t^{1/n})^n = t$, in a compatible way whenever $m \mid n$; for instance, one asks that

$$(t^{1/n})^{n/m} = t^{1/m}.$$

Then $\mathbb{C}((t^{1/m})) \subseteq \mathbb{C}((t^{1/n}))$ whenever $m \mid n$, so these fields form a directed system inside the common ambient field Ξ . These are called **Puisseux series fields**. Their union

$$\mathbb{C}\langle\langle t \rangle\rangle := \bigcup_{n \geq 1} \mathbb{C}((t^{1/n}))$$

is the field of **Puisseux series**.

If one prefers a more invariant formulation, one can also regard $\mathbb{C}\langle\langle t \rangle\rangle$ as the direct limit of the directed system $\{\mathbb{C}((t^{1/n}))\}$ indexed by divisibility. For most concrete computations, however, it is harmless and convenient to think of all these fields as honest subfields of a fixed algebraic closure Ξ .

The first pleasant fact is that each extension $\mathbb{C}((t^{1/n}))/\mathbb{C}((t))$ is a very concrete finite Galois extension.

Proposition 46.2. *Assume $\mathbb{C}$ contains all n -th roots of unity. Then $\mathbb{C}((t^{1/n}))/\mathbb{C}((t))$ is a Galois extension of degree n , and*

$$\text{Gal}(\mathbb{C}((t^{1/n}))/\mathbb{C}((t))) \cong \mu_n,$$

where μ_n denotes the group of n -th roots of unity in $\mathbb{C}$. Explicitly, for $\zeta \in \mu_n$, the corresponding automorphism sends

$$t^{1/n} \mapsto \zeta t^{1/n}.$$

Proof. The element $t^{1/n}$ satisfies the polynomial $x^n - t \in \mathbb{C}((t))[x]$, so the extension has degree at most n . On the other hand, $1, t^{1/n}, \dots, t^{(n-1)/n}$ are linearly independent over $\mathbb{C}((t))$, so the degree is exactly n . Since $\mathbb{C}$ contains all roots of $x^n - 1$, the polynomial $x^n - t$ splits in $\mathbb{C}((t^{1/n}))$ as

$$x^n - t = \prod_{\zeta \in \mu_n} (x - \zeta t^{1/n}).$$

Therefore the extension is normal and separable, hence Galois. Every $\mathbb{C}((t))$ -automorphism is determined by the image of $t^{1/n}$, and that image must be another root of $x^n - t$, namely $\zeta t^{1/n}$ for some $\zeta \in \mu_n$. $\square$

This looks very similar to the cyclotomic and finite-field situations discussed elsewhere in the notes: one obtains a tower of explicit finite Galois extensions, and the Galois groups are easy to describe.

It is also helpful to see concretely why fractional exponents naturally appear when we try to solve algebraic equations over $\mathbb{C}((t))$.

Example 46.1. Consider the equation

$$y^2 = t.$$

This equation has no solution in $\mathbb{C}((t))$, because every nonzero Laurent series in $\mathbb{C}((t))$ has integer valuation, whereas a square must have even valuation. But it does have the obvious solution $y = t^{1/2}$ in $\mathbb{C}((t^{1/2}))$. So already the simplest quadratic equation forces us to adjoin a fractional power of t .

Example 46.2. Consider instead

$$y^2 = t + t^2 = t(1 + t).$$

Formally we would like to write

$$y = t^{1/2}(1 + t)^{1/2}.$$

Using the binomial series

$$(1 + t)^{1/2} = 1 + \frac{1}{2}t - \frac{1}{8}t^2 + \cdots,$$

we are led to the Puiseux expansion

$$y = t^{1/2} + \frac{1}{2}t^{3/2} - \frac{1}{8}t^{5/2} + \cdots,$$

which lies in $\mathbb{C}((t^{1/2}))$. This is a good toy model for the general Newton–Puiseux phenomenon: the roots of algebraic equations are often expressed not by ordinary power series, but by series in fractional powers of t .

Example 46.3. Here is a useful heuristic that already hints at the Newton polygon method. Suppose we try to solve

$$y^3 - t = 0$$

by looking for a solution whose leading term is ct^α with $c \in \mathbb{C}^\times$. Comparing the lowest powers of t , we get

$$(ct^\alpha)^3 = c^3 t^{3\alpha} \approx t.$$

So we must have $3\alpha = 1$, hence $\alpha = 1/3$, and the leading coefficient must satisfy $c^3 = 1$. This suggests three solutions with leading terms

$$t^{1/3}, \quad \zeta_3 t^{1/3}, \quad \zeta_3^2 t^{1/3}.$$

In this very simple example the heuristic already gives the exact roots. In more complicated equations it does not usually finish the problem, but it tells us the correct fractional exponent to try first.

This heuristic is systematized by the Newton polygon method: one records the valuations of the coefficients of a polynomial and reads off from the lower convex hull the possible leading exponents of its roots.

These examples suggest the philosophy behind the theorem below: if one allows all fractional powers $t^{1/n}$ at once, then one should expect to have enough room to solve arbitrary algebraic equations over $\mathbb{C}((t))$.

The deep theorem, called the **Newton–Puiseux theorem**, says that in characteristic 0 the field of Puiseux series is already algebraically closed.

Theorem 46.3 (Newton–Puisseux, statement only). *Over a field of characteristic 0 (in particular over $\mathbb{C}$), the field*

$$\bigcup_{n \geq 1} \mathbb{C}((t^{1/n}))$$

is algebraically closed.

We will not prove this theorem here, since a full proof requires a careful analysis of Newton polygons or an equivalent recursive construction of fractional-power series solutions. But even without the proof, the statement is very instructive: unlike the algebraic closure of $\mathbb{Q}$ or $\mathbb{Q}_p$, this algebraic closure is built from a completely explicit tower of finite Galois extensions.

Remark 46.3. *The analogy with the finite-field example is worth emphasizing.*

- For $\mathbb{F}_p$, the algebraic closure is the union of finite fields $\mathbb{F}_{p^n}$, and the Galois groups are generated by Frobenius.
- For $\mathbb{C}((t))$, the Newton–Puisseux theorem says that the algebraic closure is the union of the fields $\mathbb{C}((t^{1/n}))$, and each finite step has cyclic Galois group given by multiplying $t^{1/n}$ by roots of unity.

In both cases, the algebraic closure is approached through a family of very concrete finite Galois extensions.

Exercise 46.1. *Work inside a fixed algebraic closure of $\mathbb{C}((t))$.*

1. *Show that $x^4 - t$ is irreducible over $\mathbb{C}((t))$, and deduce that adjoining one root produces an extension of degree 4.*
2. *Show that all four roots lie in $\mathbb{C}((t^{1/4}))$ and describe explicitly the Galois group of $\mathbb{C}((t^{1/4}))/\mathbb{C}((t))$.*
3. *Compare this with the finite-field example above: what plays the role of Frobenius, and what plays the role of adjoining larger and larger finite extensions?*

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